Eco-Loop Building Agents

Building Models (.idf files)

Deliverable 2: the base baseline building file, along with the modified version generated during runtime evaluation by the closed-loop AI agent.

Contents:

`baseline.idf`

Unmodified DOE Small Office reference building - the base building file, no AI involved.

`ai_optimized_setpoints.idf`

Same building with HTGSETP_SCH / CLGSETP_SCH replaced by the schedule the closed-loop agent actually converged on during a live run (see logs/ai_decisions.csv for the full decision trail this was exported from).

Generated 2026-07-27 - github.com/srivatsa927/Eco-Loop-Building-Agents

```
! RefBldgSmallOfficeNew2004_Chicago.idf
! DOE Commercial Reference Building
! Small Office, new construction 90.1-2004
! ASHRAE Standards 90.1-2004 and 62-1999
!
! Description:  Single story, five zone office building.
! Form:         Area = 511 m2 (5,500 ft2); Number of Stories = 1; Shape = rectangle, Aspect ratio = 1.5
! Envelope:     Envelope thermal properties vary with climate according to ASHRAE Standard 90.1-2004.
!               Opaque constructions: mass walls; attic roof; slab-on-grade floor
!               Windows: window-to-wall ratio = 21.2%, equal distribution of punched windows
!               Infiltration = 0.4 cfm/ft2 above grade wall area at 0.3 in wc (75 Pa) adjusted to 0.016 in wc (4 Pa).
!                              25% of full value when ventilation system on.
! HVAC:         PSZ-AC, gas furnace
!               No economizers, per ASHRAE 90.1-2004
!
! Int. gains:   lights = 10.76 W/m2 (1.0 W/ft2) (building area method);
!               elec. plug loads = 10.76 W/m2 (1.0 W/ft2)
!               gas plug load = 0 W/m2 (0 W/ft2)
!               people = 28 total; 5.38/100 m2 (5.0/1000 ft2)
!
! Detached Shading:                 None
! Daylight:                         None
! Natural Ventilation:              None
! Zonal Equipment:                  None
! Air Primary Loops:                PSZ
! Plant Loops:                      SHWSys1
! System Equipment Autosize:        Yes
! Purchased Cooling:                None
! Purchased Heating:                None
! Coils:                            Coil:Cooling:DX:SingleSpeed; Coil:Heating:Fuel
! Pumps:                            None
! Boilers:                          None
! Chillers:                         None
!
! Reference citation for the Commercial Reference Buildings:
! Deru, M.; Field, K.; Studer, D.; Benne, K.; Griffith, B.; Torcellini, P;
! Halverson, M.; Winiarski, D.; Liu, B.; Rosenberg, M.; Huang, J.;
! Yazdanian, M.; Crawley, D. (2010).
! U.S. Department of Energy Commercial Reference Building Models of the National Building Stock.
! Washington, DC: U.S. Department of Energy, Energy Efficiency and
! Renewable Energy, Office of Building Technologies.
! ***GENERAL SIMULATION PARAMETERS***
! Number of Zones: 6

  Version,26.1;

  SimulationControl,
    YES,                     !- Do Zone Sizing Calculation
    YES,                     !- Do System Sizing Calculation
    YES,                     !- Do Plant Sizing Calculation
    NO,                      !- Run Simulation for Sizing Periods
    YES,                     !- Run Simulation for Weather File Run Periods
    No,                      !- Do HVAC Sizing Simulation for Sizing Periods
    1;                       !- Maximum Number of HVAC Sizing Simulation Passes

  Building,
    Ref Bldg Small Office New2004_v1.3_5.0,  !- Name
    0.0000,                  !- North Axis {deg}
    City,                    !- Terrain
    0.0400,                  !- Loads Convergence Tolerance Value {W}
    0.2000,                  !- Temperature Convergence Tolerance Value {deltaC}
    FullInteriorAndExterior, !- Solar Distribution
    25,                      !- Maximum Number of Warmup Days
    6;                       !- Minimum Number of Warmup Days

  RunPeriod,
```

```
    14,                      !- Begin Day of Month
    ,                        !- Begin Year
    7,                       !- End Month
    17,                      !- End Day of Month
    ,                        !- End Year
    Monday,                  !- Day of Week for Start Day
    No,                      !- Use Weather File Holidays and Special Days
    No,                      !- Use Weather File Daylight Saving Period
    No,                      !- Apply Weekend Holiday Rule
    Yes,                     !- Use Weather File Rain Indicators
    Yes;                     !- Use Weather File Snow Indicators

! ***HOLIDAYS & DAYLIGHT SAVINGS***

  RunPeriodControl:DaylightSavingTime,
    2nd Sunday in March,     !- Start Date
    1st Sunday in November;  !- End Date

  RunPeriodControl:SpecialDays,
    New Years Day,           !- Name
    January 1,               !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Veterans Day,            !- Name
    November 11,             !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Christmas,               !- Name
    December 25,             !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Independence Day,        !- Name
    July 4,                  !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    MLK Day,                 !- Name
    3rd Monday in January,   !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Presidents Day,          !- Name
    3rd Monday in February,  !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Memorial Day,            !- Name
    Last Monday in May,      !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Labor Day,               !- Name
    1st Monday in September, !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type
```

```
    2nd Monday in October,   !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Thanksgiving,            !- Name
    4th Thursday in November,!- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

! ***SCHEDULE TYPES***

  ScheduleTypeLimits,
    Any Number;              !- Name

  ScheduleTypeLimits,
    Fraction,                !- Name
    0.0,                     !- Lower Limit Value
    1.0,                     !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    Temperature,             !- Name
    -60,                     !- Lower Limit Value
    200,                     !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    On/Off,                  !- Name
    0,                       !- Lower Limit Value
    1,                       !- Upper Limit Value
    DISCRETE;                !- Numeric Type

  ScheduleTypeLimits,
    Control Type,            !- Name
    0,                       !- Lower Limit Value
    4,                       !- Upper Limit Value
    DISCRETE;                !- Numeric Type

  ScheduleTypeLimits,
    Humidity,                !- Name
    10,                      !- Lower Limit Value
    90,                      !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    Number;                  !- Name

! ***ALWAYS ON SCHEDULE***

  Schedule:Compact,
    ALWAYS_ON,               !- Name
    On/Off,                  !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,1;          !- Field 3

! ***MISC SIMULATION PARAMETERS***

  SurfaceConvectionAlgorithm:Inside,TARP;

  SurfaceConvectionAlgorithm:Outside,DOE-2;

  HeatBalanceAlgorithm,ConductionTransferFunction,200.0000;

  ZoneAirHeatBalanceAlgorithm,
```

```
  Sizing:Parameters,
    1.2,                     !- Heating Sizing Factor
    1.2,                     !- Cooling Sizing Factor
    6;                       !- Timesteps in Averaging Window

  ConvergenceLimits,
    2,                       !- Minimum System Timestep {minutes}
    25;                      !- Maximum HVAC Iterations

  ShadowCalculation,
    PolygonClipping,         !- Shading Calculation Method
    Periodic,                !- Shading Calculation Update Frequency Method
    7,                       !- Shading Calculation Update Frequency
    15000;                   !- Maximum Figures in Shadow Overlap Calculations

  Timestep,6;

! WeatherFileName=USA_IL_Chicago-OHare_TMY2.epw

  Site:Location,
    USA IL-CHICAGO-OHARE,    !- Name
    41.77,                   !- Latitude {deg}
    -87.75,                  !- Longitude {deg}
    -6.00,                   !- Time Zone {hr}
    190;                     !- Elevation {m}

 ! CHICAGO_IL_USA Annual Heating 99.6%, MaxDB=-20.6Â°C

  SizingPeriod:DesignDay,
    CHICAGO Ann Htg 99.6% Condns DB,  !- Name
    1,                       !- Month
    21,                      !- Day of Month
    WinterDesignDay,         !- Day Type
    -20.6,                   !- Maximum Dry-Bulb Temperature {C}
    0.0,                     !- Daily Dry-Bulb Temperature Range {deltaC}
    ,                        !- Dry-Bulb Temperature Range Modifier Type
    ,                        !- Dry-Bulb Temperature Range Modifier Day Schedule Name
    Wetbulb,                 !- Humidity Condition Type
    -20.6,                   !- Wetbulb or DewPoint at Maximum Dry-Bulb {C}
    ,                        !- Humidity Condition Day Schedule Name
    ,                        !- Humidity Ratio at Maximum Dry-Bulb {kgWater/kgDryAir}
    ,                        !- Enthalpy at Maximum Dry-Bulb {J/kg}
    ,                        !- Daily Wet-Bulb Temperature Range {deltaC}
    99063.,                  !- Barometric Pressure {Pa}
    4.9,                     !- Wind Speed {m/s}
    270,                     !- Wind Direction {deg}
    No,                      !- Rain Indicator
    No,                      !- Snow Indicator
    No,                      !- Daylight Saving Time Indicator
    ASHRAEClearSky,          !- Solar Model Indicator
    ,                        !- Beam Solar Day Schedule Name
    ,                        !- Diffuse Solar Day Schedule Name
    ,                        !- ASHRAE Clear Sky Optical Depth for Beam Irradiance (taub) {dimensionless}
    ,                        !- ASHRAE Clear Sky Optical Depth for Diffuse Irradiance (taud) {dimensionless}
    0.00;                    !- Sky Clearness

 ! CHICAGO_IL_USA Annual Cooling (WB=>MDB) .4%, MDB=31.2Â°C WB=25.5Â°C

  SizingPeriod:DesignDay,
    CHICAGO Ann Clg .4% Condns WB=>MDB,  !- Name
    7,                       !- Month
    21,                      !- Day of Month
    SummerDesignDay,         !- Day Type
    31.2,                    !- Maximum Dry-Bulb Temperature {C}
    10.7,                    !- Daily Dry-Bulb Temperature Range {deltaC}
```

```
    Wetbulb,                 !- Humidity Condition Type
    25.5,                    !- Wetbulb or DewPoint at Maximum Dry-Bulb {C}
    ,                        !- Humidity Condition Day Schedule Name
    ,                        !- Humidity Ratio at Maximum Dry-Bulb {kgWater/kgDryAir}
    ,                        !- Enthalpy at Maximum Dry-Bulb {J/kg}
    ,                        !- Daily Wet-Bulb Temperature Range {deltaC}
    99063.,                  !- Barometric Pressure {Pa}
    5.3,                     !- Wind Speed {m/s}
    230,                     !- Wind Direction {deg}
    No,                      !- Rain Indicator
    No,                      !- Snow Indicator
    No,                      !- Daylight Saving Time Indicator
    ASHRAEClearSky,          !- Solar Model Indicator
    ,                        !- Beam Solar Day Schedule Name
    ,                        !- Diffuse Solar Day Schedule Name
    ,                        !- ASHRAE Clear Sky Optical Depth for Beam Irradiance (taub) {dimensionless}
    ,                        !- ASHRAE Clear Sky Optical Depth for Diffuse Irradiance (taud) {dimensionless}
    1.00;                    !- Sky Clearness

  Site:WaterMainsTemperature,
    CORRELATION,             !- Calculation Method
    ,                        !- Temperature Schedule Name
    9.69,                    !- Annual Average Outdoor Air Temperature {C}
    28.10;                   !- Maximum Difference In Monthly Average Outdoor Air Temperatures {deltaC}

  Site:GroundTemperature:BuildingSurface,19.527,19.502,19.536,19.598,20.002,21.640,22.225,22.375,21.449,20.121,19.802,19.633;

! ***OPAQUE CONSTRUCTIONS AND MATERIALS***
! Exterior Walls

  Construction,
    Mass Non-res Ext Wall,   !- Name
    1IN Stucco,              !- Outside Layer
    8IN Concrete HW,         !- Layer 2
    Mass NonRes Wall Insulation,  !- Layer 3
    1/2IN Gypsum;            !- Layer 4

  Material,
    Mass NonRes Wall Insulation,  !- Name
    MediumRough,             !- Roughness
    0.0495494599433393,      !- Thickness {m}
    0.049,                   !- Conductivity {W/m-K}
    265.0000,                !- Density {kg/m3}
    836.8000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Roof

  Construction,
    Attic Non-res Floor,     !- Name
    1/2IN Gypsum,            !- Outside Layer
    AtticFloor NonRes Insulation,  !- Layer 2
    1/2IN Gypsum;            !- Layer 3

  Construction,
    Attic Non-res Roof,      !- Name
    Roof Membrane,           !- Outside Layer
    Metal Decking;           !- Layer 2

  Material,
    AtticFloor NonRes Insulation,  !- Name
    MediumRough,             !- Roughness
    0.236804989096202,       !- Thickness {m}
    0.049,                   !- Conductivity {W/m-K}
```

```
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Slab on grade, unheated

  Construction,
    ext-slab,                !- Name
    HW CONCRETE,             !- Outside Layer
    CP02 CARPET PAD;         !- Layer 2

! Interior Walls

  Construction,
    int-walls,               !- Name
    1/2IN Gypsum,            !- Outside Layer
    1/2IN Gypsum;            !- Layer 2

! ***WINDOW/DOOR CONSTRUCTIONS AND MATERIALS***

  Construction,
    Window Non-res Fixed,    !- Name
    NonRes Fixed Assembly Window;  !- Outside Layer

  WindowMaterial:SimpleGlazingSystem,
    NonRes Fixed Assembly Window,  !- Name
    3.23646,                 !- U-Factor {W/m2-K}
    0.39,                    !- Solar Heat Gain Coefficient
    ;                        !- Visible Transmittance

! ***COMMON CONSTRUCTIONS AND MATERIALS***

  Construction,
    InteriorFurnishings,     !- Name
    Std Wood 6inch;          !- Outside Layer

  Material,
    Std Wood 6inch,          !- Name
    MediumSmooth,            !- Roughness
    0.15,                    !- Thickness {m}
    0.12,                    !- Conductivity {W/m-K}
    540.0000,                !- Density {kg/m3}
    1210,                    !- Specific Heat {J/kg-K}
    0.9000000,               !- Thermal Absorptance
    0.7000000,               !- Solar Absorptance
    0.7000000;               !- Visible Absorptance

  Material,
    Wood Siding,             !- Name
    MediumSmooth,            !- Roughness
    0.0100,                  !- Thickness {m}
    0.1100,                  !- Conductivity {W/m-K}
    544.6200,                !- Density {kg/m3}
    1210.0000,               !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7800,                  !- Solar Absorptance
    0.7800;                  !- Visible Absorptance

  Material,
    1/2IN Gypsum,            !- Name
    Smooth,                  !- Roughness
    0.0127,                  !- Thickness {m}
    0.1600,                  !- Conductivity {W/m-K}
    784.9000,                !- Density {kg/m3}
    830.0000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
```

```
Material,
  1IN Stucco,               !- Name
  Smooth,                   !- Roughness
  0.0253,                   !- Thickness {m}
  0.6918,                   !- Conductivity {W/m-K}
  1858.0000,                !- Density {kg/m3}
  837.0000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.9200,                   !- Solar Absorptance
  0.9200;                   !- Visible Absorptance

Material,
  8IN CONCRETE HW,          !- Name
  Rough,                    !- Roughness
  0.2032,                   !- Thickness {m}
  1.3110,                   !- Conductivity {W/m-K}
  2240.0000,                !- Density {kg/m3}
  836.8000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material,
  Metal Siding,             !- Name
  Smooth,                   !- Roughness
  0.0015,                   !- Thickness {m}
  44.9600,                  !- Conductivity {W/m-K}
  7688.8600,                !- Density {kg/m3}
  410.0000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.2000,                   !- Solar Absorptance
  0.2000;                   !- Visible Absorptance

Material,
  HW CONCRETE,              !- Name
  Rough,                    !- Roughness
  0.1016,                   !- Thickness {m}
  1.3110,                   !- Conductivity {W/m-K}
  2240.0000,                !- Density {kg/m3}
  836.8000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material:NoMass,
  CP02 CARPET PAD,          !- Name
  VeryRough,                !- Roughness
  0.2165,                   !- Thermal Resistance {m2-K/W}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.8000;                   !- Visible Absorptance

Material,
  Roof Membrane,            !- Name
  VeryRough,                !- Roughness
  0.0095,                   !- Thickness {m}
  0.1600,                   !- Conductivity {W/m-K}
  1121.2900,                !- Density {kg/m3}
  1460.0000,                !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material,
  Metal Decking,            !- Name
```

```
    45.0060,                 !- Conductivity {W/m-K}
    7680.0000,               !- Density {kg/m3}
    418.4000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.3000;                  !- Visible Absorptance

  Material,
    Metal Roofing,           !- Name
    MediumSmooth,            !- Roughness
    0.0015,                  !- Thickness {m}
    45.0060,                 !- Conductivity {W/m-K}
    7680.0000,               !- Density {kg/m3}
    418.4000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.3000;                  !- Visible Absorptance

  Material,
    MAT-CC05 4 HW CONCRETE,  !- Name
    Rough,                   !- Roughness
    0.1016,                  !- Thickness {m}
    1.3110,                  !- Conductivity {W/m-K}
    2240.0000,               !- Density {kg/m3}
    836.8000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Acoustic tile for drop ceiling

  Material,
    Std AC02,                !- Name
    MediumSmooth,            !- Roughness
    1.2700000E-02,           !- Thickness {m}
    5.7000000E-02,           !- Conductivity {W/m-K}
    288.0000,                !- Density {kg/m3}
    1339.000,                !- Specific Heat {J/kg-K}
    0.9000000,               !- Thermal Absorptance
    0.7000000,               !- Solar Absorptance
    0.2000000;               !- Visible Absorptance

  Material:NoMass,
    MAT-AIR-WALL,            !- Name
    Rough,                   !- Roughness
    0.2079491,               !- Thermal Resistance {m2-K/W}
    0.9,                     !- Thermal Absorptance
    0.7;                     !- Solar Absorptance

! ZONE LIST:
! Attic
! Core_ZN
! Perimeter_ZN_1
! Perimeter_ZN_2
! Perimeter_ZN_3
! Perimeter_ZN_4
! ***ZONES***

  Zone,
    Attic,                   !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
```

```
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    No;                      !- Part of Total Floor Area

  Zone,
    Core_ZN,                 !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_1,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_2,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_3,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
```

```
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

! ***WALLS***

  BuildingSurface:Detailed,
    Attic_floor_core,        !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_ceiling,         !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_east,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_north,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
```

```
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_west,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_east,         !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    3,                       !- Number of Vertices
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    18.4600,9.2300,6.3300;  !- X,Y,Z ==> Vertex 3 {m}

  BuildingSurface:Detailed,
    Attic_roof_north,        !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    9.2300,9.2300,6.3300,  !- X,Y,Z ==> Vertex 3 {m}
    18.4600,9.2300,6.3300;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_south,        !- Name
```

```
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    18.4600,9.2300,6.3300,  !- X,Y,Z ==> Vertex 3 {m}
    9.2300,9.2300,6.3300;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_west,         !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    3,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    -0.6000,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    9.2300,9.2300,6.3300;  !- X,Y,Z ==> Vertex 3 {m}

  BuildingSurface:Detailed,
    Attic_soffit_east,       !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_north,      !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_south,      !- Name
    Floor,                   !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    -0.6000,-0.6000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_west,       !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    -0.6000,-0.6000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_ceiling,         !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_core,        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_floor,           !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_east,       !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_north,      !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_south,      !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_west,       !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_ceiling,  !- Name
    Ceiling,                 !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_east,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_south,      !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_south,  !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_ceiling,  !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_east,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_east,!- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_south,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_east,       !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_ceiling,  !- Name
    Ceiling,                 !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_east,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_north,  !- Name
    Wall,                    !- Surface Type
    Mass Non-res Ext Wall,   !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_south,  !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_north,      !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_ceiling,  !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_west,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_east,!- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_west,       !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_south,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_west,!- Name
    Wall,                    !- Surface Type
    Mass Non-res Ext Wall,   !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

! ***WINDOWS***

  FenestrationSurface:Detailed,
```

```
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    1.3900,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    1.3900,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    3.2200,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    3.2200,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    6.0100,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    6.0100,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    7.8400,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    7.8400,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    10.6200,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    10.6200,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    12.4500,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    12.4500,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    15.2400,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    15.2400,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    17.0700,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    17.0700,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_5,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
```

```
    19.8500,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    21.6800,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    21.6800,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_6,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    24.4700,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    24.4700,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    26.3000,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    26.3000,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_door,  !- Name
    GlassDoor,               !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    12.930,0.0000,2.1340,  !- X,Y,Z ==> Vertex 1 {m}
    12.930,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    14.760,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    14.760,0.0000,2.1340;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,1.3900,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,1.3900,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,3.2200,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,3.2200,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,6.0100,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,6.0100,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,7.8400,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,7.8400,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_3,  !- Name
```

```
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,10.6200,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,10.6200,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,12.4500,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,12.4500,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,15.2400,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,15.2400,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,17.0700,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,17.0700,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    26.3000,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    26.3000,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    24.4700,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    24.4700,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    21.6800,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    21.6800,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    19.8500,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    19.8500,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
```

```
    15.2400,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    15.2400,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    12.4500,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    12.4500,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    10.6200,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    10.6200,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_5,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    7.8400,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    7.8400,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    6.0100,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    6.0100,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_6,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    3.2200,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    3.2200,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    1.3900,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    1.3900,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,17.0700,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,17.0700,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,15.2400,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,15.2400,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_2,  !- Name
    Window,                  !- Surface Type
```

```
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,12.4500,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,12.4500,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,10.6200,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,10.6200,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,7.8400,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,7.8400,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,6.0100,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,6.0100,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,3.2200,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,3.2200,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,1.3900,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,1.3900,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

! ***GEOMETRY RULES***

  GlobalGeometryRules,
    UpperLeftCorner,         !- Starting Vertex Position
    Counterclockwise,        !- Vertex Entry Direction
    Relative,                !- Coordinate System
    Relative;                !- Daylighting Reference Point Coordinate System

! ***PEOPLE***

  People,
    Core_ZN People,          !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
```

```
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_1 People,   !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_2 People,   !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_3 People,   !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
```

```
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_4 People,   !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

! ***LIGHTS***

  Lights,
    Core_ZN_Lights,          !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_1_Lights,   !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_2_Lights,   !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
```

```
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_3_Lights,   !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_4_Lights,   !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

! ***EQUIPMENT GAINS***

  ElectricEquipment,
    Core_ZN_MiscPlug_Equip,  !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_1_MiscPlug_Equip,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_2_MiscPlug_Equip,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
```

```
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_3_MiscPlug_Equip,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_4_MiscPlug_Equip,  !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

! ***EXTERIOR LOADS***

  Exterior:Lights,
    Exterior Facade Lighting,!- Name
    ALWAYS_ON,               !- Schedule Name
    2303,                    !- Design Level {W}
    AstronomicalClock,       !- Control Option
    Exterior Facade Lighting;!- End-Use Subcategory

! ***INFILTRATION***
! Infiltration through ceiling/roof

  ZoneInfiltration:DesignFlowRate,
    Core_ZN_Infiltration,    !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    AirChanges/Hour,         !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    ,                        !- Flow Rate per Exterior Surface Area {m3/s-m2}
    0.36,                    !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_1_Infiltration,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
```

```
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_2_Infiltration,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_3_Infiltration,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_4_Infiltration,  !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

! ***INTERNAL MASS***

  InternalMass,
    Core_ZN Internal Mass,   !- Name
    InteriorFurnishings,     !- Construction Name
    Core_ZN,                 !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    299.3148;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_1 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_1,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    226.9000;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_2 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_2,          !- Zone or ZoneList Name
```

```
  InternalMass,
    Perimeter_ZN_3 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    226.9000;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_4 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_4,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    134.6000;                !- Surface Area {m2}

! ***INTERNAL GAINS SCHEDULES***

  Schedule:Compact,
    INFIL_QUARTER_ON_SCH,    !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 06:00,1.0,        !- Field 3
    Until: 22:00,0.25,       !- Field 5
    Until: 24:00,1.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 06:00,1.0,        !- Field 10
    Until: 18:00,0.25,       !- Field 12
    Until: 24:00,1.0,        !- Field 14
    For: Sunday Holidays AllOtherDays, !- Field 16
    Until: 24:00,1.0;        !- Field 17

  Schedule:Compact,
    BLDG_OCC_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: SummerDesignDay,    !- Field 2
    Until: 06:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.05,       !- Field 7
    For: Weekdays,           !- Field 9
    Until: 06:00,0.0,        !- Field 10
    Until: 07:00,0.1,        !- Field 12
    Until: 08:00,0.2,        !- Field 14
    Until: 12:00,0.95,       !- Field 16
    Until: 13:00,0.5,        !- Field 18
    Until: 17:00,0.95,       !- Field 20
    Until: 18:00,0.3,        !- Field 22
    Until: 20:00,0.1,        !- Field 24
    Until: 24:00,0.05,       !- Field 26
    For: Saturday,           !- Field 28
    Until: 06:00,0.0,        !- Field 29
    Until: 08:00,0.1,        !- Field 31
    Until: 12:00,0.3,        !- Field 33
    Until: 17:00,0.1,        !- Field 35
    Until: 24:00,0.0,        !- Field 37
    For: AllOtherDays,       !- Field 39
    Until: 24:00,0.0;        !- Field 40

  Schedule:Compact,
    BLDG_LIGHT_SCH,          !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays,           !- Field 2
    Until: 05:00,0.05,       !- Field 3
    Until: 07:00,0.1,        !- Field 5
```

```
    Until: 18:00,0.5,        !- Field 11
    Until: 20:00,0.3,        !- Field 13
    Until: 22:00,0.2,        !- Field 15
    Until: 23:00,0.1,        !- Field 17
    Until: 24:00,0.05,       !- Field 19
    For: Saturday,           !- Field 21
    Until: 06:00,0.05,       !- Field 22
    Until: 08:00,0.1,        !- Field 24
    Until: 12:00,0.3,        !- Field 26
    Until: 17:00,0.15,       !- Field 28
    Until: 24:00,0.05,       !- Field 30
    For: SummerDesignDay,    !- Field 32
    Until: 24:00,1.0,        !- Field 33
    For: WinterDesignDay,    !- Field 35
    Until: 24:00,0.0,        !- Field 36
    For: AllOtherDays,       !- Field 38
    Until: 24:00,0.05;       !- Field 39

  Schedule:Compact,
    BLDG_EQUIP_SCH,          !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays,           !- Field 2
    Until: 08:00,0.40,       !- Field 3
    Until: 12:00,0.90,       !- Field 5
    Until: 13:00,0.80,       !- Field 7
    Until: 17:00,0.90,       !- Field 9
    Until: 18:00,0.50,       !- Field 11
    Until: 24:00,0.40,       !- Field 13
    For: Saturday,           !- Field 15
    Until: 06:00,0.30,       !- Field 16
    Until: 08:00,0.4,        !- Field 18
    Until: 12:00,0.5,        !- Field 20
    Until: 17:00,0.35,       !- Field 22
    Until: 24:00,0.30,       !- Field 24
    For: SummerDesignDay,    !- Field 26
    Until: 24:00,1.0,        !- Field 27
    For: WinterDesignDay,    !- Field 29
    Until: 24:00,0.0,        !- Field 30
    For: AllOtherDays,       !- Field 32
    Until: 24:00,0.30;       !- Field 33

  Schedule:Compact,
    ACTIVITY_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,120;        !- Field 3

  Schedule:Compact,
    WORK_EFF_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.0;        !- Field 3

  Schedule:Compact,
    AIR_VELO_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.2;        !- Field 3

  Schedule:Compact,
    CLOTHING_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
```

```
    Until: 24:00,1.0,        !- Field 3
    Through: 09/30,          !- Field 5
    For: AllDays,            !- Field 6
    Until: 24:00,0.5,        !- Field 7
    Through: 12/31,          !- Field 9
    For: AllDays,            !- Field 10
    Until: 24:00,1.0;        !- Field 11

  ZoneInfiltration:DesignFlowRate,
    Attic_Infiltration,      !- Name
    Attic,                   !- Zone or ZoneList or Space or SpaceList Name
    ALWAYS_ON,               !- Schedule Name
    AirChanges/Hour,         !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    ,                        !- Flow Rate per Exterior Surface Area {m3/s-m2}
    1.0,                     !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

! ***HVAC EQUIPMENT***

  Fan:SystemModel,
    PSZ-AC:1_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Air Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    AUTOSIZE,                !- Design Maximum Air Flow Rate {m3/s}
    Discrete,                !- Speed Control Method
    0.0,                     !- Electric Power Minimum Flow Rate Fraction
    622.0,                   !- Design Pressure Rise {Pa}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Air Stream Fraction
    AUTOSIZE,                !- Design Electric Power Consumption {W}
    TotalEfficiencyAndPressure,  !- Design Power Sizing Method
    ,                        !- Electric Power Per Unit Flow Rate {W/(m3/s)}
    ,                        !- Electric Power Per Unit Flow Rate Per Unit Pressure {W/((m3/s)-Pa)}
    0.53625,                 !- Fan Total Efficiency
    ,                        !- Electric Power Function of Flow Fraction Curve Name
    ,                        !- Night Ventilation Mode Pressure Rise {Pa}
    ,                        !- Night Ventilation Mode Flow Fraction
    ,                        !- Motor Loss Zone Name
    ,                        !- Motor Loss Radiative Fraction
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:2_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Air Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:3_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
```

```
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:4_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Air Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:5_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Air Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Coil:Heating:Fuel,
    PSZ-AC:1_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Air Outlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:2_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Air Outlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:3_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Air Outlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:4_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Air Outlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode;  !- Temperature Setpoint Node Name
```

```
    PSZ-AC:5_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Air Outlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode;  !- Temperature Setpoint Node Name

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:1_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
    ,                        !- Nominal Time for Condensate Removal to Begin {s}
    ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:2_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
    ,                        !- Nominal Time for Condensate Removal to Begin {s}
    ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:3_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
```

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Coil:Cooling:DX:SingleSpeed,
  PSZ-AC:4_CoolC DXCoil,   !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
  AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
  3.66668442928701,        !- Gross Rated Cooling COP {W/W}
  AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
  ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Air Inlet Node Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Air Outlet Node Name
  Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
  ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
  Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
  ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
  Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
  ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
  ,                        !- Nominal Time for Condensate Removal to Begin {s}
  ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

Coil:Cooling:DX:SingleSpeed,
  PSZ-AC:5_CoolC DXCoil,   !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
  AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
  3.66668442928701,        !- Gross Rated Cooling COP {W/W}
  AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
  ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Air Inlet Node Name
  PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Air Outlet Node Name
  Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
  ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
  Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
  ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
  Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
  ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
  ,                        !- Nominal Time for Condensate Removal to Begin {s}
  ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

ZoneHVAC:EquipmentList,
  Core_ZN Equipment,       !- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Core_ZN Direct Air ADU,  !- Zone Equipment 1 Name
  1,                       !- Zone Equipment 1 Cooling Sequence
  1,                       !- Zone Equipment 1 Heating or No-Load Sequence
  ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
  ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

ZoneHVAC:EquipmentList,
  Perimeter_ZN_1 Equipment,!- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Perimeter_ZN_1 Direct Air ADU,  !- Zone Equipment 1 Name
  1,                       !- Zone Equipment 1 Cooling Sequence
  1,                       !- Zone Equipment 1 Heating or No-Load Sequence
  ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
  ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

ZoneHVAC:EquipmentList,
  Perimeter_ZN_2 Equipment,!- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Perimeter_ZN_2 Direct Air ADU,  !- Zone Equipment 1 Name
```

```
    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

  ZoneHVAC:EquipmentList,
    Perimeter_ZN_3 Equipment,!- Name
    SequentialLoad,          !- Load Distribution Scheme
    ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
    Perimeter_ZN_3 Direct Air ADU,  !- Zone Equipment 1 Name
    1,                       !- Zone Equipment 1 Cooling Sequence
    1,                       !- Zone Equipment 1 Heating or No-Load Sequence
    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

  ZoneHVAC:EquipmentList,
    Perimeter_ZN_4 Equipment,!- Name
    SequentialLoad,          !- Load Distribution Scheme
    ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
    Perimeter_ZN_4 Direct Air ADU,  !- Zone Equipment 1 Name
    1,                       !- Zone Equipment 1 Cooling Sequence
    1,                       !- Zone Equipment 1 Heating or No-Load Sequence
    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

! ***SIZING & CONTROLS***

  Sizing:Zone,
    Core_ZN,                 !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,    !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                 !- Zone Heating Design Supply Air Temperature {C}
    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Core_ZN,         !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Core_ZN,         !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
```

```
  Sizing:Zone,
    Perimeter_ZN_1,           !- Zone or ZoneList Name
    SupplyAirTemperature,     !- Zone Cooling Design Supply Air Temperature Input Method
    13.9000,                  !- Zone Cooling Design Supply Air Temperature {C}
    ,                         !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,     !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                  !- Zone Heating Design Supply Air Temperature {C}
    ,                         !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                   !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                   !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_1,   !- Design Specification Outdoor Air Object Name
    ,                         !- Zone Heating Sizing Factor
    ,                         !- Zone Cooling Sizing Factor
    DesignDay,                !- Cooling Design Air Flow Method
    ,                         !- Cooling Design Air Flow Rate {m3/s}
    ,                         !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Cooling Minimum Air Flow {m3/s}
    ,                         !- Cooling Minimum Air Flow Fraction
    DesignDay,                !- Heating Design Air Flow Method
    ,                         !- Heating Design Air Flow Rate {m3/s}
    ,                         !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Heating Maximum Air Flow {m3/s}
    ,                         !- Heating Maximum Air Flow Fraction
    ,                         !- Design Specification Zone Air Distribution Object Name
    No,                       !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,         !- Dedicated Outdoor Air System Control Strategy
    autosize,                 !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                 !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                         !- Zone Load Sizing Method
    ,                         !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                         !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                         !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                         !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                         !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                         !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_1,   !- Name
    Flow/Person,              !- Outdoor Air Method
    0.01,                     !- Outdoor Air Flow per Person {m3/s-person}
    ,                         !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                         !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_2,           !- Zone or ZoneList Name
    SupplyAirTemperature,     !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                  !- Zone Cooling Design Supply Air Temperature {C}
    ,                         !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,     !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                  !- Zone Heating Design Supply Air Temperature {C}
    ,                         !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                   !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                   !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_2,   !- Design Specification Outdoor Air Object Name
    ,                         !- Zone Heating Sizing Factor
    ,                         !- Zone Cooling Sizing Factor
    DesignDay,                !- Cooling Design Air Flow Method
    ,                         !- Cooling Design Air Flow Rate {m3/s}
    ,                         !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Cooling Minimum Air Flow {m3/s}
    ,                         !- Cooling Minimum Air Flow Fraction
    DesignDay,                !- Heating Design Air Flow Method
    ,                         !- Heating Design Air Flow Rate {m3/s}
    ,                         !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Heating Maximum Air Flow {m3/s}
    ,                         !- Heating Maximum Air Flow Fraction
```

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    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_2,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    13.9000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,    !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                 !- Zone Heating Design Supply Air Temperature {C}
    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_3,  !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_3,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_4,          !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
```

```
    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_4,  !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_4,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  ZoneControl:Thermostat,
    Core_ZN Thermostat,      !- Name
    Core_ZN,                 !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Core_ZN DualSPSched;     !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_1 Thermostat,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_1 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_2 Thermostat,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_2 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_3 Thermostat,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_3 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_4 Thermostat,  !- Name
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    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_4 DualSPSched;  !- Control 1 Name

  ThermostatSetpoint:DualSetpoint,
    Core_ZN DualSPSched,     !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_1 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_2 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_3 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_4 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrCore_ZN,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Core_ZN,                 !- Control Zone Name
    Core_ZN Air Node,        !- Zone Node Name
    Core_ZN Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_1,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_1,          !- Control Zone Name
    Perimeter_ZN_1 Air Node, !- Zone Node Name
    Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_2,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_2,          !- Control Zone Name
    Perimeter_ZN_2 Air Node, !- Zone Node Name
    Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_3,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_3,          !- Control Zone Name
    Perimeter_ZN_3 Air Node, !- Zone Node Name
    Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
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  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_4,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_4,          !- Control Zone Name
    Perimeter_ZN_4 Air Node, !- Zone Node Name
    Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  Sizing:System,
    PSZ-AC:1,                !- AirLoop Name
    Sensible,                !- Type of Load to Size On
    AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                     !- Central Heating Maximum System Air Flow Ratio
    7.0,                     !- Preheat Design Temperature {C}
    0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Precool Design Temperature {C}
    0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
    ,                        !- Cooling Supply Air Flow Rate {m3/s}
    ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,               !- Heating Supply Air Flow Rate Method
    ,                        !- Heating Supply Air Flow Rate {m3/s}
    ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                        !- System Outdoor Air Method
    1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,   !- Cooling Design Capacity Method
    autosize,                !- Cooling Design Capacity {W}
    ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,   !- Heating Design Capacity Method
    autosize,                !- Heating Design Capacity {W}
    ,                        !- Heating Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Heating Design Capacity
    VAV;                     !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:2,                !- AirLoop Name
    Sensible,                !- Type of Load to Size On
    AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                     !- Central Heating Maximum System Air Flow Ratio
    7.0,                     !- Preheat Design Temperature {C}
    0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Precool Design Temperature {C}
    0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
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    ,                         !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                         !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,                !- Heating Supply Air Flow Rate Method
    ,                         !- Heating Supply Air Flow Rate {m3/s}
    ,                         !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                         !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                         !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                         !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                         !- System Outdoor Air Method
    1.0,                      !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,    !- Cooling Design Capacity Method
    autosize,                 !- Cooling Design Capacity {W}
    ,                         !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                         !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,    !- Heating Design Capacity Method
    autosize,                 !- Heating Design Capacity {W}
    ,                         !- Heating Design Capacity Per Floor Area {W/m2}
    ,                         !- Fraction of Autosized Heating Design Capacity
    VAV;                      !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:3,                 !- AirLoop Name
    Sensible,                 !- Type of Load to Size On
    AUTOSIZE,                 !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                      !- Central Heating Maximum System Air Flow Ratio
    7.0,                      !- Preheat Design Temperature {C}
    0.008,                    !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                  !- Precool Design Temperature {C}
    0.008,                    !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                  !- Central Cooling Design Supply Air Temperature {C}
    40.0,                     !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,            !- Type of Zone Sum to Use
    No,                       !- 100% Outdoor Air in Cooling
    No,                       !- 100% Outdoor Air in Heating
    0.0085,                   !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                   !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,                !- Cooling Supply Air Flow Rate Method
    ,                         !- Cooling Supply Air Flow Rate {m3/s}
    ,                         !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                         !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                         !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,                !- Heating Supply Air Flow Rate Method
    ,                         !- Heating Supply Air Flow Rate {m3/s}
    ,                         !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                         !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                         !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                         !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                         !- System Outdoor Air Method
    1.0,                      !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,    !- Cooling Design Capacity Method
    autosize,                 !- Cooling Design Capacity {W}
    ,                         !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                         !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,    !- Heating Design Capacity Method
    autosize,                 !- Heating Design Capacity {W}
    ,                         !- Heating Design Capacity Per Floor Area {W/m2}
    ,                         !- Fraction of Autosized Heating Design Capacity
    VAV;                      !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:4,                 !- AirLoop Name
    Sensible,                 !- Type of Load to Size On
    AUTOSIZE,                 !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                      !- Central Heating Maximum System Air Flow Ratio
    7.0,                      !- Preheat Design Temperature {C}
    0.008,                    !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
```

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    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
    ,                        !- Cooling Supply Air Flow Rate {m3/s}
    ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,               !- Heating Supply Air Flow Rate Method
    ,                        !- Heating Supply Air Flow Rate {m3/s}
    ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                        !- System Outdoor Air Method
    1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,   !- Cooling Design Capacity Method
    autosize,                !- Cooling Design Capacity {W}
    ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,   !- Heating Design Capacity Method
    autosize,                !- Heating Design Capacity {W}
    ,                        !- Heating Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Heating Design Capacity
    VAV;                     !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:5,                !- AirLoop Name
    Sensible,                !- Type of Load to Size On
    AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                     !- Central Heating Maximum System Air Flow Ratio
    7.0,                     !- Preheat Design Temperature {C}
    0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Precool Design Temperature {C}
    0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
    ,                        !- Cooling Supply Air Flow Rate {m3/s}
    ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,               !- Heating Supply Air Flow Rate Method
    ,                        !- Heating Supply Air Flow Rate {m3/s}
    ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                        !- System Outdoor Air Method
    1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,   !- Cooling Design Capacity Method
    autosize,                !- Cooling Design Capacity {W}
    ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,   !- Heating Design Capacity Method
    autosize,                !- Heating Design Capacity {W}
    ,                        !- Heating Design Capacity Per Floor Area {W/m2}
```

```
  Controller:OutdoorAir,
    PSZ-AC:1_OA_Controller,  !- Name
    PSZ-AC:1_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:1_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:2_OA_Controller,  !- Name
    PSZ-AC:2_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:2 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:2_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:3_OA_Controller,  !- Name
    PSZ-AC:3_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:3 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:3_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name
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    PSZ-AC:4_OA_Controller,  !- Name
    PSZ-AC:4_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:4 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:4_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:5_OA_Controller,  !- Name
    PSZ-AC:5_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:5 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:5_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Curve:Quadratic,
    Cool-PLF-fPLR,           !- Name
    0.90949556,              !- Coefficient1 Constant
    0.09864773,              !- Coefficient2 x
    -0.00819488,             !- Coefficient3 x**2
    0,                       !- Minimum Value of x
    1,                       !- Maximum Value of x
    0.7,                     !- Minimum Curve Output
    1;                       !- Maximum Curve Output

  Curve:Cubic,
    ConstantCubic,           !- Name
    1,                       !- Coefficient1 Constant
    0,                       !- Coefficient2 x
    0,                       !- Coefficient3 x**2
    0,                       !- Coefficient4 x**3
    -100,                    !- Minimum Value of x
    100;                     !- Maximum Value of x

  Curve:Biquadratic,
    Cool-Cap-fT,             !- Name
    0.9712123,               !- Coefficient1 Constant
    -0.015275502,            !- Coefficient2 x
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```
    -0.0000068364,           !- Coefficient5 y**2
    -0.0002905956,           !- Coefficient6 x*y
    -100,                    !- Minimum Value of x
    100,                     !- Maximum Value of x
    -100,                    !- Minimum Value of y
    100;                     !- Maximum Value of y

  Curve:Biquadratic,
    Cool-EIR-fT,             !- Name
    0.28687133,              !- Coefficient1 Constant
    0.023902164,             !- Coefficient2 x
    -0.000810648,            !- Coefficient3 x**2
    0.013458546,             !- Coefficient4 y
    0.0003389364,            !- Coefficient5 y**2
    -0.0004870044,           !- Coefficient6 x*y
    -100,                    !- Minimum Value of x
    100,                     !- Maximum Value of x
    -100,                    !- Minimum Value of y
    100;                     !- Maximum Value of y

! ***AIR LOOPS***

  AirLoopHVAC,
    PSZ-AC:1,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:1 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:1 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:1 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:1 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:1 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:2,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:2 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:2 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:2 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:2 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:2 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:2 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:3,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:3 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:3 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:3 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:3 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:3 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:4,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:4 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:4 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:4 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
```

```
    PSZ-AC:4 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:5,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:5 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:5 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:5 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:5 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:5 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  CoilSystem:Cooling:DX,
    PSZ-AC:1_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:1_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:2_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:2_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:3_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:3_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:4_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:4_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:5_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:5_CoolC DXCoil;   !- Cooling Coil Name

! ***CONNECTIONS***

  ZoneHVAC:EquipmentConnections,
    Core_ZN,                 !- Zone Name
    Core_ZN Equipment,       !- Zone Conditioning Equipment List Name
    Core_ZN Inlet Nodes,     !- Zone Air Inlet Node or NodeList Name
    ,                        !- Zone Air Exhaust Node or NodeList Name
```

```
ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_1,          !- Zone Name
  Perimeter_ZN_1 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_1 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_1 Air Node, !- Zone Air Node Name
  Perimeter_ZN_1 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_2,          !- Zone Name
  Perimeter_ZN_2 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_2 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_2 Air Node, !- Zone Air Node Name
  Perimeter_ZN_2 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_3,          !- Zone Name
  Perimeter_ZN_3 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_3 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_3 Air Node, !- Zone Air Node Name
  Perimeter_ZN_3 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_4,          !- Zone Name
  Perimeter_ZN_4 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_4 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_4 Air Node, !- Zone Air Node Name
  Perimeter_ZN_4 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

NodeList,
  Core_ZN Inlet Nodes,     !- Name
  Core_ZN Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
  PSZ-AC:1_OANode List,    !- Name
  PSZ-AC:1_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:2_OANode List,    !- Name
  PSZ-AC:2_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:3_OANode List,    !- Name
  PSZ-AC:3_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:4_OANode List,    !- Name
  PSZ-AC:4_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:5_OANode List,    !- Name
  PSZ-AC:5_OAInlet Node;   !- Node 1 Name

NodeList,
  Perimeter_ZN_1 Inlet Nodes,  !- Name
  Perimeter_ZN_1 Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
  Perimeter_ZN_2 Inlet Nodes,  !- Name
  Perimeter_ZN_2 Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
```

```
NodeList,
  Perimeter_ZN_4 Inlet Nodes,  !- Name
  Perimeter_ZN_4 Direct Air Inlet Node Name;  !- Node 1 Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Core_ZN Direct Air,      !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Core_ZN Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Core_ZN Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Core_ZN Direct Air ADU,  !- Name
  Core_ZN Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Core_ZN Direct Air,      !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_1 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_1 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_1 Direct Air ADU,  !- Name
  Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_1 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_2 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_2 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_2 Direct Air ADU,  !- Name
  Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_2 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_3 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_3 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
```

```
ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_3 Direct Air ADU,  !- Name
  Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_3 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_4 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_4 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_4 Direct Air ADU,  !- Name
  Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_4 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AvailabilityManagerAssignmentList,
  PSZ-AC:1 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:1 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:2 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:2 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:3 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:3 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:4 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:4 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:5 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:5 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManager:NightCycle,
  PSZ-AC:1 Availability Manager,  !- Name
  ALWAYS_ON,               !- Applicability Schedule Name
  HVACOperationSchd,       !- Fan Schedule Name
  CycleOnAny,              !- Control Type
  1.0,                     !- Thermostat Tolerance {deltaC}
  FixedRunTime,            !- Cycling Run Time Control Type
  1800;                    !- Cycling Run Time {s}

AvailabilityManager:NightCycle,
  PSZ-AC:2 Availability Manager,  !- Name
  ALWAYS_ON,               !- Applicability Schedule Name
  HVACOperationSchd,       !- Fan Schedule Name
  CycleOnAny,              !- Control Type
```

```
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:3 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:4 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:5 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  BranchList,
    PSZ-AC:1 Air Loop Branches,  !- Name
    PSZ-AC:1 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:2 Air Loop Branches,  !- Name
    PSZ-AC:2 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:3 Air Loop Branches,  !- Name
    PSZ-AC:3 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:4 Air Loop Branches,  !- Name
    PSZ-AC:4 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:5 Air Loop Branches,  !- Name
    PSZ-AC:5 Air Loop Main Branch;  !- Branch 1 Name

  Branch,
    PSZ-AC:1 Air Loop Main Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
    PSZ-AC:1_OA,             !- Component 1 Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Component 1 Outlet Node Name
    CoilSystem:Cooling:DX,   !- Component 2 Object Type
    PSZ-AC:1_CoolC,          !- Component 2 Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Component 2 Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Component 2 Outlet Node Name
    Coil:Heating:Fuel,       !- Component 3 Object Type
    PSZ-AC:1_HeatC,          !- Component 3 Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Component 3 Inlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Component 3 Outlet Node Name
    Fan:SystemModel,         !- Component 4 Object Type
    PSZ-AC:1_Fan,            !- Component 4 Name
```

```
Branch,
  PSZ-AC:2 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:2_OA,             !- Component 1 Name
  PSZ-AC:2 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:2_CoolC,          !- Component 2 Name
  PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:2_HeatC,          !- Component 3 Name
  PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:2_Fan,            !- Component 4 Name
  PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:2 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:3 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:3_OA,             !- Component 1 Name
  PSZ-AC:3 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:3_CoolC,          !- Component 2 Name
  PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:3_HeatC,          !- Component 3 Name
  PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:3_Fan,            !- Component 4 Name
  PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:3 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:4 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:4_OA,             !- Component 1 Name
  PSZ-AC:4 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:4_CoolC,          !- Component 2 Name
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:4_HeatC,          !- Component 3 Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:4_Fan,            !- Component 4 Name
  PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:4 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:5 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:5_OA,             !- Component 1 Name
```

```
    CoilSystem:Cooling:DX,   !- Component 2 Object Type
    PSZ-AC:5_CoolC,          !- Component 2 Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Component 2 Inlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Component 2 Outlet Node Name
    Coil:Heating:Fuel,       !- Component 3 Object Type
    PSZ-AC:5_HeatC,          !- Component 3 Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Component 3 Inlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Component 3 Outlet Node Name
    Fan:ConstantVolume,      !- Component 4 Object Type
    PSZ-AC:5_Fan,            !- Component 4 Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Component 4 Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:1_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:1_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:2_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:2_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:3_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:3_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:4_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:4_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:5_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:5_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:1_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:1_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:2_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:2_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:3_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:3_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:4_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:4_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:5_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:5_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:1_OA,             !- Name
    PSZ-AC:1_OA_Controllers, !- Controller List Name
```

```
  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:2_OA,             !- Name
    PSZ-AC:2_OA_Controllers, !- Controller List Name
    PSZ-AC:2_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:3_OA,             !- Name
    PSZ-AC:3_OA_Controllers, !- Controller List Name
    PSZ-AC:3_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:4_OA,             !- Name
    PSZ-AC:4_OA_Controllers, !- Controller List Name
    PSZ-AC:4_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:5_OA,             !- Name
    PSZ-AC:5_OA_Controllers, !- Controller List Name
    PSZ-AC:5_OA_Equipment;   !- Outdoor Air Equipment List Name

  OutdoorAir:NodeList,
    PSZ-AC:1_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:2_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:3_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:4_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:5_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:Node,
    PSZ-AC:1_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:2_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:3_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:4_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:5_CoolCOA Ref node;  !- Name

  OutdoorAir:Mixer,
    PSZ-AC:1_OAMixing Box,   !- Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:1_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:1_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:1 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:2_OAMixing Box,   !- Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:2_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:2_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:2 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:3_OAMixing Box,   !- Name
```

```
    PSZ-AC:3_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:3 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:4_OAMixing Box,   !- Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:4_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:4_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:4 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:5_OAMixing Box,   !- Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:5_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:5_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:5 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  SetpointManager:MixedAir,
    PSZ-AC:1_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:1_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:1_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
```

```
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:3_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:3_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
```

```
AirLoopHVAC:SupplyPath,
  PSZ-AC:1,                !- Name
  PSZ-AC:1 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:1 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:2,                !- Name
  PSZ-AC:2 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:2 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:3,                !- Name
  PSZ-AC:3 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:3 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:4,                !- Name
  PSZ-AC:4 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:4 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:5,                !- Name
  PSZ-AC:5 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:5 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:1 Supply Air Splitter,  !- Name
  PSZ-AC:1 Zone Equipment Inlet Node,  !- Inlet Node Name
  Core_ZN Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:2 Supply Air Splitter,  !- Name
  PSZ-AC:2 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_1 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:3 Supply Air Splitter,  !- Name
  PSZ-AC:3 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_2 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:4 Supply Air Splitter,  !- Name
  PSZ-AC:4 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_3 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:5 Supply Air Splitter,  !- Name
  PSZ-AC:5 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_4 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ReturnPath,
  PSZ-AC:1 Return Air Path,!- Name
  PSZ-AC:1 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
  AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
  PSZ-AC:1 Return Air Mixer;  !- Component 1 Name

AirLoopHVAC:ReturnPath,
  PSZ-AC:2 Return Air Path,!- Name
  PSZ-AC:2 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
  AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
```

```
  AirLoopHVAC:ReturnPath,
    PSZ-AC:3 Return Air Path,!- Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:3 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ReturnPath,
    PSZ-AC:4 Return Air Path,!- Name
    PSZ-AC:4 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:4 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ReturnPath,
    PSZ-AC:5 Return Air Path,!- Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:5 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:1 Return Air Mixer,  !- Name
    PSZ-AC:1 Zone Equipment Outlet Node,  !- Outlet Node Name
    Core_ZN Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:2 Return Air Mixer,  !- Name
    PSZ-AC:2 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_1 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:3 Return Air Mixer,  !- Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_2 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:4 Return Air Mixer,  !- Name
    PSZ-AC:4 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_3 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:5 Return Air Mixer,  !- Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_4 Return Air Node Name;  !- Inlet 1 Node Name

! ***SCHEDULES***

  Schedule:Compact,
    CLGSETP_SCH,             !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 06:00,26.7,       !- Field 3
    Until: 22:00,24.0,       !- Field 5
    Until: 24:00,26.7,       !- Field 7
    For: Saturday,           !- Field 9
    Until: 06:00,26.7,       !- Field 10
    Until: 18:00,24.0,       !- Field 12
    Until: 24:00,26.7,       !- Field 14
    For: AllOtherDays,       !- Field 16
    Until: 24:00,26.7;       !- Field 17

  Schedule:Compact,
    HTGSETP_SCH,             !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays,           !- Field 2
    Until: 06:00,15.6,       !- Field 3
```

```
    For: Saturday,           !- Field 9
    Until: 06:00,15.6,       !- Field 10
    Until: 18:00,21.0,       !- Field 12
    Until: 24:00,15.6,       !- Field 14
    For: WinterDesignDay,    !- Field 16
    Until: 24:00,21.0,       !- Field 17
    For: AllOtherDays,       !- Field 19
    Until: 24:00,15.6;       !- Field 20

  Schedule:Compact,
    MinOA_Sched,             !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 07:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 07:00,0.0,        !- Field 10
    Until: 18:00,1.0,        !- Field 12
    Until: 24:00,0.0,        !- Field 14
    For: AllOtherDays,       !- Field 16
    Until: 24:00,0.0;        !- Field 17

  Schedule:Compact,
    Dual Zone Control Type Sched,  !- Name
    Control Type,            !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,4;          !- Field 3

  Schedule:Compact,
    HVACOperationSchd,       !- Name
    On/Off,                  !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 06:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 06:00,0.0,        !- Field 10
    Until: 18:00,1.0,        !- Field 12
    Until: 24:00,0.0,        !- Field 14
    For: AllOtherDays,       !- Field 16
    Until: 24:00,0.0;        !- Field 17

! ***SWH EQUIPMENT***

  WaterHeater:Mixed,
    SWHSys1 Water Heater,    !- Name
    0.1514,                  !- Tank Volume {m3}
    SWHSys1 Water Heater Setpoint Temperature Schedule Name,  !- Setpoint Temperature Schedule Name
    2.0,                     !- Deadband Temperature Difference {deltaC}
    82.2222,                 !- Maximum Temperature Limit {C}
    Cycle,                   !- Heater Control Type
    845000,                  !- Heater Maximum Capacity {W}
    ,                        !- Heater Minimum Capacity {W}
    ,                        !- Heater Ignition Minimum Flow Rate {m3/s}
    ,                        !- Heater Ignition Delay {s}
    NATURALGAS,              !- Heater Fuel Type
    0.8,                     !- Heater Thermal Efficiency
    ,                        !- Part Load Factor Curve Name
    20,                      !- Off Cycle Parasitic Fuel Consumption Rate {W}
    NATURALGAS,              !- Off Cycle Parasitic Fuel Type
    0.8,                     !- Off Cycle Parasitic Heat Fraction to Tank
    ,                        !- On Cycle Parasitic Fuel Consumption Rate {W}
```

```
    SCHEDULE,                !- Ambient Temperature Indicator
    SWHSys1 Water Heater Ambient Temperature Schedule Name,  !- Ambient Temperature Schedule Name
    ,                        !- Ambient Temperature Zone Name
    ,                        !- Ambient Temperature Outdoor Air Node Name
    6.0,                     !- Off Cycle Loss Coefficient to Ambient Temperature {W/K}
    ,                        !- Off Cycle Loss Fraction to Zone
    6.0,                     !- On Cycle Loss Coefficient to Ambient Temperature {W/K}
    ,                        !- On Cycle Loss Fraction to Zone
    ,                        !- Peak Use Flow Rate {m3/s}
    ,                        !- Use Flow Rate Fraction Schedule Name
    ,                        !- Cold Water Supply Temperature Schedule Name
    SWHSys1 Pump-SWHSys1 Water HeaterNode,  !- Use Side Inlet Node Name
    SWHSys1 Supply Equipment Outlet Node,  !- Use Side Outlet Node Name
    1.0,                     !- Use Side Effectiveness
    ,                        !- Source Side Inlet Node Name
    ,                        !- Source Side Outlet Node Name
    1.0,                     !- Source Side Effectiveness
    AUTOSIZE,                !- Use Side Design Flow Rate {m3/s}
    AUTOSIZE,                !- Source Side Design Flow Rate {m3/s}
    1.5;                     !- Indirect Water Heating Recovery Time {hr}

  Pump:ConstantSpeed,
    SWHSys1 Pump,            !- Name
    SWHSys1 Supply Inlet Node,  !- Inlet Node Name
    SWHSys1 Pump-SWHSys1 Water HeaterNodeviaConnector,  !- Outlet Node Name
    AUTOSIZE,                !- Design Flow Rate {m3/s}
    0.001,                   !- Design Pump Head {Pa}
    AUTOSIZE,                !- Design Power Consumption {W}
    1,                       !- Motor Efficiency
    0.0,                     !- Fraction of Motor Inefficiencies to Fluid Stream
    Intermittent,            !- Pump Control Type
    ;                        !- Pump Flow Rate Schedule Name

  WaterUse:Equipment,
    Core_ZN Water Equipment, !- Name
    ,                        !- End-Use Subcategory
    3.15e-006,               !- Peak Flow Rate {m3/s}
    BLDG_SWH_SCH,            !- Flow Rate Fraction Schedule Name
    Water Equipment Temp Sched,  !- Target Temperature Schedule Name
    Water Equipment Hot Supply Temp Sched,  !- Hot Water Supply Temperature Schedule Name
    ,                        !- Cold Water Supply Temperature Schedule Name
    Core_ZN,                 !- Zone Name
    Water Equipment Sensible fract sched,  !- Sensible Fraction Schedule Name
    Water Equipment Latent fract sched;  !- Latent Fraction Schedule Name

  PlantEquipmentList,
    SWHSys1 Equipment List,  !- Name
    WaterHeater:Mixed,       !- Equipment 1 Object Type
    SWHSys1 Water Heater;    !- Equipment 1 Name

! ***SWH SIZING & CONTROLS***

  Sizing:Plant,
    SWHSys1,                 !- Plant or Condenser Loop Name
    Heating,                 !- Loop Type
    60,                      !- Design Loop Exit Temperature {C}
    5.0;                     !- Loop Design Temperature Difference {deltaC}

  SetpointManager:Scheduled,
    SWHSys1 Loop Setpoint Manager,  !- Name
    Temperature,             !- Control Variable
    SWHSys1-Loop-Temp-Schedule,  !- Schedule Name
    SWHSys1 Supply Outlet Node;  !- Setpoint Node or NodeList Name

  PlantEquipmentOperationSchemes,
    SWHSys1 Loop Operation Scheme List,  !- Name
```

```
    ALWAYS_ON;                !- Control Scheme 1 Schedule Name

  PlantEquipmentOperation:HeatingLoad,
    SWHSys1 Operation Scheme,!- Name
    0.0,                     !- Load Range 1 Lower Limit {W}
    1000000000000000,        !- Load Range 1 Upper Limit {W}
    SWHSys1 Equipment List;  !- Range 1 Equipment List Name

! ***SWH LOOP***

  PlantLoop,
    SWHSys1,                 !- Name
    Water,                   !- Fluid Type
    ,                        !- User Defined Fluid Type
    SWHSys1 Loop Operation Scheme List,  !- Plant Equipment Operation Scheme Name
    SWHSys1 Supply Outlet Node,  !- Loop Temperature Setpoint Node Name
    60.0,                    !- Maximum Loop Temperature {C}
    10.0,                    !- Minimum Loop Temperature {C}
    AUTOSIZE,                !- Maximum Loop Flow Rate {m3/s}
    0.0,                     !- Minimum Loop Flow Rate {m3/s}
    AUTOSIZE,                !- Plant Loop Volume {m3}
    SWHSys1 Supply Inlet Node,  !- Plant Side Inlet Node Name
    SWHSys1 Supply Outlet Node,  !- Plant Side Outlet Node Name
    SWHSys1 Supply Branches, !- Plant Side Branch List Name
    SWHSys1 Supply Connectors,  !- Plant Side Connector List Name
    SWHSys1 Demand Inlet Node,  !- Demand Side Inlet Node Name
    SWHSys1 Demand Outlet Node,  !- Demand Side Outlet Node Name
    SWHSys1 Demand Branches, !- Demand Side Branch List Name
    SWHSys1 Demand Connectors,  !- Demand Side Connector List Name
    Optimal;                 !- Load Distribution Scheme

! ***SWH CONNECTIONS***

  BranchList,
    SWHSys1 Demand Branches, !- Name
    SWHSys1 Demand Inlet Branch,  !- Branch 1 Name
    SWHSys1 Demand Load Branch 1,  !- Branch 2 Name
    SWHSys1 Demand Bypass Branch,  !- Branch 3 Name
    SWHSys1 Demand Outlet Branch;  !- Branch 4 Name

  BranchList,
    SWHSys1 Supply Branches, !- Name
    SWHSys1 Supply Inlet Branch,  !- Branch 1 Name
    SWHSys1 Supply Equipment Branch,  !- Branch 2 Name
    SWHSys1 Supply Equipment Bypass Branch,  !- Branch 3 Name
    SWHSys1 Supply Outlet Branch;  !- Branch 4 Name

  Branch,
    SWHSys1 Demand Bypass Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Demand Bypass Pipe,  !- Component 1 Name
    SWHSys1 Demand Bypass Pipe Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Bypass Pipe Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Demand Inlet Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Demand Inlet Pipe,  !- Component 1 Name
    SWHSys1 Demand Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Inlet Pipe-SWHSys1 Demand Mixer;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Demand Load Branch 1,  !- Name
    ,                        !- Pressure Drop Curve Name
```

```
    Core_ZN Water Equipment Water Inlet Node,  !- Component 1 Inlet Node Name
    Core_ZN Water Equipment Water Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Demand Outlet Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Demand Outlet Pipe,  !- Component 1 Name
    SWHSys1 Demand Mixer-SWHSys1 Demand Outlet Pipe,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Equipment Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    WaterHeater:Mixed,       !- Component 1 Object Type
    SWHSys1 Water Heater,    !- Component 1 Name
    SWHSys1 Pump-SWHSys1 Water HeaterNode,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Equipment Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Equipment Bypass Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Supply Equipment Bypass Pipe,  !- Component 1 Name
    SWHSys1 Supply Equip Bypass Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Equip Bypass Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Inlet Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pump:ConstantSpeed,      !- Component 1 Object Type
    SWHSys1 Pump,            !- Component 1 Name
    SWHSys1 Supply Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Pump-SWHSys1 Water HeaterNodeviaConnector;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Outlet Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Supply Outlet Pipe,  !- Component 1 Name
    SWHSys1 Supply Mixer-SWHSys1 Supply Outlet Pipe,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Outlet Node;  !- Component 1 Outlet Node Name

  ConnectorList,
    SWHSys1 Demand Connectors,  !- Name
    Connector:Splitter,      !- Connector 1 Object Type
    SWHSys1 Demand Splitter, !- Connector 1 Name
    Connector:Mixer,         !- Connector 2 Object Type
    SWHSys1 Demand Mixer;    !- Connector 2 Name

  ConnectorList,
    SWHSys1 Supply Connectors,  !- Name
    Connector:Splitter,      !- Connector 1 Object Type
    SWHSys1 Supply Splitter, !- Connector 1 Name
    Connector:Mixer,         !- Connector 2 Object Type
    SWHSys1 Supply Mixer;    !- Connector 2 Name

  WaterUse:Connections,
    Core_ZN Water Equipment, !- Name
    Core_ZN Water Equipment Water Inlet Node,  !- Inlet Node Name
    Core_ZN Water Equipment Water Outlet Node,  !- Outlet Node Name
    ,                        !- Supply Water Storage Tank Name
    ,                        !- Reclamation Water Storage Tank Name
    ,                        !- Hot Water Supply Temperature Schedule Name
    ,                        !- Cold Water Supply Temperature Schedule Name
    ,                        !- Drain Water Heat Exchanger Type
```

```
    Core_ZN Water Equipment; !- Water Use Equipment 1 Name

  Connector:Splitter,
    SWHSys1 Demand Splitter, !- Name
    SWHSys1 Demand Inlet Branch,  !- Inlet Branch Name
    SWHSys1 Demand Load Branch 1,  !- Outlet Branch 1 Name
    SWHSys1 Demand Bypass Branch;  !- Outlet Branch 2 Name

  Connector:Splitter,
    SWHSys1 Supply Splitter, !- Name
    SWHSys1 Supply Inlet Branch,  !- Inlet Branch Name
    SWHSys1 Supply Equipment Branch,  !- Outlet Branch 1 Name
    SWHSys1 Supply Equipment Bypass Branch;  !- Outlet Branch 2 Name

  Connector:Mixer,
    SWHSys1 Demand Mixer,    !- Name
    SWHSys1 Demand Outlet Branch,  !- Outlet Branch Name
    SWHSys1 Demand Load Branch 1,  !- Inlet Branch 1 Name
    SWHSys1 Demand Bypass Branch;  !- Inlet Branch 2 Name

  Connector:Mixer,
    SWHSys1 Supply Mixer,    !- Name
    SWHSys1 Supply Outlet Branch,  !- Outlet Branch Name
    SWHSys1 Supply Equipment Branch,  !- Inlet Branch 1 Name
    SWHSys1 Supply Equipment Bypass Branch;  !- Inlet Branch 2 Name

  Pipe:Adiabatic,
    SWHSys1 Demand Bypass Pipe,  !- Name
    SWHSys1 Demand Bypass Pipe Inlet Node,  !- Inlet Node Name
    SWHSys1 Demand Bypass Pipe Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Demand Inlet Pipe,  !- Name
    SWHSys1 Demand Inlet Node,  !- Inlet Node Name
    SWHSys1 Demand Inlet Pipe-SWHSys1 Demand Mixer;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Demand Outlet Pipe,  !- Name
    SWHSys1 Demand Mixer-SWHSys1 Demand Outlet Pipe,  !- Inlet Node Name
    SWHSys1 Demand Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Supply Equipment Bypass Pipe,  !- Name
    SWHSys1 Supply Equip Bypass Inlet Node,  !- Inlet Node Name
    SWHSys1 Supply Equip Bypass Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Supply Outlet Pipe,  !- Name
    SWHSys1 Supply Mixer-SWHSys1 Supply Outlet Pipe,  !- Inlet Node Name
    SWHSys1 Supply Outlet Node;  !- Outlet Node Name

! ***SWH SCHEDULES***

  Schedule:Compact,
    BLDG_SWH_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 05:00,0.05,       !- Field 3
    Until: 06:00,0.08,       !- Field 5
    Until: 07:00,0.07,       !- Field 7
    Until: 08:00,0.19,       !- Field 9
    Until: 09:00,0.35,       !- Field 11
    Until: 10:00,0.38,       !- Field 13
    Until: 11:00,0.39,       !- Field 15
    Until: 12:00,0.47,       !- Field 17
```

```
    Until: 15:00,0.34,       !- Field 23
    Until: 16:00,0.33,       !- Field 25
    Until: 17:00,0.44,       !- Field 27
    Until: 18:00,0.26,       !- Field 29
    Until: 19:00,0.21,       !- Field 31
    Until: 20:00,0.15,       !- Field 33
    Until: 21:00,0.17,       !- Field 35
    Until: 22:00,0.08,       !- Field 37
    Until: 24:00,0.05,       !- Field 39
    For: Saturday WinterDesignDay, !- Field 41
    Until: 05:00,0.05,       !- Field 42
    Until: 06:00,0.08,       !- Field 44
    Until: 07:00,0.07,       !- Field 46
    Until: 08:00,0.11,       !- Field 48
    Until: 09:00,0.15,       !- Field 50
    Until: 10:00,0.21,       !- Field 52
    Until: 11:00,0.19,       !- Field 54
    Until: 12:00,0.23,       !- Field 56
    Until: 13:00,0.20,       !- Field 58
    Until: 14:00,0.19,       !- Field 60
    Until: 15:00,0.15,       !- Field 62
    Until: 16:00,0.13,       !- Field 64
    Until: 17:00,0.14,       !- Field 66
    Until: 21:00,0.07,       !- Field 68
    Until: 22:00,0.09,       !- Field 70
    Until: 24:00,0.05,       !- Field 72
    For: AllOtherDays,       !- Field 74
    Until: 05:00,0.04,       !- Field 75
    Until: 06:00,0.07,       !- Field 77
    Until: 11:00,0.04,       !- Field 79
    Until: 13:00,0.06,       !- Field 81
    Until: 14:00,0.09,       !- Field 83
    Until: 15:00,0.06,       !- Field 85
    Until: 21:00,0.04,       !- Field 87
    Until: 22:00,0.07,       !- Field 89
    Until: 24:00,0.04;       !- Field 91

  Schedule:Compact,
    Water Equipment Latent fract sched,  !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.05;       !- Field 3

  Schedule:Compact,
    Water Equipment Sensible fract sched,  !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.2;        !- Field 3

  Schedule:Compact,
    SWHSys1 Water Heater Ambient Temperature Schedule Name,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,22.0;       !- Field 3

  Schedule:Compact,
    Water Equipment Temp Sched,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,43.3;       !- Field 3

  Schedule:Compact,
```

```
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,43.3;       !- Field 3

  Schedule:Compact,
    SWHSys1 Water Heater Setpoint Temperature Schedule Name,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,60.0;       !- Field 3

  Schedule:Compact,
    SWHSys1-Loop-Temp-Schedule,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,60.0;       !- Field 3

! ***ECONOMICS***
!   IN_EIAMonthlyRateGas, Source EIA historical Nov2003 thru Oct2004
! Indiana 1999 state average electricity emissions factors based on eGRID, 1065, AirData
! PSI_CS_CommercialElectricService, source http://www.cinergypsi.com/pdfs/RateCS.pdf, effective 2004-05-24
! PSI_LLF_LowLoadFactorService,source http://www.cinergypsi.com/pdfs/RATELLF.pdf, effective 2004-05-24

  UtilityCost:Tariff,
    PSI_LLF_LowLoadFactorService,  !- Name
    Electricity:Facility,    !- Output Meter Name
    kWh,                     !- Conversion Factor Choice
    ,                        !- Energy Conversion Factor
    ,                        !- Demand Conversion Factor
    ,                        !- Time of Use Period Schedule Name
    ,                        !- Season Schedule Name
    ,                        !- Month Schedule Name
    HalfHour,                !- Demand Window Length
    15.00,                   !- Monthly Charge or Variable Name
    ,                        !- Minimum Monthly Charge or Variable Name
    ,                        !- Real Time Pricing Charge Schedule Name
    ,                        !- Customer Baseline Load Schedule Name
    Comm Elect;              !- Group Name

  UtilityCost:Charge:Block,
    AnnualEnergyCharge,      !- Utility Cost Charge Block Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    ,                        !- Remaining Into Variable
    ,                        !- Block Size Multiplier Value or Variable Name
    300,                     !- Block Size 1 Value or Variable Name
    0.108222,                !- Block 1 Cost per Unit Value or Variable Name
    700,                     !- Block Size 2 Value or Variable Name
    0.087021,                !- Block 2 Cost per Unit Value or Variable Name
    1500,                    !- Block Size 3 Value or Variable Name
    0.078420,                !- Block 3 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 4 Value or Variable Name
    0.058320;                !- Block 4 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Block,
    AnnualDemandBaseCharge,  !- Utility Cost Charge Block Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    ,                        !- Remaining Into Variable
    TotalDemand,             !- Block Size Multiplier Value or Variable Name
    190,                     !- Block Size 1 Value or Variable Name
```

```
    0.051773,                !- Block 2 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 3 Value or Variable Name
    0.046965;                !- Block 3 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    FuelCostAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.002028;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    QualPollutionControlAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000536;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    SoxNoxRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.001127;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    DSMRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000370;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    PurchPowerTrackerAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000031;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    MidwestISOAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000216;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    CleanCoalRiderEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000833;                !- Cost per Unit Value or Variable Name

  UtilityCost:Qualify,
    MinDemand75kw,           !- Utility Cost Qualify Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    TotalDemand,             !- Variable Name
    Minimum,                 !- Qualify Type
    75,                      !- Threshold Value or Variable Name
```

```
    12;                      !- Number of Months

  UtilityCost:Charge:Simple,
    TaxofeightPercent,       !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    SubTotal,                !- Source Variable
    Annual,                  !- Season
    Taxes,                   !- Category Variable Name
    0.08;                    !- Cost per Unit Value or Variable Name

!end  PSI_LLF_LowLoadFactorService

  UtilityCost:Tariff,
    PSI_CS_CommercialElectricService,  !- Name
    Electricity:Facility,    !- Output Meter Name
    kWh,                     !- Conversion Factor Choice
    ,                        !- Energy Conversion Factor
    ,                        !- Demand Conversion Factor
    ,                        !- Time of Use Period Schedule Name
    ,                        !- Season Schedule Name
    ,                        !- Month Schedule Name
    ,                        !- Demand Window Length
    9.40,                    !- Monthly Charge or Variable Name
    ,                        !- Minimum Monthly Charge or Variable Name
    ,                        !- Real Time Pricing Charge Schedule Name
    ,                        !- Customer Baseline Load Schedule Name
    Comm Elect;              !- Group Name

  UtilityCost:Charge:Block,
    AnnualEnergyCharge,      !- Utility Cost Charge Block Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    ,                        !- Remaining Into Variable
    ,                        !- Block Size Multiplier Value or Variable Name
    300,                     !- Block Size 1 Value or Variable Name
    0.082409,                !- Block 1 Cost per Unit Value or Variable Name
    700,                     !- Block Size 2 Value or Variable Name
    0.072873,                !- Block 2 Cost per Unit Value or Variable Name
    1500,                    !- Block Size 3 Value or Variable Name
    0.061696,                !- Block 3 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 4 Value or Variable Name
    0.041179;                !- Block 4 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    FuelCostAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.002028;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    QualPollutionControlAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000536;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    SoxNoxRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
```

```
UtilityCost:Charge:Simple,
  DSMRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  totalEnergy,             !- Source Variable
  Annual,                  !- Season
  EnergyCharges,           !- Category Variable Name
  0.000021;                !- Cost per Unit Value or Variable Name

UtilityCost:Charge:Simple,
  PurchPowerTrackerAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  totalEnergy,             !- Source Variable
  Annual,                  !- Season
  EnergyCharges,           !- Category Variable Name
  0.000034;                !- Cost per Unit Value or Variable Name

UtilityCost:Charge:Simple,
  MidwestISOAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  totalEnergy,             !- Source Variable
  Annual,                  !- Season
  EnergyCharges,           !- Category Variable Name
  -0.000203;               !- Cost per Unit Value or Variable Name

UtilityCost:Charge:Simple,
  CleanCoalRiderEnergyCharge,  !- Utility Cost Charge Simple Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  totalEnergy,             !- Source Variable
  Annual,                  !- Season
  EnergyCharges,           !- Category Variable Name
  0.000807;                !- Cost per Unit Value or Variable Name

UtilityCost:Qualify,
  MaxDemand75kw,           !- Utility Cost Qualify Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  TotalDemand,             !- Variable Name
  Maximum,                 !- Qualify Type
  75,                      !- Threshold Value or Variable Name
  Annual,                  !- Season
  Count,                   !- Threshold Test
  1;                       !- Number of Months

UtilityCost:Charge:Simple,
  TaxofeightPercent,       !- Utility Cost Charge Simple Name
  PSI_CS_CommercialElectricService,  !- Tariff Name
  SubTotal,                !- Source Variable
  Annual,                  !- Season
  Taxes,                   !- Category Variable Name
  0.08;                    !- Cost per Unit Value or Variable Name

UtilityCost:Tariff,
  IN_EIAMonthlyRateGas,    !- Name
  NaturalGas:Facility,     !- Output Meter Name
  MCF,                     !- Conversion Factor Choice
  ,                        !- Energy Conversion Factor
  ,                        !- Demand Conversion Factor
  ,                        !- Time of Use Period Schedule Name
  ,                        !- Season Schedule Name
  ,                        !- Month Schedule Name
  ,                        !- Demand Window Length
  0.0,                     !- Monthly Charge or Variable Name
  ,                        !- Minimum Monthly Charge or Variable Name
  ,                        !- Real Time Pricing Charge Schedule Name
  ,                        !- Customer Baseline Load Schedule Name
  Comm Gas;                !- Group Name
```

```
    MonthlyRateGasCharge,    !- Utility Cost Charge Simple Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    IN_MonthlyGasRates;      !- Cost per Unit Value or Variable Name

  UtilityCost:Variable,
    IN_MonthlyGasRates,      !- Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    Currency,                !- Variable Type
    8.22,                    !- January Value
    7.51,                    !- February Value
    8.97,                    !- March Value
    9.01,                    !- April Value
    9.16,                    !- May Value
    10.44,                   !- June Value
    10.32,                   !- July Value
    10.13,                   !- August Value
    9.20,                    !- September Value
    8.18,                    !- October Value
    7.83,                    !- November Value
    7.63;                    !- December Value

  UtilityCost:Charge:Simple,
    TaxofEightPercent,       !- Utility Cost Charge Simple Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    SubTotal,                !- Source Variable
    Annual,                  !- Season
    Taxes,                   !- Category Variable Name
    0.08;                    !- Cost per Unit Value or Variable Name

! ***GENERAL REPORTING***

  OutputControl:ReportingTolerances,
    0.556,                   !- Tolerance for Time Heating Setpoint Not Met {deltaC}
    0.556;                   !- Tolerance for Time Cooling Setpoint Not Met {deltaC}

  Output:SQLite,
    Simple;                  !- Option Type

  Output:VariableDictionary,IDF,Unsorted;

  Output:Surfaces:List,Details;

  Output:Surfaces:Drawing,DXF;

  Output:Constructions,Constructions;

! ***REPORT METERS/VARIABLES***

  Output:Meter,Electricity:Facility,HOURLY;

  Output:Meter,Fans:Electricity,HOURLY;

  Output:Meter,Cooling:Electricity,HOURLY;

  Output:Meter,Heating:Electricity,HOURLY;

  Output:Meter,InteriorLights:Electricity,HOURLY;

  Output:Meter,InteriorEquipment:Electricity,HOURLY;

  Output:Meter,NaturalGas:Facility,HOURLY;

  Output:Meter,Heating:NaturalGas,HOURLY;
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  Output:Meter,Water Heater:WaterSystems:NaturalGas,HOURLY;

  Output:Variable,*,Site Outdoor Air Drybulb Temperature,HOURLY;

  Output:Variable,*,Site Outdoor Air Humidity Ratio,HOURLY;

  Output:Variable,*,Site Outdoor Air Relative Humidity,HOURLY;

  Output:Variable,*,Air System Outdoor Air Flow Fraction,HOURLY;

  Output:Variable,*,Air System Simulation Cycle On Off Status,HOURLY;

  Output:Variable,*,Air System Outdoor Air Economizer Status,HOURLY;

  Output:Variable,*,Air System Total Heating Energy,HOURLY;

  Output:Variable,*,Air System Total Cooling Energy,HOURLY;

  Output:Variable,*,Air System Fan Electricity Energy,HOURLY;

  Output:Variable,*,Zone Mean Air Temperature,hourly;

! ***REPORT TABLES***

  OutputControl:Table:Style,
    HTML;                    !- Column Separator

  Output:Table:SummaryReports,
    AnnualBuildingUtilityPerformanceSummary,  !- Report 1 Name
    InputVerificationandResultsSummary,  !- Report 2 Name
    ClimaticDataSummary,     !- Report 3 Name
    EnvelopeSummary,         !- Report 4 Name
    EquipmentSummary,        !- Report 5 Name
    ComponentSizingSummary,  !- Report 6 Name
    HVACSizingSummary,       !- Report 7 Name
    SystemSummary;           !- Report 8 Name

  Output:Table:Monthly,
    Emissions Data Summary,  !- Name
    4,                       !- Digits After Decimal
    CO2:Facility,            !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    NOx:Facility,            !- Variable or Meter 2 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 2
    SO2:Facility,            !- Variable or Meter 3 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 3
    PM:Facility,             !- Variable or Meter 4 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 4
    Hg:Facility,             !- Variable or Meter 5 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 5
    WaterEnvironmentalFactors:Facility,  !- Variable or Meter 6 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 6
    Carbon Equivalent:Facility,  !- Variable or Meter 7 Name
    SumOrAverage;            !- Aggregation Type for Variable or Meter 7

  Output:Table:Monthly,
    Components of Peak Electrical Demand,  !- Name
    3,                       !- Digits After Decimal
    Electricity:Facility,    !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Electricity:Facility,    !- Variable or Meter 2 Name
    Maximum,                 !- Aggregation Type for Variable or Meter 2
    InteriorLights:Electricity,  !- Variable or Meter 3 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 3
    InteriorEquipment:Electricity,  !- Variable or Meter 4 Name
```

```
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 5
    Heating:Electricity,     !- Variable or Meter 6 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 6
    Cooling:Electricity,     !- Variable or Meter 7 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 7
    ExteriorLights:Electricity,  !- Variable or Meter 8 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 8
    Pumps:Electricity,       !- Variable or Meter 9 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 9
    HeatRejection:Electricity,  !- Variable or Meter 10 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 10
    ExteriorEquipment:Electricity,  !- Variable or Meter 11 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 11
    Humidification:Electricity,  !- Variable or Meter 12 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 12
    HeatRecovery:Electricity,!- Variable or Meter 13 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 13
    WaterSystems:Electricity,!- Variable or Meter 14 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 14
    Refrigeration:Electricity,  !- Variable or Meter 15 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 15
    Generators:Electricity,  !- Variable or Meter 16 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 16
    ElectricityProduced:Facility,  !- Variable or Meter 17 Name
    ValueWhenMaximumOrMinimum;  !- Aggregation Type for Variable or Meter 17

  Output:Table:Monthly,
    Heating Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Heating Coil NaturalGas Rate,  !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Heating Coil NaturalGas Rate,  !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:Monthly,
    Cooling Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Cooling Coil Electricity Rate,  !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Cooling Coil Electricity Rate,  !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:Monthly,
    Fan Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Fan Electricity Rate,    !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Fan Electricity Rate,    !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:TimeBins,
    *,                       !- Key Value
    Zone Air Relative Humidity,  !- Variable Name
    60,                      !- Interval Start
    10,                      !- Interval Size
    4;                       !- Interval Count

  Output:Table:TimeBins,
    *,                       !- Key Value
    Air System Outdoor Air Flow Fraction,  !- Variable Name
    0.00,                    !- Interval Start
    0.20,                    !- Interval Size
    5;                       !- Interval Count

  Output:Table:TimeBins,
    *,                       !- Key Value
```

```
    1,                       !- Interval Size
    4;                       !- Interval Count

! ***ENVIRONMENTAL FACTORS REPORTING***

  Output:EnvironmentalImpactFactors,
    Monthly;                 !- Reporting Frequency

  EnvironmentalImpactFactors,
    0.663,                   !- District Heating Water Efficiency
    4.18,                    !- District Cooling COP {W/W}
    0.585,                   !- District Heating Steam Conversion Efficiency
    80.7272,                 !- Total Carbon Equivalent Emission Factor From N2O {kg/kg}
    6.2727,                  !- Total Carbon Equivalent Emission Factor From CH4 {kg/kg}
    0.2727;                  !- Total Carbon Equivalent Emission Factor From CO2 {kg/kg}

! Indiana electricity source and emission factors based on Deru and Torcellini 2007

  FuelFactors,
    Electricity,             !- Existing Fuel Resource Name
    3.546,                   !- Source Energy Factor {J/J}
    ,                        !- Source Energy Schedule Name
    3.417E+02,               !- CO2 Emission Factor {g/MJ}
    ,                        !- CO2 Emission Factor Schedule Name
    1.186E-01,               !- CO Emission Factor {g/MJ}
    ,                        !- CO Emission Factor Schedule Name
    7.472E-01,               !- CH4 Emission Factor {g/MJ}
    ,                        !- CH4 Emission Factor Schedule Name
    6.222E-01,               !- NOx Emission Factor {g/MJ}
    ,                        !- NOx Emission Factor Schedule Name
    8.028E-03,               !- N2O Emission Factor {g/MJ}
    ,                        !- N2O Emission Factor Schedule Name
    1.872E+00,               !- SO2 Emission Factor {g/MJ}
    ,                        !- SO2 Emission Factor Schedule Name
    0.0,                     !- PM Emission Factor {g/MJ}
    ,                        !- PM Emission Factor Schedule Name
    1.739E-02,               !- PM10 Emission Factor {g/MJ}
    ,                        !- PM10 Emission Factor Schedule Name
    0.0,                     !- PM2.5 Emission Factor {g/MJ}
    ,                        !- PM2.5 Emission Factor Schedule Name
    0.0,                     !- NH3 Emission Factor {g/MJ}
    ,                        !- NH3 Emission Factor Schedule Name
    1.019E-02,               !- NMVOC Emission Factor {g/MJ}
    ,                        !- NMVOC Emission Factor Schedule Name
    5.639E-06,               !- Hg Emission Factor {g/MJ}
    ,                        !- Hg Emission Factor Schedule Name
    2.778E-05,               !- Pb Emission Factor {g/MJ}
    ,                        !- Pb Emission Factor Schedule Name
    0.4309556,               !- Water Emission Factor {L/MJ}
    ,                        !- Water Emission Factor Schedule Name
    0,                       !- Nuclear High Level Emission Factor {g/MJ}
    ,                        !- Nuclear High Level Emission Factor Schedule Name
    0;                       !- Nuclear Low Level Emission Factor {m3/MJ}

! Deru and Torcellini 2007
! Source Energy and Emission Factors for Energy Use in Buildings
! NREL/TP-550-38617
! source factor and Higher Heating Values from Table 5
! post-combustion emission factors for boiler from Table 9 (with factor of 1000 correction for natural gas)

  FuelFactors,
    NaturalGas,              !- Existing Fuel Resource Name
    1.092,                   !- Source Energy Factor {J/J}
    ,                        !- Source Energy Schedule Name
    5.21E+01,                !- CO2 Emission Factor {g/MJ}
    ,                        !- CO2 Emission Factor Schedule Name
```

```
    1.06E-03,                !- CH4 Emission Factor {g/MJ}
    ,                        !- CH4 Emission Factor Schedule Name
    4.73E-02,                !- NOx Emission Factor {g/MJ}
    ,                        !- NOx Emission Factor Schedule Name
    1.06E-03,                !- N2O Emission Factor {g/MJ}
    ,                        !- N2O Emission Factor Schedule Name
    2.68E-04,                !- SO2 Emission Factor {g/MJ}
    ,                        !- SO2 Emission Factor Schedule Name
    0.0,                     !- PM Emission Factor {g/MJ}
    ,                        !- PM Emission Factor Schedule Name
    3.59E-03,                !- PM10 Emission Factor {g/MJ}
    ,                        !- PM10 Emission Factor Schedule Name
    0.0,                     !- PM2.5 Emission Factor {g/MJ}
    ,                        !- PM2.5 Emission Factor Schedule Name
    0,                       !- NH3 Emission Factor {g/MJ}
    ,                        !- NH3 Emission Factor Schedule Name
    2.61E-03,                !- NMVOC Emission Factor {g/MJ}
    ,                        !- NMVOC Emission Factor Schedule Name
    1.11E-07,                !- Hg Emission Factor {g/MJ}
    ,                        !- Hg Emission Factor Schedule Name
    2.13E-07,                !- Pb Emission Factor {g/MJ}
    ,                        !- Pb Emission Factor Schedule Name
    0,                       !- Water Emission Factor {L/MJ}
    ,                        !- Water Emission Factor Schedule Name
    0,                       !- Nuclear High Level Emission Factor {g/MJ}
    ,                        !- Nuclear High Level Emission Factor Schedule Name
    0;                       !- Nuclear Low Level Emission Factor {m3/MJ}


  Output:EnergyManagementSystem,
    Verbose,                 !- Actuator Availability Dictionary Reporting
    Verbose,                 !- Internal Variable Availability Dictionary Reporting
    Verbose;                 !- EMS Runtime Language Debug Output Level

  Output:Variable,*,Zone Mean Air Temperature,Hourly;
  Output:Variable,*,Zone Air Relative Humidity,Hourly;
  Output:Variable,*,Zone Thermal Comfort Fanger Model PMV,Hourly;
  Output:Variable,*,Site Outdoor Air Drybulb Temperature,Hourly;
  Output:Variable,*,Facility Total Electricity Demand Rate,Hourly;
```

```
! RefBldgSmallOfficeNew2004_Chicago.idf
! DOE Commercial Reference Building
! Small Office, new construction 90.1-2004
! ASHRAE Standards 90.1-2004 and 62-1999
!
! Description:  Single story, five zone office building.
! Form:         Area = 511 m2 (5,500 ft2); Number of Stories = 1; Shape = rectangle, Aspect ratio = 1.5
! Envelope:     Envelope thermal properties vary with climate according to ASHRAE Standard 90.1-2004.
!               Opaque constructions: mass walls; attic roof; slab-on-grade floor
!               Windows: window-to-wall ratio = 21.2%, equal distribution of punched windows
!               Infiltration = 0.4 cfm/ft2 above grade wall area at 0.3 in wc (75 Pa) adjusted to 0.016 in wc (4 Pa).
!                              25% of full value when ventilation system on.
! HVAC:         PSZ-AC, gas furnace
!               No economizers, per ASHRAE 90.1-2004
!
! Int. gains:   lights = 10.76 W/m2 (1.0 W/ft2) (building area method);
!               elec. plug loads = 10.76 W/m2 (1.0 W/ft2)
!               gas plug load = 0 W/m2 (0 W/ft2)
!               people = 28 total; 5.38/100 m2 (5.0/1000 ft2)
!
! Detached Shading:                 None
! Daylight:                         None
! Natural Ventilation:              None
! Zonal Equipment:                  None
! Air Primary Loops:                PSZ
! Plant Loops:                      SHWSys1
! System Equipment Autosize:        Yes
! Purchased Cooling:                None
! Purchased Heating:                None
! Coils:                            Coil:Cooling:DX:SingleSpeed; Coil:Heating:Fuel
! Pumps:                            None
! Boilers:                          None
! Chillers:                         None
!
! Reference citation for the Commercial Reference Buildings:
! Deru, M.; Field, K.; Studer, D.; Benne, K.; Griffith, B.; Torcellini, P;
! Halverson, M.; Winiarski, D.; Liu, B.; Rosenberg, M.; Huang, J.;
! Yazdanian, M.; Crawley, D. (2010).
! U.S. Department of Energy Commercial Reference Building Models of the National Building Stock.
! Washington, DC: U.S. Department of Energy, Energy Efficiency and
! Renewable Energy, Office of Building Technologies.
! ***GENERAL SIMULATION PARAMETERS***
! Number of Zones: 6

  Version,26.1;

  SimulationControl,
    YES,                     !- Do Zone Sizing Calculation
    YES,                     !- Do System Sizing Calculation
    YES,                     !- Do Plant Sizing Calculation
    NO,                      !- Run Simulation for Sizing Periods
    YES,                     !- Run Simulation for Weather File Run Periods
    No,                      !- Do HVAC Sizing Simulation for Sizing Periods
    1;                       !- Maximum Number of HVAC Sizing Simulation Passes

  Building,
    Ref Bldg Small Office New2004_v1.3_5.0,  !- Name
    0.0000,                  !- North Axis {deg}
    City,                    !- Terrain
    0.0400,                  !- Loads Convergence Tolerance Value {W}
    0.2000,                  !- Temperature Convergence Tolerance Value {deltaC}
    FullInteriorAndExterior, !- Solar Distribution
    25,                      !- Maximum Number of Warmup Days
    6;                       !- Minimum Number of Warmup Days

  RunPeriod,
```

```
    14,                       !- Begin Day of Month
    ,                         !- Begin Year
    7,                        !- End Month
    17,                       !- End Day of Month
    ,                         !- End Year
    Monday,                   !- Day of Week for Start Day
    No,                       !- Use Weather File Holidays and Special Days
    No,                       !- Use Weather File Daylight Saving Period
    No,                       !- Apply Weekend Holiday Rule
    Yes,                      !- Use Weather File Rain Indicators
    Yes;                      !- Use Weather File Snow Indicators

! ***HOLIDAYS & DAYLIGHT SAVINGS***

  RunPeriodControl:DaylightSavingTime,
    2nd Sunday in March,     !- Start Date
    1st Sunday in November;  !- End Date

  RunPeriodControl:SpecialDays,
    New Years Day,            !- Name
    January 1,                !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Veterans Day,             !- Name
    November 11,              !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Christmas,                !- Name
    December 25,              !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Independence Day,         !- Name
    July 4,                   !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    MLK Day,                  !- Name
    3rd Monday in January,    !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Presidents Day,           !- Name
    3rd Monday in February,   !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Memorial Day,             !- Name
    Last Monday in May,       !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type

  RunPeriodControl:SpecialDays,
    Labor Day,                !- Name
    1st Monday in September, !- Start Date
    1,                        !- Duration {days}
    Holiday;                  !- Special Day Type
```

```
    2nd Monday in October,   !- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

  RunPeriodControl:SpecialDays,
    Thanksgiving,            !- Name
    4th Thursday in November,!- Start Date
    1,                       !- Duration {days}
    Holiday;                 !- Special Day Type

! ***SCHEDULE TYPES***

  ScheduleTypeLimits,
    Any Number;              !- Name

  ScheduleTypeLimits,
    Fraction,                !- Name
    0.0,                     !- Lower Limit Value
    1.0,                     !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    Temperature,             !- Name
    -60,                     !- Lower Limit Value
    200,                     !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    On/Off,                  !- Name
    0,                       !- Lower Limit Value
    1,                       !- Upper Limit Value
    DISCRETE;                !- Numeric Type

  ScheduleTypeLimits,
    Control Type,            !- Name
    0,                       !- Lower Limit Value
    4,                       !- Upper Limit Value
    DISCRETE;                !- Numeric Type

  ScheduleTypeLimits,
    Humidity,                !- Name
    10,                      !- Lower Limit Value
    90,                      !- Upper Limit Value
    CONTINUOUS;              !- Numeric Type

  ScheduleTypeLimits,
    Number;                  !- Name

! ***ALWAYS ON SCHEDULE***

  Schedule:Compact,
    ALWAYS_ON,               !- Name
    On/Off,                  !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,1;          !- Field 3

! ***MISC SIMULATION PARAMETERS***

  SurfaceConvectionAlgorithm:Inside,TARP;

  SurfaceConvectionAlgorithm:Outside,DOE-2;

  HeatBalanceAlgorithm,ConductionTransferFunction,200.0000;

  ZoneAirHeatBalanceAlgorithm,
```

```
  Sizing:Parameters,
    1.2,                     !- Heating Sizing Factor
    1.2,                     !- Cooling Sizing Factor
    6;                       !- Timesteps in Averaging Window

  ConvergenceLimits,
    2,                       !- Minimum System Timestep {minutes}
    25;                      !- Maximum HVAC Iterations

  ShadowCalculation,
    PolygonClipping,         !- Shading Calculation Method
    Periodic,                !- Shading Calculation Update Frequency Method
    7,                       !- Shading Calculation Update Frequency
    15000;                   !- Maximum Figures in Shadow Overlap Calculations

  Timestep,6;

! WeatherFileName=USA_IL_Chicago-OHare_TMY2.epw

  Site:Location,
    USA IL-CHICAGO-OHARE,    !- Name
    41.77,                   !- Latitude {deg}
    -87.75,                  !- Longitude {deg}
    -6.00,                   !- Time Zone {hr}
    190;                     !- Elevation {m}

 ! CHICAGO_IL_USA Annual Heating 99.6%, MaxDB=-20.6Â°C

  SizingPeriod:DesignDay,
    CHICAGO Ann Htg 99.6% Condns DB,  !- Name
    1,                       !- Month
    21,                      !- Day of Month
    WinterDesignDay,         !- Day Type
    -20.6,                   !- Maximum Dry-Bulb Temperature {C}
    0.0,                     !- Daily Dry-Bulb Temperature Range {deltaC}
    ,                        !- Dry-Bulb Temperature Range Modifier Type
    ,                        !- Dry-Bulb Temperature Range Modifier Day Schedule Name
    Wetbulb,                 !- Humidity Condition Type
    -20.6,                   !- Wetbulb or DewPoint at Maximum Dry-Bulb {C}
    ,                        !- Humidity Condition Day Schedule Name
    ,                        !- Humidity Ratio at Maximum Dry-Bulb {kgWater/kgDryAir}
    ,                        !- Enthalpy at Maximum Dry-Bulb {J/kg}
    ,                        !- Daily Wet-Bulb Temperature Range {deltaC}
    99063.,                  !- Barometric Pressure {Pa}
    4.9,                     !- Wind Speed {m/s}
    270,                     !- Wind Direction {deg}
    No,                      !- Rain Indicator
    No,                      !- Snow Indicator
    No,                      !- Daylight Saving Time Indicator
    ASHRAEClearSky,          !- Solar Model Indicator
    ,                        !- Beam Solar Day Schedule Name
    ,                        !- Diffuse Solar Day Schedule Name
    ,                        !- ASHRAE Clear Sky Optical Depth for Beam Irradiance (taub) {dimensionless}
    ,                        !- ASHRAE Clear Sky Optical Depth for Diffuse Irradiance (taud) {dimensionless}
    0.00;                    !- Sky Clearness

 ! CHICAGO_IL_USA Annual Cooling (WB=>MDB) .4%, MDB=31.2Â°C WB=25.5Â°C

  SizingPeriod:DesignDay,
    CHICAGO Ann Clg .4% Condns WB=>MDB,  !- Name
    7,                       !- Month
    21,                      !- Day of Month
    SummerDesignDay,         !- Day Type
    31.2,                    !- Maximum Dry-Bulb Temperature {C}
    10.7,                    !- Daily Dry-Bulb Temperature Range {deltaC}
```

```
    Wetbulb,                 !- Humidity Condition Type
    25.5,                    !- Wetbulb or DewPoint at Maximum Dry-Bulb {C}
    ,                        !- Humidity Condition Day Schedule Name
    ,                        !- Humidity Ratio at Maximum Dry-Bulb {kgWater/kgDryAir}
    ,                        !- Enthalpy at Maximum Dry-Bulb {J/kg}
    ,                        !- Daily Wet-Bulb Temperature Range {deltaC}
    99063.,                  !- Barometric Pressure {Pa}
    5.3,                     !- Wind Speed {m/s}
    230,                     !- Wind Direction {deg}
    No,                      !- Rain Indicator
    No,                      !- Snow Indicator
    No,                      !- Daylight Saving Time Indicator
    ASHRAEClearSky,          !- Solar Model Indicator
    ,                        !- Beam Solar Day Schedule Name
    ,                        !- Diffuse Solar Day Schedule Name
    ,                        !- ASHRAE Clear Sky Optical Depth for Beam Irradiance (taub) {dimensionless}
    ,                        !- ASHRAE Clear Sky Optical Depth for Diffuse Irradiance (taud) {dimensionless}
    1.00;                    !- Sky Clearness

  Site:WaterMainsTemperature,
    CORRELATION,             !- Calculation Method
    ,                        !- Temperature Schedule Name
    9.69,                    !- Annual Average Outdoor Air Temperature {C}
    28.10;                   !- Maximum Difference In Monthly Average Outdoor Air Temperatures {deltaC}

  Site:GroundTemperature:BuildingSurface,19.527,19.502,19.536,19.598,20.002,21.640,22.225,22.375,21.449,20.121,19.802,19.633;

! ***OPAQUE CONSTRUCTIONS AND MATERIALS***
! Exterior Walls

  Construction,
    Mass Non-res Ext Wall,   !- Name
    1IN Stucco,              !- Outside Layer
    8IN Concrete HW,         !- Layer 2
    Mass NonRes Wall Insulation,  !- Layer 3
    1/2IN Gypsum;            !- Layer 4

  Material,
    Mass NonRes Wall Insulation,  !- Name
    MediumRough,             !- Roughness
    0.0495494599433393,      !- Thickness {m}
    0.049,                   !- Conductivity {W/m-K}
    265.0000,                !- Density {kg/m3}
    836.8000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Roof

  Construction,
    Attic Non-res Floor,     !- Name
    1/2IN Gypsum,            !- Outside Layer
    AtticFloor NonRes Insulation,  !- Layer 2
    1/2IN Gypsum;            !- Layer 3

  Construction,
    Attic Non-res Roof,      !- Name
    Roof Membrane,           !- Outside Layer
    Metal Decking;           !- Layer 2

  Material,
    AtticFloor NonRes Insulation,  !- Name
    MediumRough,             !- Roughness
    0.236804989096202,       !- Thickness {m}
    0.049,                   !- Conductivity {W/m-K}
```

```
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Slab on grade, unheated

  Construction,
    ext-slab,                !- Name
    HW CONCRETE,             !- Outside Layer
    CP02 CARPET PAD;         !- Layer 2

! Interior Walls

  Construction,
    int-walls,               !- Name
    1/2IN Gypsum,            !- Outside Layer
    1/2IN Gypsum;            !- Layer 2

! ***WINDOW/DOOR CONSTRUCTIONS AND MATERIALS***

  Construction,
    Window Non-res Fixed,    !- Name
    NonRes Fixed Assembly Window;  !- Outside Layer

  WindowMaterial:SimpleGlazingSystem,
    NonRes Fixed Assembly Window,  !- Name
    3.23646,                 !- U-Factor {W/m2-K}
    0.39,                    !- Solar Heat Gain Coefficient
    ;                        !- Visible Transmittance

! ***COMMON CONSTRUCTIONS AND MATERIALS***

  Construction,
    InteriorFurnishings,     !- Name
    Std Wood 6inch;          !- Outside Layer

  Material,
    Std Wood 6inch,          !- Name
    MediumSmooth,            !- Roughness
    0.15,                    !- Thickness {m}
    0.12,                    !- Conductivity {W/m-K}
    540.0000,                !- Density {kg/m3}
    1210,                    !- Specific Heat {J/kg-K}
    0.9000000,               !- Thermal Absorptance
    0.7000000,               !- Solar Absorptance
    0.7000000;               !- Visible Absorptance

  Material,
    Wood Siding,             !- Name
    MediumSmooth,            !- Roughness
    0.0100,                  !- Thickness {m}
    0.1100,                  !- Conductivity {W/m-K}
    544.6200,                !- Density {kg/m3}
    1210.0000,               !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7800,                  !- Solar Absorptance
    0.7800;                  !- Visible Absorptance

  Material,
    1/2IN Gypsum,            !- Name
    Smooth,                  !- Roughness
    0.0127,                  !- Thickness {m}
    0.1600,                  !- Conductivity {W/m-K}
    784.9000,                !- Density {kg/m3}
    830.0000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
```

```
Material,
  1IN Stucco,               !- Name
  Smooth,                   !- Roughness
  0.0253,                   !- Thickness {m}
  0.6918,                   !- Conductivity {W/m-K}
  1858.0000,                !- Density {kg/m3}
  837.0000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.9200,                   !- Solar Absorptance
  0.9200;                   !- Visible Absorptance

Material,
  8IN CONCRETE HW,          !- Name
  Rough,                    !- Roughness
  0.2032,                   !- Thickness {m}
  1.3110,                   !- Conductivity {W/m-K}
  2240.0000,                !- Density {kg/m3}
  836.8000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material,
  Metal Siding,             !- Name
  Smooth,                   !- Roughness
  0.0015,                   !- Thickness {m}
  44.9600,                  !- Conductivity {W/m-K}
  7688.8600,                !- Density {kg/m3}
  410.0000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.2000,                   !- Solar Absorptance
  0.2000;                   !- Visible Absorptance

Material,
  HW CONCRETE,              !- Name
  Rough,                    !- Roughness
  0.1016,                   !- Thickness {m}
  1.3110,                   !- Conductivity {W/m-K}
  2240.0000,                !- Density {kg/m3}
  836.8000,                 !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material:NoMass,
  CP02 CARPET PAD,          !- Name
  VeryRough,                !- Roughness
  0.2165,                   !- Thermal Resistance {m2-K/W}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.8000;                   !- Visible Absorptance

Material,
  Roof Membrane,            !- Name
  VeryRough,                !- Roughness
  0.0095,                   !- Thickness {m}
  0.1600,                   !- Conductivity {W/m-K}
  1121.2900,                !- Density {kg/m3}
  1460.0000,                !- Specific Heat {J/kg-K}
  0.9000,                   !- Thermal Absorptance
  0.7000,                   !- Solar Absorptance
  0.7000;                   !- Visible Absorptance

Material,
  Metal Decking,            !- Name
```

```
    45.0060,                 !- Conductivity {W/m-K}
    7680.0000,               !- Density {kg/m3}
    418.4000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.3000;                  !- Visible Absorptance

  Material,
    Metal Roofing,           !- Name
    MediumSmooth,            !- Roughness
    0.0015,                  !- Thickness {m}
    45.0060,                 !- Conductivity {W/m-K}
    7680.0000,               !- Density {kg/m3}
    418.4000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.3000;                  !- Visible Absorptance

  Material,
    MAT-CC05 4 HW CONCRETE,  !- Name
    Rough,                   !- Roughness
    0.1016,                  !- Thickness {m}
    1.3110,                  !- Conductivity {W/m-K}
    2240.0000,               !- Density {kg/m3}
    836.8000,                !- Specific Heat {J/kg-K}
    0.9000,                  !- Thermal Absorptance
    0.7000,                  !- Solar Absorptance
    0.7000;                  !- Visible Absorptance

! Acoustic tile for drop ceiling

  Material,
    Std AC02,                !- Name
    MediumSmooth,            !- Roughness
    1.2700000E-02,           !- Thickness {m}
    5.7000000E-02,           !- Conductivity {W/m-K}
    288.0000,                !- Density {kg/m3}
    1339.000,                !- Specific Heat {J/kg-K}
    0.9000000,               !- Thermal Absorptance
    0.7000000,               !- Solar Absorptance
    0.2000000;               !- Visible Absorptance

  Material:NoMass,
    MAT-AIR-WALL,            !- Name
    Rough,                   !- Roughness
    0.2079491,               !- Thermal Resistance {m2-K/W}
    0.9,                     !- Thermal Absorptance
    0.7;                     !- Solar Absorptance

! ZONE LIST:
! Attic
! Core_ZN
! Perimeter_ZN_1
! Perimeter_ZN_2
! Perimeter_ZN_3
! Perimeter_ZN_4
! ***ZONES***

  Zone,
    Attic,                   !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
```

```
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    No;                      !- Part of Total Floor Area

  Zone,
    Core_ZN,                 !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_1,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_2,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
    Perimeter_ZN_3,          !- Name
    0.0000,                  !- Direction of Relative North {deg}
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

  Zone,
```

```
    0.0000,                  !- X Origin {m}
    0.0000,                  !- Y Origin {m}
    0.0000,                  !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    ,                        !- Ceiling Height {m}
    ,                        !- Volume {m3}
    autocalculate,           !- Floor Area {m2}
    ,                        !- Zone Inside Convection Algorithm
    ,                        !- Zone Outside Convection Algorithm
    Yes;                     !- Part of Total Floor Area

! ***WALLS***

  BuildingSurface:Detailed,
    Attic_floor_core,        !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_ceiling,         !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_east,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_north,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
```

```
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_floor_perimeter_west,  !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_ceiling,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_east,         !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    3,                       !- Number of Vertices
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    18.4600,9.2300,6.3300;  !- X,Y,Z ==> Vertex 3 {m}

  BuildingSurface:Detailed,
    Attic_roof_north,        !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    9.2300,9.2300,6.3300,  !- X,Y,Z ==> Vertex 3 {m}
    18.4600,9.2300,6.3300;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_south,        !- Name
```

```
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    18.4600,9.2300,6.3300,  !- X,Y,Z ==> Vertex 3 {m}
    9.2300,9.2300,6.3300;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_roof_west,         !- Name
    Roof,                    !- Surface Type
    Attic Non-res Roof,      !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    3,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    -0.6000,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    9.2300,9.2300,6.3300;  !- X,Y,Z ==> Vertex 3 {m}

  BuildingSurface:Detailed,
    Attic_soffit_east,       !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_north,      !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    28.2900,19.0600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_south,      !- Name
    Floor,                   !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    28.2900,-0.6000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    -0.6000,-0.6000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Attic_soffit_west,       !- Name
    Floor,                   !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Attic,                   !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    -0.6000,19.0600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    -0.6000,-0.6000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_ceiling,         !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_core,        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_floor,           !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_east,       !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_north,      !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_south,      !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Core_ZN_wall_west,       !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Core_ZN,                 !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_ceiling,  !- Name
    Ceiling,                 !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_east,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_south,      !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_south,  !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_1_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_1,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_south,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_ceiling,  !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_east,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_east,!- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_south,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_east,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_2_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_2,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_east,       !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_ceiling,  !- Name
    Ceiling,                 !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_east,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_2_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    22.6900,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_north,  !- Name
    Wall,                    !- Surface Type
    Mass Non-res Ext Wall,   !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    27.6900,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_south,  !- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_north,      !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    22.6900,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    22.6900,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_3_wall_west,!- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_3,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_4_wall_north,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_ceiling,  !- Name
    Ceiling,                 !- Surface Type
    Attic Non-res Floor,     !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Attic_floor_perimeter_west,  !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_floor,    !- Name
    Floor,                   !- Surface Type
    ext-slab,                !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Ground,                  !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,0.0000;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_east,!- Name
    Wall,                    !- Surface Type
```

```
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Core_ZN_wall_west,       !- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,5.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,13.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_north,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_3_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    5.0000,13.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    5.0000,13.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,18.4600,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_south,  !- Name
    Wall,                    !- Surface Type
    int-walls,               !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Surface,                 !- Outside Boundary Condition
    Perimeter_ZN_1_wall_west,!- Outside Boundary Condition Object
    NoSun,                   !- Sun Exposure
    NoWind,                  !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,0.0000,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    5.0000,5.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    5.0000,5.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

  BuildingSurface:Detailed,
    Perimeter_ZN_4_wall_west,!- Name
    Wall,                    !- Surface Type
    Mass Non-res Ext Wall,   !- Construction Name
    Perimeter_ZN_4,          !- Zone Name
    ,                        !- Space Name
    Outdoors,                !- Outside Boundary Condition
    ,                        !- Outside Boundary Condition Object
    SunExposed,              !- Sun Exposure
    WindExposed,             !- Wind Exposure
    AutoCalculate,           !- View Factor to Ground
    4,                       !- Number of Vertices
    0.0000,18.4600,3.0500,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,18.4600,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,0.0000,3.0500;  !- X,Y,Z ==> Vertex 4 {m}

! ***WINDOWS***

  FenestrationSurface:Detailed,
```

```
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    1.3900,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    1.3900,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    3.2200,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    3.2200,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    6.0100,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    6.0100,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    7.8400,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    7.8400,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    10.6200,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    10.6200,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    12.4500,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    12.4500,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    15.2400,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    15.2400,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    17.0700,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    17.0700,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_5,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
```

```
    19.8500,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    21.6800,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    21.6800,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_Window_6,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    24.4700,0.0000,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    24.4700,0.0000,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    26.3000,0.0000,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    26.3000,0.0000,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_1_wall_south_door,  !- Name
    GlassDoor,               !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_1_wall_south,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    12.930,0.0000,2.1340,  !- X,Y,Z ==> Vertex 1 {m}
    12.930,0.0000,0.0000,  !- X,Y,Z ==> Vertex 2 {m}
    14.760,0.0000,0.0000,  !- X,Y,Z ==> Vertex 3 {m}
    14.760,0.0000,2.1340;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,1.3900,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,1.3900,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,3.2200,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,3.2200,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,6.0100,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,6.0100,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,7.8400,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,7.8400,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_3,  !- Name
```

```
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,10.6200,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,10.6200,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,12.4500,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,12.4500,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_2_wall_east_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_2_wall_east,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    27.6900,15.2400,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    27.6900,15.2400,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    27.6900,17.0700,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    27.6900,17.0700,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    26.3000,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    26.3000,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    24.4700,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    24.4700,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_2,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    21.6800,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    21.6800,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    19.8500,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    19.8500,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
```

```
    15.2400,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    15.2400,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    12.4500,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    12.4500,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    10.6200,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    10.6200,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_5,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    7.8400,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    7.8400,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    6.0100,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    6.0100,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_3_wall_north_Window_6,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_3_wall_north,  !- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    3.2200,18.4600,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    3.2200,18.4600,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    1.3900,18.4600,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    1.3900,18.4600,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_1,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,17.0700,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,17.0700,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,15.2400,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,15.2400,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_2,  !- Name
    Window,                  !- Surface Type
```

```
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,12.4500,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,12.4500,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,10.6200,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,10.6200,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_3,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,7.8400,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,7.8400,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,6.0100,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,6.0100,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

  FenestrationSurface:Detailed,
    Perimeter_ZN_4_wall_west_Window_4,  !- Name
    Window,                  !- Surface Type
    Window Non-res Fixed,    !- Construction Name
    Perimeter_ZN_4_wall_west,!- Building Surface Name
    ,                        !- Outside Boundary Condition Object
    AutoCalculate,           !- View Factor to Ground
    ,                        !- Frame and Divider Name
    1.0000,                  !- Multiplier
    4,                       !- Number of Vertices
    0.0000,3.2200,2.4240,  !- X,Y,Z ==> Vertex 1 {m}
    0.0000,3.2200,0.9000,  !- X,Y,Z ==> Vertex 2 {m}
    0.0000,1.3900,0.9000,  !- X,Y,Z ==> Vertex 3 {m}
    0.0000,1.3900,2.4240;  !- X,Y,Z ==> Vertex 4 {m}

! ***GEOMETRY RULES***

  GlobalGeometryRules,
    UpperLeftCorner,         !- Starting Vertex Position
    Counterclockwise,        !- Vertex Entry Direction
    Relative,                !- Coordinate System
    Relative;                !- Daylighting Reference Point Coordinate System

! ***PEOPLE***

  People,
    Core_ZN People,          !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
```

```
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_1 People,   !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_2 People,   !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_3 People,   !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
```

```
    FANGER;                  !- Thermal Comfort Model 1 Type

  People,
    Perimeter_ZN_4 People,   !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_OCC_SCH,            !- Number of People Schedule Name
    Area/Person,             !- Number of People Calculation Method
    ,                        !- Number of People
    ,                        !- People per Floor Area {person/m2}
    18.58,                   !- Floor Area per Person {m2/person}
    0.3000,                  !- Fraction Radiant
    AUTOCALCULATE,           !- Sensible Heat Fraction
    ACTIVITY_SCH,            !- Activity Level Schedule Name
    ,                        !- Carbon Dioxide Generation Rate {m3/s-W}
    No,                      !- Enable ASHRAE 55 Comfort Warnings
    EnclosureAveraged,       !- Mean Radiant Temperature Calculation Type
    ,                        !- Surface Name/Angle Factor List Name
    WORK_EFF_SCH,            !- Work Efficiency Schedule Name
    ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
    ,                        !- Clothing Insulation Calculation Method Schedule Name
    CLOTHING_SCH,            !- Clothing Insulation Schedule Name
    AIR_VELO_SCH,            !- Air Velocity Schedule Name
    FANGER;                  !- Thermal Comfort Model 1 Type

! ***LIGHTS***

  Lights,
    Core_ZN_Lights,          !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_1_Lights,   !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_2_Lights,   !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
```

```
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_3_Lights,   !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

  Lights,
    Perimeter_ZN_4_Lights,   !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_LIGHT_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Lighting Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Return Air Fraction
    0.7000,                  !- Fraction Radiant
    0.2000,                  !- Fraction Visible
    1.0000,                  !- Fraction Replaceable
    General,                 !- End-Use Subcategory
    No;                      !- Return Air Fraction Calculated from Plenum Temperature

! ***EQUIPMENT GAINS***

  ElectricEquipment,
    Core_ZN_MiscPlug_Equip,  !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_1_MiscPlug_Equip,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_2_MiscPlug_Equip,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
```

```
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_3_MiscPlug_Equip,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

  ElectricEquipment,
    Perimeter_ZN_4_MiscPlug_Equip,  !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    BLDG_EQUIP_SCH,          !- Schedule Name
    Watts/Area,              !- Design Level Calculation Method
    ,                        !- Design Level {W}
    10.76,                   !- Watts per Floor Area {W/m2}
    ,                        !- Watts per Person {W/person}
    0.0000,                  !- Fraction Latent
    0.5000,                  !- Fraction Radiant
    0.0000,                  !- Fraction Lost
    MiscPlug;                !- End-Use Subcategory

! ***EXTERIOR LOADS***

  Exterior:Lights,
    Exterior Facade Lighting,!- Name
    ALWAYS_ON,               !- Schedule Name
    2303,                    !- Design Level {W}
    AstronomicalClock,       !- Control Option
    Exterior Facade Lighting;!- End-Use Subcategory

! ***INFILTRATION***
! Infiltration through ceiling/roof

  ZoneInfiltration:DesignFlowRate,
    Core_ZN_Infiltration,    !- Name
    Core_ZN,                 !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    AirChanges/Hour,         !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    ,                        !- Flow Rate per Exterior Surface Area {m3/s-m2}
    0.36,                    !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_1_Infiltration,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
```

```
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_2_Infiltration,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_3_Infiltration,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

  ZoneInfiltration:DesignFlowRate,
    Perimeter_ZN_4_Infiltration,  !- Name
    Perimeter_ZN_4,          !- Zone or ZoneList or Space or SpaceList Name
    INFIL_QUARTER_ON_SCH,    !- Schedule Name
    Flow/ExteriorArea,       !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    0.000302,                !- Flow Rate per Exterior Surface Area {m3/s-m2}
    ,                        !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

! ***INTERNAL MASS***

  InternalMass,
    Core_ZN Internal Mass,   !- Name
    InteriorFurnishings,     !- Construction Name
    Core_ZN,                 !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    299.3148;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_1 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_1,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    226.9000;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_2 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_2,          !- Zone or ZoneList Name
```

```
  InternalMass,
    Perimeter_ZN_3 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    226.9000;                !- Surface Area {m2}

  InternalMass,
    Perimeter_ZN_4 Internal Mass,  !- Name
    InteriorFurnishings,     !- Construction Name
    Perimeter_ZN_4,          !- Zone or ZoneList Name
    ,                        !- Space or SpaceList Name
    134.6000;                !- Surface Area {m2}

! ***INTERNAL GAINS SCHEDULES***

  Schedule:Compact,
    INFIL_QUARTER_ON_SCH,    !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 06:00,1.0,        !- Field 3
    Until: 22:00,0.25,       !- Field 5
    Until: 24:00,1.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 06:00,1.0,        !- Field 10
    Until: 18:00,0.25,       !- Field 12
    Until: 24:00,1.0,        !- Field 14
    For: Sunday Holidays AllOtherDays, !- Field 16
    Until: 24:00,1.0;        !- Field 17

  Schedule:Compact,
    BLDG_OCC_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: SummerDesignDay,    !- Field 2
    Until: 06:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.05,       !- Field 7
    For: Weekdays,           !- Field 9
    Until: 06:00,0.0,        !- Field 10
    Until: 07:00,0.1,        !- Field 12
    Until: 08:00,0.2,        !- Field 14
    Until: 12:00,0.95,       !- Field 16
    Until: 13:00,0.5,        !- Field 18
    Until: 17:00,0.95,       !- Field 20
    Until: 18:00,0.3,        !- Field 22
    Until: 20:00,0.1,        !- Field 24
    Until: 24:00,0.05,       !- Field 26
    For: Saturday,           !- Field 28
    Until: 06:00,0.0,        !- Field 29
    Until: 08:00,0.1,        !- Field 31
    Until: 12:00,0.3,        !- Field 33
    Until: 17:00,0.1,        !- Field 35
    Until: 24:00,0.0,        !- Field 37
    For: AllOtherDays,       !- Field 39
    Until: 24:00,0.0;        !- Field 40

  Schedule:Compact,
    BLDG_LIGHT_SCH,          !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays,           !- Field 2
    Until: 05:00,0.05,       !- Field 3
    Until: 07:00,0.1,        !- Field 5
```

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    Until: 18:00,0.5,        !- Field 11
    Until: 20:00,0.3,        !- Field 13
    Until: 22:00,0.2,        !- Field 15
    Until: 23:00,0.1,        !- Field 17
    Until: 24:00,0.05,       !- Field 19
    For: Saturday,           !- Field 21
    Until: 06:00,0.05,       !- Field 22
    Until: 08:00,0.1,        !- Field 24
    Until: 12:00,0.3,        !- Field 26
    Until: 17:00,0.15,       !- Field 28
    Until: 24:00,0.05,       !- Field 30
    For: SummerDesignDay,    !- Field 32
    Until: 24:00,1.0,        !- Field 33
    For: WinterDesignDay,    !- Field 35
    Until: 24:00,0.0,        !- Field 36
    For: AllOtherDays,       !- Field 38
    Until: 24:00,0.05;       !- Field 39

  Schedule:Compact,
    BLDG_EQUIP_SCH,          !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays,           !- Field 2
    Until: 08:00,0.40,       !- Field 3
    Until: 12:00,0.90,       !- Field 5
    Until: 13:00,0.80,       !- Field 7
    Until: 17:00,0.90,       !- Field 9
    Until: 18:00,0.50,       !- Field 11
    Until: 24:00,0.40,       !- Field 13
    For: Saturday,           !- Field 15
    Until: 06:00,0.30,       !- Field 16
    Until: 08:00,0.4,        !- Field 18
    Until: 12:00,0.5,        !- Field 20
    Until: 17:00,0.35,       !- Field 22
    Until: 24:00,0.30,       !- Field 24
    For: SummerDesignDay,    !- Field 26
    Until: 24:00,1.0,        !- Field 27
    For: WinterDesignDay,    !- Field 29
    Until: 24:00,0.0,        !- Field 30
    For: AllOtherDays,       !- Field 32
    Until: 24:00,0.30;       !- Field 33

  Schedule:Compact,
    ACTIVITY_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,120;        !- Field 3

  Schedule:Compact,
    WORK_EFF_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.0;        !- Field 3

  Schedule:Compact,
    AIR_VELO_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.2;        !- Field 3

  Schedule:Compact,
    CLOTHING_SCH,            !- Name
    Any Number,              !- Schedule Type Limits Name
```

```
    Until: 24:00,1.0,        !- Field 3
    Through: 09/30,          !- Field 5
    For: AllDays,            !- Field 6
    Until: 24:00,0.5,        !- Field 7
    Through: 12/31,          !- Field 9
    For: AllDays,            !- Field 10
    Until: 24:00,1.0;        !- Field 11

  ZoneInfiltration:DesignFlowRate,
    Attic_Infiltration,      !- Name
    Attic,                   !- Zone or ZoneList or Space or SpaceList Name
    ALWAYS_ON,               !- Schedule Name
    AirChanges/Hour,         !- Design Flow Rate Calculation Method
    ,                        !- Design Flow Rate {m3/s}
    ,                        !- Flow Rate per Floor Area {m3/s-m2}
    ,                        !- Flow Rate per Exterior Surface Area {m3/s-m2}
    1.0,                     !- Air Changes per Hour {1/hr}
    1.0000,                  !- Constant Term Coefficient
    0.0000,                  !- Temperature Term Coefficient
    0.0000,                  !- Velocity Term Coefficient
    0.0000;                  !- Velocity Squared Term Coefficient

! ***HVAC EQUIPMENT***

  Fan:SystemModel,
    PSZ-AC:1_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Air Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    AUTOSIZE,                !- Design Maximum Air Flow Rate {m3/s}
    Discrete,                !- Speed Control Method
    0.0,                     !- Electric Power Minimum Flow Rate Fraction
    622.0,                   !- Design Pressure Rise {Pa}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Air Stream Fraction
    AUTOSIZE,                !- Design Electric Power Consumption {W}
    TotalEfficiencyAndPressure,  !- Design Power Sizing Method
    ,                        !- Electric Power Per Unit Flow Rate {W/(m3/s)}
    ,                        !- Electric Power Per Unit Flow Rate Per Unit Pressure {W/((m3/s)-Pa)}
    0.53625,                 !- Fan Total Efficiency
    ,                        !- Electric Power Function of Flow Fraction Curve Name
    ,                        !- Night Ventilation Mode Pressure Rise {Pa}
    ,                        !- Night Ventilation Mode Flow Fraction
    ,                        !- Motor Loss Zone Name
    ,                        !- Motor Loss Radiative Fraction
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:2_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Air Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:3_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
```

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    PSZ-AC:3 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:4_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Air Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Fan:ConstantVolume,
    PSZ-AC:5_Fan,            !- Name
    HVACOperationSchd,       !- Availability Schedule Name
    0.53625,                 !- Fan Total Efficiency
    622.0,                   !- Pressure Rise {Pa}
    AUTOSIZE,                !- Maximum Flow Rate {m3/s}
    0.825,                   !- Motor Efficiency
    1.0,                     !- Motor In Airstream Fraction
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Air Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Air Outlet Node Name
    Fan Energy;              !- End-Use Subcategory

  Coil:Heating:Fuel,
    PSZ-AC:1_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Air Outlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:2_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Air Outlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:3_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Air Outlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode;  !- Temperature Setpoint Node Name

  Coil:Heating:Fuel,
    PSZ-AC:4_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Air Outlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode;  !- Temperature Setpoint Node Name
```

```
    PSZ-AC:5_HeatC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    NaturalGas,              !- Fuel Type
    0.8,                     !- Burner Efficiency
    AUTOSIZE,                !- Nominal Capacity {W}
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Air Inlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Air Outlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode;  !- Temperature Setpoint Node Name

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:1_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
    ,                        !- Nominal Time for Condensate Removal to Begin {s}
    ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:2_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
    ,                        !- Nominal Time for Condensate Removal to Begin {s}
    ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

  Coil:Cooling:DX:SingleSpeed,
    PSZ-AC:3_CoolC DXCoil,   !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
    AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
    3.66668442928701,        !- Gross Rated Cooling COP {W/W}
    AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
    ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Air Inlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Air Outlet Node Name
    Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
    ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
    Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
    ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
    Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
    ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
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Coil:Cooling:DX:SingleSpeed,
  PSZ-AC:4_CoolC DXCoil,   !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
  AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
  3.66668442928701,        !- Gross Rated Cooling COP {W/W}
  AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
  ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Air Inlet Node Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Air Outlet Node Name
  Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
  ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
  Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
  ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
  Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
  ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
  ,                        !- Nominal Time for Condensate Removal to Begin {s}
  ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

Coil:Cooling:DX:SingleSpeed,
  PSZ-AC:5_CoolC DXCoil,   !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  AUTOSIZE,                !- Gross Rated Total Cooling Capacity {W}
  AUTOSIZE,                !- Gross Rated Sensible Heat Ratio
  3.66668442928701,        !- Gross Rated Cooling COP {W/W}
  AUTOSIZE,                !- Rated Air Flow Rate {m3/s}
  ,                        !- 2017 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  934.4,                   !- 2023 Rated Evaporator Fan Power Per Volume Flow Rate {W/(m3/s)}
  PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Air Inlet Node Name
  PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Air Outlet Node Name
  Cool-Cap-fT,             !- Total Cooling Capacity Function of Temperature Curve Name
  ConstantCubic,           !- Total Cooling Capacity Function of Flow Fraction Curve Name
  Cool-EIR-fT,             !- Energy Input Ratio Function of Temperature Curve Name
  ConstantCubic,           !- Energy Input Ratio Function of Flow Fraction Curve Name
  Cool-PLF-fPLR,           !- Part Load Fraction Correlation Curve Name
  ,                        !- Minimum Outdoor Dry-Bulb Temperature for Compressor Operation {C}
  ,                        !- Nominal Time for Condensate Removal to Begin {s}
  ;                        !- Ratio of Initial Moisture Evaporation Rate and Steady State Latent Capacity {dimensionless}

ZoneHVAC:EquipmentList,
  Core_ZN Equipment,       !- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Core_ZN Direct Air ADU,  !- Zone Equipment 1 Name
  1,                       !- Zone Equipment 1 Cooling Sequence
  1,                       !- Zone Equipment 1 Heating or No-Load Sequence
  ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
  ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

ZoneHVAC:EquipmentList,
  Perimeter_ZN_1 Equipment,!- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Perimeter_ZN_1 Direct Air ADU,  !- Zone Equipment 1 Name
  1,                       !- Zone Equipment 1 Cooling Sequence
  1,                       !- Zone Equipment 1 Heating or No-Load Sequence
  ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
  ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

ZoneHVAC:EquipmentList,
  Perimeter_ZN_2 Equipment,!- Name
  SequentialLoad,          !- Load Distribution Scheme
  ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
  Perimeter_ZN_2 Direct Air ADU,  !- Zone Equipment 1 Name
```

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    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

  ZoneHVAC:EquipmentList,
    Perimeter_ZN_3 Equipment,!- Name
    SequentialLoad,          !- Load Distribution Scheme
    ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
    Perimeter_ZN_3 Direct Air ADU,  !- Zone Equipment 1 Name
    1,                       !- Zone Equipment 1 Cooling Sequence
    1,                       !- Zone Equipment 1 Heating or No-Load Sequence
    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

  ZoneHVAC:EquipmentList,
    Perimeter_ZN_4 Equipment,!- Name
    SequentialLoad,          !- Load Distribution Scheme
    ZoneHVAC:AirDistributionUnit,  !- Zone Equipment 1 Object Type
    Perimeter_ZN_4 Direct Air ADU,  !- Zone Equipment 1 Name
    1,                       !- Zone Equipment 1 Cooling Sequence
    1,                       !- Zone Equipment 1 Heating or No-Load Sequence
    ,                        !- Zone Equipment 1 Sequential Cooling Fraction Schedule Name
    ;                        !- Zone Equipment 1 Sequential Heating Fraction Schedule Name

! ***SIZING & CONTROLS***

  Sizing:Zone,
    Core_ZN,                 !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,    !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                 !- Zone Heating Design Supply Air Temperature {C}
    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Core_ZN,         !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Core_ZN,         !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
```

```
  Sizing:Zone,
    Perimeter_ZN_1,           !- Zone or ZoneList Name
    SupplyAirTemperature,     !- Zone Cooling Design Supply Air Temperature Input Method
    13.9000,                  !- Zone Cooling Design Supply Air Temperature {C}
    ,                         !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,     !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                  !- Zone Heating Design Supply Air Temperature {C}
    ,                         !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                   !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                   !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_1,   !- Design Specification Outdoor Air Object Name
    ,                         !- Zone Heating Sizing Factor
    ,                         !- Zone Cooling Sizing Factor
    DesignDay,                !- Cooling Design Air Flow Method
    ,                         !- Cooling Design Air Flow Rate {m3/s}
    ,                         !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Cooling Minimum Air Flow {m3/s}
    ,                         !- Cooling Minimum Air Flow Fraction
    DesignDay,                !- Heating Design Air Flow Method
    ,                         !- Heating Design Air Flow Rate {m3/s}
    ,                         !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Heating Maximum Air Flow {m3/s}
    ,                         !- Heating Maximum Air Flow Fraction
    ,                         !- Design Specification Zone Air Distribution Object Name
    No,                       !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,         !- Dedicated Outdoor Air System Control Strategy
    autosize,                 !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                 !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                         !- Zone Load Sizing Method
    ,                         !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                         !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                         !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                         !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                         !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                         !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_1,   !- Name
    Flow/Person,              !- Outdoor Air Method
    0.01,                     !- Outdoor Air Flow per Person {m3/s-person}
    ,                         !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                         !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_2,           !- Zone or ZoneList Name
    SupplyAirTemperature,     !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                  !- Zone Cooling Design Supply Air Temperature {C}
    ,                         !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,     !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                  !- Zone Heating Design Supply Air Temperature {C}
    ,                         !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                   !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                   !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_2,   !- Design Specification Outdoor Air Object Name
    ,                         !- Zone Heating Sizing Factor
    ,                         !- Zone Cooling Sizing Factor
    DesignDay,                !- Cooling Design Air Flow Method
    ,                         !- Cooling Design Air Flow Rate {m3/s}
    ,                         !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Cooling Minimum Air Flow {m3/s}
    ,                         !- Cooling Minimum Air Flow Fraction
    DesignDay,                !- Heating Design Air Flow Method
    ,                         !- Heating Design Air Flow Rate {m3/s}
    ,                         !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                         !- Heating Maximum Air Flow {m3/s}
    ,                         !- Heating Maximum Air Flow Fraction
```

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    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_2,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    13.9000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
    SupplyAirTemperature,    !- Zone Heating Design Supply Air Temperature Input Method
    40.0000,                 !- Zone Heating Design Supply Air Temperature {C}
    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_3,  !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_3,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  Sizing:Zone,
    Perimeter_ZN_4,          !- Zone or ZoneList Name
    SupplyAirTemperature,    !- Zone Cooling Design Supply Air Temperature Input Method
    14.0000,                 !- Zone Cooling Design Supply Air Temperature {C}
    ,                        !- Zone Cooling Design Supply Air Temperature Difference {deltaC}
```

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    ,                        !- Zone Heating Design Supply Air Temperature Difference {deltaC}
    0.0085,                  !- Zone Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Zone Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    SZ DSOA Perimeter_ZN_4,  !- Design Specification Outdoor Air Object Name
    ,                        !- Zone Heating Sizing Factor
    ,                        !- Zone Cooling Sizing Factor
    DesignDay,               !- Cooling Design Air Flow Method
    ,                        !- Cooling Design Air Flow Rate {m3/s}
    ,                        !- Cooling Minimum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Cooling Minimum Air Flow {m3/s}
    ,                        !- Cooling Minimum Air Flow Fraction
    DesignDay,               !- Heating Design Air Flow Method
    ,                        !- Heating Design Air Flow Rate {m3/s}
    ,                        !- Heating Maximum Air Flow per Zone Floor Area {m3/s-m2}
    ,                        !- Heating Maximum Air Flow {m3/s}
    ,                        !- Heating Maximum Air Flow Fraction
    ,                        !- Design Specification Zone Air Distribution Object Name
    No,                      !- Account for Dedicated Outdoor Air System
    NeutralSupplyAir,        !- Dedicated Outdoor Air System Control Strategy
    autosize,                !- Dedicated Outdoor Air Low Setpoint Temperature for Design {C}
    autosize,                !- Dedicated Outdoor Air High Setpoint Temperature for Design {C}
    ,                        !- Zone Load Sizing Method
    ,                        !- Zone Latent Cooling Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Dehumidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ,                        !- Zone Cooling Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}
    ,                        !- Zone Latent Heating Design Supply Air Humidity Ratio Input Method
    ,                        !- Zone Humidification Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    ;                        !- Zone Humidification Design Supply Air Humidity Ratio Difference {kgWater/kgDryAir}

  DesignSpecification:OutdoorAir,
    SZ DSOA Perimeter_ZN_4,  !- Name
    Flow/Person,             !- Outdoor Air Method
    0.01,                    !- Outdoor Air Flow per Person {m3/s-person}
    ,                        !- Outdoor Air Flow per Zone Floor Area {m3/s-m2}
    ;                        !- Outdoor Air Flow per Zone {m3/s}

  ZoneControl:Thermostat,
    Core_ZN Thermostat,      !- Name
    Core_ZN,                 !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Core_ZN DualSPSched;     !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_1 Thermostat,  !- Name
    Perimeter_ZN_1,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_1 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_2 Thermostat,  !- Name
    Perimeter_ZN_2,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_2 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_3 Thermostat,  !- Name
    Perimeter_ZN_3,          !- Zone or ZoneList Name
    Dual Zone Control Type Sched,  !- Control Type Schedule Name
    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_3 DualSPSched;  !- Control 1 Name

  ZoneControl:Thermostat,
    Perimeter_ZN_4 Thermostat,  !- Name
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    ThermostatSetpoint:DualSetpoint,  !- Control 1 Object Type
    Perimeter_ZN_4 DualSPSched;  !- Control 1 Name

  ThermostatSetpoint:DualSetpoint,
    Core_ZN DualSPSched,     !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_1 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_2 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_3 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  ThermostatSetpoint:DualSetpoint,
    Perimeter_ZN_4 DualSPSched,  !- Name
    HTGSETP_SCH,             !- Heating Setpoint Temperature Schedule Name
    CLGSETP_SCH;             !- Cooling Setpoint Temperature Schedule Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrCore_ZN,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Core_ZN,                 !- Control Zone Name
    Core_ZN Air Node,        !- Zone Node Name
    Core_ZN Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_1,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_1,          !- Control Zone Name
    Perimeter_ZN_1 Air Node, !- Zone Node Name
    Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_2,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_2,          !- Control Zone Name
    Perimeter_ZN_2 Air Node, !- Zone Node Name
    Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

  SetpointManager:SingleZone:Reheat,
    SupAirTemp MngrPerimeter_ZN_3,  !- Name
    Temperature,             !- Control Variable
    10.0,                    !- Minimum Supply Air Temperature {C}
    50.0,                    !- Maximum Supply Air Temperature {C}
    Perimeter_ZN_3,          !- Control Zone Name
    Perimeter_ZN_3 Air Node, !- Zone Node Name
    Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
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SetpointManager:SingleZone:Reheat,
  SupAirTemp MngrPerimeter_ZN_4,  !- Name
  Temperature,             !- Control Variable
  10.0,                    !- Minimum Supply Air Temperature {C}
  50.0,                    !- Maximum Supply Air Temperature {C}
  Perimeter_ZN_4,          !- Control Zone Name
  Perimeter_ZN_4 Air Node, !- Zone Node Name
  Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Zone Inlet Node Name
  PSZ-AC:5 Supply Equipment Outlet Node;  !- Setpoint Node or NodeList Name

Sizing:System,
  PSZ-AC:1,                !- AirLoop Name
  Sensible,                !- Type of Load to Size On
  AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
  1.0,                     !- Central Heating Maximum System Air Flow Ratio
  7.0,                     !- Preheat Design Temperature {C}
  0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
  12.8000,                 !- Precool Design Temperature {C}
  0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
  12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
  40.0,                    !- Central Heating Design Supply Air Temperature {C}
  NonCoincident,           !- Type of Zone Sum to Use
  No,                      !- 100% Outdoor Air in Cooling
  No,                      !- 100% Outdoor Air in Heating
  0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
  0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
  DesignDay,               !- Cooling Supply Air Flow Rate Method
  ,                        !- Cooling Supply Air Flow Rate {m3/s}
  ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
  ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
  ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
  DesignDay,               !- Heating Supply Air Flow Rate Method
  ,                        !- Heating Supply Air Flow Rate {m3/s}
  ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
  ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
  ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
  ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
  ,                        !- System Outdoor Air Method
  1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
  CoolingDesignCapacity,   !- Cooling Design Capacity Method
  autosize,                !- Cooling Design Capacity {W}
  ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
  ,                        !- Fraction of Autosized Cooling Design Capacity
  HeatingDesignCapacity,   !- Heating Design Capacity Method
  autosize,                !- Heating Design Capacity {W}
  ,                        !- Heating Design Capacity Per Floor Area {W/m2}
  ,                        !- Fraction of Autosized Heating Design Capacity
  VAV;                     !- Central Cooling Capacity Control Method

Sizing:System,
  PSZ-AC:2,                !- AirLoop Name
  Sensible,                !- Type of Load to Size On
  AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
  1.0,                     !- Central Heating Maximum System Air Flow Ratio
  7.0,                     !- Preheat Design Temperature {C}
  0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
  12.8000,                 !- Precool Design Temperature {C}
  0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
  12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
  40.0,                    !- Central Heating Design Supply Air Temperature {C}
  NonCoincident,           !- Type of Zone Sum to Use
  No,                      !- 100% Outdoor Air in Cooling
  No,                      !- 100% Outdoor Air in Heating
  0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
  0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
  DesignDay,               !- Cooling Supply Air Flow Rate Method
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    ,                          !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                          !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,                 !- Heating Supply Air Flow Rate Method
    ,                          !- Heating Supply Air Flow Rate {m3/s}
    ,                          !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                          !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                          !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                          !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                          !- System Outdoor Air Method
    1.0,                       !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,     !- Cooling Design Capacity Method
    autosize,                  !- Cooling Design Capacity {W}
    ,                          !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                          !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,     !- Heating Design Capacity Method
    autosize,                  !- Heating Design Capacity {W}
    ,                          !- Heating Design Capacity Per Floor Area {W/m2}
    ,                          !- Fraction of Autosized Heating Design Capacity
    VAV;                       !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:3,                  !- AirLoop Name
    Sensible,                  !- Type of Load to Size On
    AUTOSIZE,                  !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                       !- Central Heating Maximum System Air Flow Ratio
    7.0,                       !- Preheat Design Temperature {C}
    0.008,                     !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                   !- Precool Design Temperature {C}
    0.008,                     !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                   !- Central Cooling Design Supply Air Temperature {C}
    40.0,                      !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,             !- Type of Zone Sum to Use
    No,                        !- 100% Outdoor Air in Cooling
    No,                        !- 100% Outdoor Air in Heating
    0.0085,                    !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                    !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,                 !- Cooling Supply Air Flow Rate Method
    ,                          !- Cooling Supply Air Flow Rate {m3/s}
    ,                          !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                          !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                          !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,                 !- Heating Supply Air Flow Rate Method
    ,                          !- Heating Supply Air Flow Rate {m3/s}
    ,                          !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                          !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                          !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                          !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                          !- System Outdoor Air Method
    1.0,                       !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,     !- Cooling Design Capacity Method
    autosize,                  !- Cooling Design Capacity {W}
    ,                          !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                          !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,     !- Heating Design Capacity Method
    autosize,                  !- Heating Design Capacity {W}
    ,                          !- Heating Design Capacity Per Floor Area {W/m2}
    ,                          !- Fraction of Autosized Heating Design Capacity
    VAV;                       !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:4,                  !- AirLoop Name
    Sensible,                  !- Type of Load to Size On
    AUTOSIZE,                  !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                       !- Central Heating Maximum System Air Flow Ratio
    7.0,                       !- Preheat Design Temperature {C}
    0.008,                     !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
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    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
    ,                        !- Cooling Supply Air Flow Rate {m3/s}
    ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,               !- Heating Supply Air Flow Rate Method
    ,                        !- Heating Supply Air Flow Rate {m3/s}
    ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                        !- System Outdoor Air Method
    1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,   !- Cooling Design Capacity Method
    autosize,                !- Cooling Design Capacity {W}
    ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,   !- Heating Design Capacity Method
    autosize,                !- Heating Design Capacity {W}
    ,                        !- Heating Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Heating Design Capacity
    VAV;                     !- Central Cooling Capacity Control Method

  Sizing:System,
    PSZ-AC:5,                !- AirLoop Name
    Sensible,                !- Type of Load to Size On
    AUTOSIZE,                !- Design Outdoor Air Flow Rate {m3/s}
    1.0,                     !- Central Heating Maximum System Air Flow Ratio
    7.0,                     !- Preheat Design Temperature {C}
    0.008,                   !- Preheat Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Precool Design Temperature {C}
    0.008,                   !- Precool Design Humidity Ratio {kgWater/kgDryAir}
    12.8000,                 !- Central Cooling Design Supply Air Temperature {C}
    40.0,                    !- Central Heating Design Supply Air Temperature {C}
    NonCoincident,           !- Type of Zone Sum to Use
    No,                      !- 100% Outdoor Air in Cooling
    No,                      !- 100% Outdoor Air in Heating
    0.0085,                  !- Central Cooling Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    0.0080,                  !- Central Heating Design Supply Air Humidity Ratio {kgWater/kgDryAir}
    DesignDay,               !- Cooling Supply Air Flow Rate Method
    ,                        !- Cooling Supply Air Flow Rate {m3/s}
    ,                        !- Cooling Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Cooling Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Cooling Supply Air Flow Rate Per Unit Cooling Capacity {m3/s-W}
    DesignDay,               !- Heating Supply Air Flow Rate Method
    ,                        !- Heating Supply Air Flow Rate {m3/s}
    ,                        !- Heating Supply Air Flow Rate Per Floor Area {m3/s-m2}
    ,                        !- Heating Fraction of Autosized Heating Supply Air Flow Rate
    ,                        !- Heating Fraction of Autosized Cooling Supply Air Flow Rate
    ,                        !- Heating Supply Air Flow Rate Per Unit Heating Capacity {m3/s-W}
    ,                        !- System Outdoor Air Method
    1.0,                     !- Zone Maximum Outdoor Air Fraction {dimensionless}
    CoolingDesignCapacity,   !- Cooling Design Capacity Method
    autosize,                !- Cooling Design Capacity {W}
    ,                        !- Cooling Design Capacity Per Floor Area {W/m2}
    ,                        !- Fraction of Autosized Cooling Design Capacity
    HeatingDesignCapacity,   !- Heating Design Capacity Method
    autosize,                !- Heating Design Capacity {W}
    ,                        !- Heating Design Capacity Per Floor Area {W/m2}
```

```
  Controller:OutdoorAir,
    PSZ-AC:1_OA_Controller,  !- Name
    PSZ-AC:1_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:1_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:2_OA_Controller,  !- Name
    PSZ-AC:2_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:2 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:2_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:3_OA_Controller,  !- Name
    PSZ-AC:3_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:3 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:3_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name
```

```
    PSZ-AC:4_OA_Controller,  !- Name
    PSZ-AC:4_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:4 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:4_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Controller:OutdoorAir,
    PSZ-AC:5_OA_Controller,  !- Name
    PSZ-AC:5_OARelief Node,  !- Relief Air Outlet Node Name
    PSZ-AC:5 Supply Equipment Inlet Node,  !- Return Air Node Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:5_OAInlet Node,   !- Actuator Node Name
    AUTOSIZE,                !- Minimum Outdoor Air Flow Rate {m3/s}
    AUTOSIZE,                !- Maximum Outdoor Air Flow Rate {m3/s}
    NoEconomizer,            !- Economizer Control Type
    ModulateFlow,            !- Economizer Control Action Type
    28.0,                    !- Economizer Maximum Limit Dry-Bulb Temperature {C}
    64000.0,                 !- Economizer Maximum Limit Enthalpy {J/kg}
    ,                        !- Economizer Maximum Limit Dewpoint Temperature {C}
    ,                        !- Electronic Enthalpy Limit Curve Name
    -100.0,                  !- Economizer Minimum Limit Dry-Bulb Temperature {C}
    NoLockout,               !- Lockout Type
    FixedMinimum,            !- Minimum Limit Type
    MinOA_Sched,             !- Minimum Outdoor Air Schedule Name
    ,                        !- Minimum Fraction of Outdoor Air Schedule Name
    ,                        !- Maximum Fraction of Outdoor Air Schedule Name
    ;                        !- Mechanical Ventilation Controller Name

  Curve:Quadratic,
    Cool-PLF-fPLR,           !- Name
    0.90949556,              !- Coefficient1 Constant
    0.09864773,              !- Coefficient2 x
    -0.00819488,             !- Coefficient3 x**2
    0,                       !- Minimum Value of x
    1,                       !- Maximum Value of x
    0.7,                     !- Minimum Curve Output
    1;                       !- Maximum Curve Output

  Curve:Cubic,
    ConstantCubic,           !- Name
    1,                       !- Coefficient1 Constant
    0,                       !- Coefficient2 x
    0,                       !- Coefficient3 x**2
    0,                       !- Coefficient4 x**3
    -100,                    !- Minimum Value of x
    100;                     !- Maximum Value of x

  Curve:Biquadratic,
    Cool-Cap-fT,             !- Name
    0.9712123,               !- Coefficient1 Constant
    -0.015275502,            !- Coefficient2 x
```

```
    -0.0000068364,           !- Coefficient5 y**2
    -0.0002905956,           !- Coefficient6 x*y
    -100,                    !- Minimum Value of x
    100,                     !- Maximum Value of x
    -100,                    !- Minimum Value of y
    100;                     !- Maximum Value of y

  Curve:Biquadratic,
    Cool-EIR-fT,             !- Name
    0.28687133,              !- Coefficient1 Constant
    0.023902164,             !- Coefficient2 x
    -0.000810648,            !- Coefficient3 x**2
    0.013458546,             !- Coefficient4 y
    0.0003389364,            !- Coefficient5 y**2
    -0.0004870044,           !- Coefficient6 x*y
    -100,                    !- Minimum Value of x
    100,                     !- Maximum Value of x
    -100,                    !- Minimum Value of y
    100;                     !- Maximum Value of y

! ***AIR LOOPS***

  AirLoopHVAC,
    PSZ-AC:1,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:1 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:1 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:1 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:1 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:1 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:2,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:2 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:2 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:2 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:2 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:2 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:2 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:3,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:3 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:3 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:3 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:3 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:3 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:4,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:4 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:4 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:4 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
```

```
    PSZ-AC:4 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  AirLoopHVAC,
    PSZ-AC:5,                !- Name
    ,                        !- Controller List Name
    PSZ-AC:5 Availability Manager List,  !- Availability Manager List Name
    AUTOSIZE,                !- Design Supply Air Flow Rate {m3/s}
    PSZ-AC:5 Air Loop Branches,  !- Branch List Name
    ,                        !- Connector List Name
    PSZ-AC:5 Supply Equipment Inlet Node,  !- Supply Side Inlet Node Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Demand Side Outlet Node Name
    PSZ-AC:5 Zone Equipment Inlet Node,  !- Demand Side Inlet Node Names
    PSZ-AC:5 Supply Equipment Outlet Node;  !- Supply Side Outlet Node Names

  CoilSystem:Cooling:DX,
    PSZ-AC:1_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:1_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:2_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:2_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:3_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:3_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:4_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:4_CoolC DXCoil;   !- Cooling Coil Name

  CoilSystem:Cooling:DX,
    PSZ-AC:5_CoolC,          !- Name
    ALWAYS_ON,               !- Availability Schedule Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- DX Cooling Coil System Inlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- DX Cooling Coil System Outlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- DX Cooling Coil System Sensor Node Name
    Coil:Cooling:DX:SingleSpeed,  !- Cooling Coil Object Type
    PSZ-AC:5_CoolC DXCoil;   !- Cooling Coil Name

! ***CONNECTIONS***

  ZoneHVAC:EquipmentConnections,
    Core_ZN,                 !- Zone Name
    Core_ZN Equipment,       !- Zone Conditioning Equipment List Name
    Core_ZN Inlet Nodes,     !- Zone Air Inlet Node or NodeList Name
    ,                        !- Zone Air Exhaust Node or NodeList Name
```

```
ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_1,          !- Zone Name
  Perimeter_ZN_1 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_1 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_1 Air Node, !- Zone Air Node Name
  Perimeter_ZN_1 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_2,          !- Zone Name
  Perimeter_ZN_2 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_2 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_2 Air Node, !- Zone Air Node Name
  Perimeter_ZN_2 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_3,          !- Zone Name
  Perimeter_ZN_3 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_3 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_3 Air Node, !- Zone Air Node Name
  Perimeter_ZN_3 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

ZoneHVAC:EquipmentConnections,
  Perimeter_ZN_4,          !- Zone Name
  Perimeter_ZN_4 Equipment,!- Zone Conditioning Equipment List Name
  Perimeter_ZN_4 Inlet Nodes,  !- Zone Air Inlet Node or NodeList Name
  ,                        !- Zone Air Exhaust Node or NodeList Name
  Perimeter_ZN_4 Air Node, !- Zone Air Node Name
  Perimeter_ZN_4 Return Air Node Name;  !- Zone Return Air Node or NodeList Name

NodeList,
  Core_ZN Inlet Nodes,     !- Name
  Core_ZN Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
  PSZ-AC:1_OANode List,    !- Name
  PSZ-AC:1_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:2_OANode List,    !- Name
  PSZ-AC:2_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:3_OANode List,    !- Name
  PSZ-AC:3_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:4_OANode List,    !- Name
  PSZ-AC:4_OAInlet Node;   !- Node 1 Name

NodeList,
  PSZ-AC:5_OANode List,    !- Name
  PSZ-AC:5_OAInlet Node;   !- Node 1 Name

NodeList,
  Perimeter_ZN_1 Inlet Nodes,  !- Name
  Perimeter_ZN_1 Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
  Perimeter_ZN_2 Inlet Nodes,  !- Name
  Perimeter_ZN_2 Direct Air Inlet Node Name;  !- Node 1 Name

NodeList,
```

```
NodeList,
  Perimeter_ZN_4 Inlet Nodes,  !- Name
  Perimeter_ZN_4 Direct Air Inlet Node Name;  !- Node 1 Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Core_ZN Direct Air,      !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Core_ZN Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Core_ZN Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Core_ZN Direct Air ADU,  !- Name
  Core_ZN Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Core_ZN Direct Air,      !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_1 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_1 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_1 Direct Air ADU,  !- Name
  Perimeter_ZN_1 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_1 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_2 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_2 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
  ;                        !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_2 Direct Air ADU,  !- Name
  Perimeter_ZN_2 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_2 Direct Air,  !- Air Terminal Name
  ,                        !- Nominal Upstream Leakage Fraction
  ,                        !- Constant Downstream Leakage Fraction
  ;                        !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_3 Direct Air,  !- Name
  ALWAYS_ON,               !- Availability Schedule Name
  Perimeter_ZN_3 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                !- Maximum Air Flow Rate {m3/s}
  ,                        !- Design Specification Outdoor Air Object Name
```

```
ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_3 Direct Air ADU,  !- Name
  Perimeter_ZN_3 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_3 Direct Air,  !- Air Terminal Name
  ,                         !- Nominal Upstream Leakage Fraction
  ,                         !- Constant Downstream Leakage Fraction
  ;                         !- Design Specification Air Terminal Sizing Object Name

AirTerminal:SingleDuct:ConstantVolume:NoReheat,
  Perimeter_ZN_4 Direct Air,  !- Name
  ALWAYS_ON,                !- Availability Schedule Name
  Perimeter_ZN_4 Direct Air Inlet Node Name ATInlet,  !- Air Inlet Node Name
  Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Air Outlet Node Name
  AUTOSIZE,                 !- Maximum Air Flow Rate {m3/s}
  ,                         !- Design Specification Outdoor Air Object Name
  ;                         !- Per Person Ventilation Rate Mode

ZoneHVAC:AirDistributionUnit,
  Perimeter_ZN_4 Direct Air ADU,  !- Name
  Perimeter_ZN_4 Direct Air Inlet Node Name,  !- Air Distribution Unit Outlet Node Name
  AirTerminal:SingleDuct:ConstantVolume:NoReheat,  !- Air Terminal Object Type
  Perimeter_ZN_4 Direct Air,  !- Air Terminal Name
  ,                         !- Nominal Upstream Leakage Fraction
  ,                         !- Constant Downstream Leakage Fraction
  ;                         !- Design Specification Air Terminal Sizing Object Name

AvailabilityManagerAssignmentList,
  PSZ-AC:1 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:1 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:2 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:2 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:3 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:3 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:4 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:4 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManagerAssignmentList,
  PSZ-AC:5 Availability Manager List,  !- Name
  AvailabilityManager:NightCycle,  !- Availability Manager 1 Object Type
  PSZ-AC:5 Availability Manager;  !- Availability Manager 1 Name

AvailabilityManager:NightCycle,
  PSZ-AC:1 Availability Manager,  !- Name
  ALWAYS_ON,                !- Applicability Schedule Name
  HVACOperationSchd,        !- Fan Schedule Name
  CycleOnAny,               !- Control Type
  1.0,                      !- Thermostat Tolerance {deltaC}
  FixedRunTime,             !- Cycling Run Time Control Type
  1800;                     !- Cycling Run Time {s}

AvailabilityManager:NightCycle,
  PSZ-AC:2 Availability Manager,  !- Name
  ALWAYS_ON,                !- Applicability Schedule Name
  HVACOperationSchd,        !- Fan Schedule Name
  CycleOnAny,               !- Control Type
```

```
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:3 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:4 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  AvailabilityManager:NightCycle,
    PSZ-AC:5 Availability Manager,  !- Name
    ALWAYS_ON,               !- Applicability Schedule Name
    HVACOperationSchd,       !- Fan Schedule Name
    CycleOnAny,              !- Control Type
    1.0,                     !- Thermostat Tolerance {deltaC}
    FixedRunTime,            !- Cycling Run Time Control Type
    1800;                    !- Cycling Run Time {s}

  BranchList,
    PSZ-AC:1 Air Loop Branches,  !- Name
    PSZ-AC:1 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:2 Air Loop Branches,  !- Name
    PSZ-AC:2 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:3 Air Loop Branches,  !- Name
    PSZ-AC:3 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:4 Air Loop Branches,  !- Name
    PSZ-AC:4 Air Loop Main Branch;  !- Branch 1 Name

  BranchList,
    PSZ-AC:5 Air Loop Branches,  !- Name
    PSZ-AC:5 Air Loop Main Branch;  !- Branch 1 Name

  Branch,
    PSZ-AC:1 Air Loop Main Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
    PSZ-AC:1_OA,             !- Component 1 Name
    PSZ-AC:1 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Component 1 Outlet Node Name
    CoilSystem:Cooling:DX,   !- Component 2 Object Type
    PSZ-AC:1_CoolC,          !- Component 2 Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Component 2 Inlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Component 2 Outlet Node Name
    Coil:Heating:Fuel,       !- Component 3 Object Type
    PSZ-AC:1_HeatC,          !- Component 3 Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode,  !- Component 3 Inlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Component 3 Outlet Node Name
    Fan:SystemModel,         !- Component 4 Object Type
    PSZ-AC:1_Fan,            !- Component 4 Name
```

```
Branch,
  PSZ-AC:2 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:2_OA,             !- Component 1 Name
  PSZ-AC:2 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:2_CoolC,          !- Component 2 Name
  PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:2_HeatC,          !- Component 3 Name
  PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:2_Fan,            !- Component 4 Name
  PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:2 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:3 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:3_OA,             !- Component 1 Name
  PSZ-AC:3 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:3_CoolC,          !- Component 2 Name
  PSZ-AC:3_OA-PSZ-AC:3_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:3_HeatC,          !- Component 3 Name
  PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:3_Fan,            !- Component 4 Name
  PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:3 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:4 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:4_OA,             !- Component 1 Name
  PSZ-AC:4 Supply Equipment Inlet Node,  !- Component 1 Inlet Node Name
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Component 1 Outlet Node Name
  CoilSystem:Cooling:DX,   !- Component 2 Object Type
  PSZ-AC:4_CoolC,          !- Component 2 Name
  PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Component 2 Inlet Node Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Component 2 Outlet Node Name
  Coil:Heating:Fuel,       !- Component 3 Object Type
  PSZ-AC:4_HeatC,          !- Component 3 Name
  PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode,  !- Component 3 Inlet Node Name
  PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Component 3 Outlet Node Name
  Fan:ConstantVolume,      !- Component 4 Object Type
  PSZ-AC:4_Fan,            !- Component 4 Name
  PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Component 4 Inlet Node Name
  PSZ-AC:4 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

Branch,
  PSZ-AC:5 Air Loop Main Branch,  !- Name
  ,                        !- Pressure Drop Curve Name
  AirLoopHVAC:OutdoorAirSystem,  !- Component 1 Object Type
  PSZ-AC:5_OA,             !- Component 1 Name
```

```
    CoilSystem:Cooling:DX,   !- Component 2 Object Type
    PSZ-AC:5_CoolC,          !- Component 2 Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Component 2 Inlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Component 2 Outlet Node Name
    Coil:Heating:Fuel,       !- Component 3 Object Type
    PSZ-AC:5_HeatC,          !- Component 3 Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode,  !- Component 3 Inlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Component 3 Outlet Node Name
    Fan:ConstantVolume,      !- Component 4 Object Type
    PSZ-AC:5_Fan,            !- Component 4 Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Component 4 Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node;  !- Component 4 Outlet Node Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:1_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:1_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:2_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:2_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:3_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:3_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:4_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:4_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:ControllerList,
    PSZ-AC:5_OA_Controllers, !- Name
    Controller:OutdoorAir,   !- Controller 1 Object Type
    PSZ-AC:5_OA_Controller;  !- Controller 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:1_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:1_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:2_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:2_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:3_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:3_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:4_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:4_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem:EquipmentList,
    PSZ-AC:5_OA_Equipment,   !- Name
    OutdoorAir:Mixer,        !- Component 1 Object Type
    PSZ-AC:5_OAMixing Box;   !- Component 1 Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:1_OA,             !- Name
    PSZ-AC:1_OA_Controllers, !- Controller List Name
```

```
  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:2_OA,             !- Name
    PSZ-AC:2_OA_Controllers, !- Controller List Name
    PSZ-AC:2_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:3_OA,             !- Name
    PSZ-AC:3_OA_Controllers, !- Controller List Name
    PSZ-AC:3_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:4_OA,             !- Name
    PSZ-AC:4_OA_Controllers, !- Controller List Name
    PSZ-AC:4_OA_Equipment;   !- Outdoor Air Equipment List Name

  AirLoopHVAC:OutdoorAirSystem,
    PSZ-AC:5_OA,             !- Name
    PSZ-AC:5_OA_Controllers, !- Controller List Name
    PSZ-AC:5_OA_Equipment;   !- Outdoor Air Equipment List Name

  OutdoorAir:NodeList,
    PSZ-AC:1_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:2_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:3_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:4_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:NodeList,
    PSZ-AC:5_OANode List;    !- Node or NodeList Name 1

  OutdoorAir:Node,
    PSZ-AC:1_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:2_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:3_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:4_CoolCOA Ref node;  !- Name

  OutdoorAir:Node,
    PSZ-AC:5_CoolCOA Ref node;  !- Name

  OutdoorAir:Mixer,
    PSZ-AC:1_OAMixing Box,   !- Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:1_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:1_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:1 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:2_OAMixing Box,   !- Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:2_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:2_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:2 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:3_OAMixing Box,   !- Name
```

```
    PSZ-AC:3_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:3 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:4_OAMixing Box,   !- Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:4_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:4_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:4 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  OutdoorAir:Mixer,
    PSZ-AC:5_OAMixing Box,   !- Name
    PSZ-AC:5_OA-PSZ-AC:5_CoolCNode,  !- Mixed Air Node Name
    PSZ-AC:5_OAInlet Node,   !- Outdoor Air Stream Node Name
    PSZ-AC:5_OARelief Node,  !- Relief Air Stream Node Name
    PSZ-AC:5 Supply Equipment Inlet Node;  !- Return Air Stream Node Name

  SetpointManager:MixedAir,
    PSZ-AC:1_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_CoolC-PSZ-AC:1_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:1_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:1_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:1_HeatC-PSZ-AC:1_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:1 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:1_OA-PSZ-AC:1_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_CoolC-PSZ-AC:2_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:2_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:2_HeatC-PSZ-AC:2_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:2 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:2_OA-PSZ-AC:2_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
```

```
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_CoolC-PSZ-AC:3_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:3_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:3_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:3_HeatC-PSZ-AC:3_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:3 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:3_OA-PSZ-AC:3_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_CoolC-PSZ-AC:4_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:4_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:4_HeatC-PSZ-AC:4_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:4 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:4_OA-PSZ-AC:4_CoolCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_CoolC SAT Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:5_CoolC-PSZ-AC:5_HeatCNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_HeatC MixedAir Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Fan Outlet Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode;  !- Setpoint Node or NodeList Name

  SetpointManager:MixedAir,
    PSZ-AC:5_OAMixed Air Temp Manager,  !- Name
    Temperature,             !- Control Variable
    PSZ-AC:5 Supply Equipment Outlet Node,  !- Reference Setpoint Node Name
    PSZ-AC:5_HeatC-PSZ-AC:5_FanNode,  !- Fan Inlet Node Name
```

```
AirLoopHVAC:SupplyPath,
  PSZ-AC:1,                !- Name
  PSZ-AC:1 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:1 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:2,                !- Name
  PSZ-AC:2 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:2 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:3,                !- Name
  PSZ-AC:3 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:3 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:4,                !- Name
  PSZ-AC:4 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:4 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:SupplyPath,
  PSZ-AC:5,                !- Name
  PSZ-AC:5 Zone Equipment Inlet Node,  !- Supply Air Path Inlet Node Name
  AirLoopHVAC:ZoneSplitter,!- Component 1 Object Type
  PSZ-AC:5 Supply Air Splitter;  !- Component 1 Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:1 Supply Air Splitter,  !- Name
  PSZ-AC:1 Zone Equipment Inlet Node,  !- Inlet Node Name
  Core_ZN Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:2 Supply Air Splitter,  !- Name
  PSZ-AC:2 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_1 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:3 Supply Air Splitter,  !- Name
  PSZ-AC:3 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_2 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:4 Supply Air Splitter,  !- Name
  PSZ-AC:4 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_3 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ZoneSplitter,
  PSZ-AC:5 Supply Air Splitter,  !- Name
  PSZ-AC:5 Zone Equipment Inlet Node,  !- Inlet Node Name
  Perimeter_ZN_4 Direct Air Inlet Node Name ATInlet;  !- Outlet 1 Node Name

AirLoopHVAC:ReturnPath,
  PSZ-AC:1 Return Air Path,!- Name
  PSZ-AC:1 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
  AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
  PSZ-AC:1 Return Air Mixer;  !- Component 1 Name

AirLoopHVAC:ReturnPath,
  PSZ-AC:2 Return Air Path,!- Name
  PSZ-AC:2 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
  AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
```

```
  AirLoopHVAC:ReturnPath,
    PSZ-AC:3 Return Air Path,!- Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:3 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ReturnPath,
    PSZ-AC:4 Return Air Path,!- Name
    PSZ-AC:4 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:4 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ReturnPath,
    PSZ-AC:5 Return Air Path,!- Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Return Air Path Outlet Node Name
    AirLoopHVAC:ZoneMixer,   !- Component 1 Object Type
    PSZ-AC:5 Return Air Mixer;  !- Component 1 Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:1 Return Air Mixer,  !- Name
    PSZ-AC:1 Zone Equipment Outlet Node,  !- Outlet Node Name
    Core_ZN Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:2 Return Air Mixer,  !- Name
    PSZ-AC:2 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_1 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:3 Return Air Mixer,  !- Name
    PSZ-AC:3 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_2 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:4 Return Air Mixer,  !- Name
    PSZ-AC:4 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_3 Return Air Node Name;  !- Inlet 1 Node Name

  AirLoopHVAC:ZoneMixer,
    PSZ-AC:5 Return Air Mixer,  !- Name
    PSZ-AC:5 Zone Equipment Outlet Node,  !- Outlet Node Name
    Perimeter_ZN_4 Return Air Node Name;  !- Inlet 1 Node Name

! ***SCHEDULES***

  Schedule:Compact,
    CLGSETP_SCH,
    Temperature,
    Through: 12/31,
    For: AllDays,
    Until: 03:00,28.0,
    Until: 05:00,27.0,
    Until: 07:00,28.0,
    Until: 09:00,25.0,
    Until: 11:00,24.5,
    Until: 13:00,25.0,
    Until: 15:00,25.0,
    Until: 17:00,25.0,
    Until: 19:00,25.0,
    Until: 21:00,25.0,
    Until: 23:00,25.0,
    Until: 24:00,28.0;

  Schedule:Compact,
    HTGSETP_SCH,
    Temperature,
```

```
    Until: 03:00,16.0,
    Until: 05:00,17.0,
    Until: 07:00,16.0,
    Until: 09:00,19.0,
    Until: 11:00,19.0,
    Until: 13:00,20.0,
    Until: 15:00,20.0,
    Until: 17:00,19.5,
    Until: 19:00,19.5,
    Until: 21:00,19.5,
    Until: 23:00,20.0,
    Until: 24:00,16.0;

  Schedule:Compact,
    MinOA_Sched,             !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 07:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 07:00,0.0,        !- Field 10
    Until: 18:00,1.0,        !- Field 12
    Until: 24:00,0.0,        !- Field 14
    For: AllOtherDays,       !- Field 16
    Until: 24:00,0.0;        !- Field 17

  Schedule:Compact,
    Dual Zone Control Type Sched,  !- Name
    Control Type,            !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,4;          !- Field 3

  Schedule:Compact,
    HVACOperationSchd,       !- Name
    On/Off,                  !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 06:00,0.0,        !- Field 3
    Until: 22:00,1.0,        !- Field 5
    Until: 24:00,0.0,        !- Field 7
    For: Saturday WinterDesignDay, !- Field 9
    Until: 06:00,0.0,        !- Field 10
    Until: 18:00,1.0,        !- Field 12
    Until: 24:00,0.0,        !- Field 14
    For: AllOtherDays,       !- Field 16
    Until: 24:00,0.0;        !- Field 17

! ***SWH EQUIPMENT***

  WaterHeater:Mixed,
    SWHSys1 Water Heater,    !- Name
    0.1514,                  !- Tank Volume {m3}
    SWHSys1 Water Heater Setpoint Temperature Schedule Name,  !- Setpoint Temperature Schedule Name
    2.0,                     !- Deadband Temperature Difference {deltaC}
    82.2222,                 !- Maximum Temperature Limit {C}
    Cycle,                   !- Heater Control Type
    845000,                  !- Heater Maximum Capacity {W}
    ,                        !- Heater Minimum Capacity {W}
    ,                        !- Heater Ignition Minimum Flow Rate {m3/s}
    ,                        !- Heater Ignition Delay {s}
    NATURALGAS,              !- Heater Fuel Type
    0.8,                     !- Heater Thermal Efficiency
    ,                        !- Part Load Factor Curve Name
```

```
    0.8,                     !- Off Cycle Parasitic Heat Fraction to Tank
    ,                        !- On Cycle Parasitic Fuel Consumption Rate {W}
    NATURALGAS,              !- On Cycle Parasitic Fuel Type
    ,                        !- On Cycle Parasitic Heat Fraction to Tank
    SCHEDULE,                !- Ambient Temperature Indicator
    SWHSys1 Water Heater Ambient Temperature Schedule Name,  !- Ambient Temperature Schedule Name
    ,                        !- Ambient Temperature Zone Name
    ,                        !- Ambient Temperature Outdoor Air Node Name
    6.0,                     !- Off Cycle Loss Coefficient to Ambient Temperature {W/K}
    ,                        !- Off Cycle Loss Fraction to Zone
    6.0,                     !- On Cycle Loss Coefficient to Ambient Temperature {W/K}
    ,                        !- On Cycle Loss Fraction to Zone
    ,                        !- Peak Use Flow Rate {m3/s}
    ,                        !- Use Flow Rate Fraction Schedule Name
    ,                        !- Cold Water Supply Temperature Schedule Name
    SWHSys1 Pump-SWHSys1 Water HeaterNode,  !- Use Side Inlet Node Name
    SWHSys1 Supply Equipment Outlet Node,  !- Use Side Outlet Node Name
    1.0,                     !- Use Side Effectiveness
    ,                        !- Source Side Inlet Node Name
    ,                        !- Source Side Outlet Node Name
    1.0,                     !- Source Side Effectiveness
    AUTOSIZE,                !- Use Side Design Flow Rate {m3/s}
    AUTOSIZE,                !- Source Side Design Flow Rate {m3/s}
    1.5;                     !- Indirect Water Heating Recovery Time {hr}

  Pump:ConstantSpeed,
    SWHSys1 Pump,            !- Name
    SWHSys1 Supply Inlet Node,  !- Inlet Node Name
    SWHSys1 Pump-SWHSys1 Water HeaterNodeviaConnector,  !- Outlet Node Name
    AUTOSIZE,                !- Design Flow Rate {m3/s}
    0.001,                   !- Design Pump Head {Pa}
    AUTOSIZE,                !- Design Power Consumption {W}
    1,                       !- Motor Efficiency
    0.0,                     !- Fraction of Motor Inefficiencies to Fluid Stream
    Intermittent,            !- Pump Control Type
    ;                        !- Pump Flow Rate Schedule Name

  WaterUse:Equipment,
    Core_ZN Water Equipment, !- Name
    ,                        !- End-Use Subcategory
    3.15e-006,               !- Peak Flow Rate {m3/s}
    BLDG_SWH_SCH,            !- Flow Rate Fraction Schedule Name
    Water Equipment Temp Sched,  !- Target Temperature Schedule Name
    Water Equipment Hot Supply Temp Sched,  !- Hot Water Supply Temperature Schedule Name
    ,                        !- Cold Water Supply Temperature Schedule Name
    Core_ZN,                 !- Zone Name
    Water Equipment Sensible fract sched,  !- Sensible Fraction Schedule Name
    Water Equipment Latent fract sched;  !- Latent Fraction Schedule Name

  PlantEquipmentList,
    SWHSys1 Equipment List,  !- Name
    WaterHeater:Mixed,       !- Equipment 1 Object Type
    SWHSys1 Water Heater;    !- Equipment 1 Name

! ***SWH SIZING & CONTROLS***

  Sizing:Plant,
    SWHSys1,                 !- Plant or Condenser Loop Name
    Heating,                 !- Loop Type
    60,                      !- Design Loop Exit Temperature {C}
    5.0;                     !- Loop Design Temperature Difference {deltaC}

  SetpointManager:Scheduled,
    SWHSys1 Loop Setpoint Manager,  !- Name
    Temperature,             !- Control Variable
    SWHSys1-Loop-Temp-Schedule,  !- Schedule Name
```

```
  PlantEquipmentOperationSchemes,
    SWHSys1 Loop Operation Scheme List,  !- Name
    PlantEquipmentOperation:HeatingLoad,  !- Control Scheme 1 Object Type
    SWHSys1 Operation Scheme,!- Control Scheme 1 Name
    ALWAYS_ON;               !- Control Scheme 1 Schedule Name

  PlantEquipmentOperation:HeatingLoad,
    SWHSys1 Operation Scheme,!- Name
    0.0,                     !- Load Range 1 Lower Limit {W}
    1000000000000000,        !- Load Range 1 Upper Limit {W}
    SWHSys1 Equipment List;  !- Range 1 Equipment List Name

! ***SWH LOOP***

  PlantLoop,
    SWHSys1,                 !- Name
    Water,                   !- Fluid Type
    ,                        !- User Defined Fluid Type
    SWHSys1 Loop Operation Scheme List,  !- Plant Equipment Operation Scheme Name
    SWHSys1 Supply Outlet Node,  !- Loop Temperature Setpoint Node Name
    60.0,                    !- Maximum Loop Temperature {C}
    10.0,                    !- Minimum Loop Temperature {C}
    AUTOSIZE,                !- Maximum Loop Flow Rate {m3/s}
    0.0,                     !- Minimum Loop Flow Rate {m3/s}
    AUTOSIZE,                !- Plant Loop Volume {m3}
    SWHSys1 Supply Inlet Node,  !- Plant Side Inlet Node Name
    SWHSys1 Supply Outlet Node,  !- Plant Side Outlet Node Name
    SWHSys1 Supply Branches, !- Plant Side Branch List Name
    SWHSys1 Supply Connectors,  !- Plant Side Connector List Name
    SWHSys1 Demand Inlet Node,  !- Demand Side Inlet Node Name
    SWHSys1 Demand Outlet Node,  !- Demand Side Outlet Node Name
    SWHSys1 Demand Branches, !- Demand Side Branch List Name
    SWHSys1 Demand Connectors,  !- Demand Side Connector List Name
    Optimal;                 !- Load Distribution Scheme

! ***SWH CONNECTIONS***

  BranchList,
    SWHSys1 Demand Branches, !- Name
    SWHSys1 Demand Inlet Branch,  !- Branch 1 Name
    SWHSys1 Demand Load Branch 1,  !- Branch 2 Name
    SWHSys1 Demand Bypass Branch,  !- Branch 3 Name
    SWHSys1 Demand Outlet Branch;  !- Branch 4 Name

  BranchList,
    SWHSys1 Supply Branches, !- Name
    SWHSys1 Supply Inlet Branch,  !- Branch 1 Name
    SWHSys1 Supply Equipment Branch,  !- Branch 2 Name
    SWHSys1 Supply Equipment Bypass Branch,  !- Branch 3 Name
    SWHSys1 Supply Outlet Branch;  !- Branch 4 Name

  Branch,
    SWHSys1 Demand Bypass Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Demand Bypass Pipe,  !- Component 1 Name
    SWHSys1 Demand Bypass Pipe Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Bypass Pipe Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Demand Inlet Branch,  !- Name
    ,                        !- Pressure Drop Curve Name
    Pipe:Adiabatic,          !- Component 1 Object Type
    SWHSys1 Demand Inlet Pipe,  !- Component 1 Name
    SWHSys1 Demand Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Inlet Pipe-SWHSys1 Demand Mixer;  !- Component 1 Outlet Node Name
```

```
    SWHSys1 Demand Load Branch 1,  !- Name
    ,                         !- Pressure Drop Curve Name
    WaterUse:Connections,     !- Component 1 Object Type
    Core_ZN Water Equipment, !- Component 1 Name
    Core_ZN Water Equipment Water Inlet Node,  !- Component 1 Inlet Node Name
    Core_ZN Water Equipment Water Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Demand Outlet Branch,  !- Name
    ,                         !- Pressure Drop Curve Name
    Pipe:Adiabatic,           !- Component 1 Object Type
    SWHSys1 Demand Outlet Pipe,  !- Component 1 Name
    SWHSys1 Demand Mixer-SWHSys1 Demand Outlet Pipe,  !- Component 1 Inlet Node Name
    SWHSys1 Demand Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Equipment Branch,  !- Name
    ,                         !- Pressure Drop Curve Name
    WaterHeater:Mixed,        !- Component 1 Object Type
    SWHSys1 Water Heater,     !- Component 1 Name
    SWHSys1 Pump-SWHSys1 Water HeaterNode,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Equipment Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Equipment Bypass Branch,  !- Name
    ,                         !- Pressure Drop Curve Name
    Pipe:Adiabatic,           !- Component 1 Object Type
    SWHSys1 Supply Equipment Bypass Pipe,  !- Component 1 Name
    SWHSys1 Supply Equip Bypass Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Equip Bypass Outlet Node;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Inlet Branch,  !- Name
    ,                         !- Pressure Drop Curve Name
    Pump:ConstantSpeed,       !- Component 1 Object Type
    SWHSys1 Pump,             !- Component 1 Name
    SWHSys1 Supply Inlet Node,  !- Component 1 Inlet Node Name
    SWHSys1 Pump-SWHSys1 Water HeaterNodeviaConnector;  !- Component 1 Outlet Node Name

  Branch,
    SWHSys1 Supply Outlet Branch,  !- Name
    ,                         !- Pressure Drop Curve Name
    Pipe:Adiabatic,           !- Component 1 Object Type
    SWHSys1 Supply Outlet Pipe,  !- Component 1 Name
    SWHSys1 Supply Mixer-SWHSys1 Supply Outlet Pipe,  !- Component 1 Inlet Node Name
    SWHSys1 Supply Outlet Node;  !- Component 1 Outlet Node Name

  ConnectorList,
    SWHSys1 Demand Connectors,  !- Name
    Connector:Splitter,       !- Connector 1 Object Type
    SWHSys1 Demand Splitter, !- Connector 1 Name
    Connector:Mixer,          !- Connector 2 Object Type
    SWHSys1 Demand Mixer;     !- Connector 2 Name

  ConnectorList,
    SWHSys1 Supply Connectors,  !- Name
    Connector:Splitter,       !- Connector 1 Object Type
    SWHSys1 Supply Splitter, !- Connector 1 Name
    Connector:Mixer,          !- Connector 2 Object Type
    SWHSys1 Supply Mixer;     !- Connector 2 Name

  WaterUse:Connections,
    Core_ZN Water Equipment, !- Name
    Core_ZN Water Equipment Water Inlet Node,  !- Inlet Node Name
    Core_ZN Water Equipment Water Outlet Node,  !- Outlet Node Name
    ,                         !- Supply Water Storage Tank Name
```

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    ,                        !- Cold Water Supply Temperature Schedule Name
    ,                        !- Drain Water Heat Exchanger Type
    ,                        !- Drain Water Heat Exchanger Destination
    ,                        !- Drain Water Heat Exchanger U-Factor Times Area {W/K}
    Core_ZN Water Equipment; !- Water Use Equipment 1 Name

  Connector:Splitter,
    SWHSys1 Demand Splitter, !- Name
    SWHSys1 Demand Inlet Branch,  !- Inlet Branch Name
    SWHSys1 Demand Load Branch 1,  !- Outlet Branch 1 Name
    SWHSys1 Demand Bypass Branch;  !- Outlet Branch 2 Name

  Connector:Splitter,
    SWHSys1 Supply Splitter, !- Name
    SWHSys1 Supply Inlet Branch,  !- Inlet Branch Name
    SWHSys1 Supply Equipment Branch,  !- Outlet Branch 1 Name
    SWHSys1 Supply Equipment Bypass Branch;  !- Outlet Branch 2 Name

  Connector:Mixer,
    SWHSys1 Demand Mixer,    !- Name
    SWHSys1 Demand Outlet Branch,  !- Outlet Branch Name
    SWHSys1 Demand Load Branch 1,  !- Inlet Branch 1 Name
    SWHSys1 Demand Bypass Branch;  !- Inlet Branch 2 Name

  Connector:Mixer,
    SWHSys1 Supply Mixer,    !- Name
    SWHSys1 Supply Outlet Branch,  !- Outlet Branch Name
    SWHSys1 Supply Equipment Branch,  !- Inlet Branch 1 Name
    SWHSys1 Supply Equipment Bypass Branch;  !- Inlet Branch 2 Name

  Pipe:Adiabatic,
    SWHSys1 Demand Bypass Pipe,  !- Name
    SWHSys1 Demand Bypass Pipe Inlet Node,  !- Inlet Node Name
    SWHSys1 Demand Bypass Pipe Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Demand Inlet Pipe,  !- Name
    SWHSys1 Demand Inlet Node,  !- Inlet Node Name
    SWHSys1 Demand Inlet Pipe-SWHSys1 Demand Mixer;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Demand Outlet Pipe,  !- Name
    SWHSys1 Demand Mixer-SWHSys1 Demand Outlet Pipe,  !- Inlet Node Name
    SWHSys1 Demand Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Supply Equipment Bypass Pipe,  !- Name
    SWHSys1 Supply Equip Bypass Inlet Node,  !- Inlet Node Name
    SWHSys1 Supply Equip Bypass Outlet Node;  !- Outlet Node Name

  Pipe:Adiabatic,
    SWHSys1 Supply Outlet Pipe,  !- Name
    SWHSys1 Supply Mixer-SWHSys1 Supply Outlet Pipe,  !- Inlet Node Name
    SWHSys1 Supply Outlet Node;  !- Outlet Node Name

! ***SWH SCHEDULES***

  Schedule:Compact,
    BLDG_SWH_SCH,            !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: Weekdays SummerDesignDay, !- Field 2
    Until: 05:00,0.05,       !- Field 3
    Until: 06:00,0.08,       !- Field 5
    Until: 07:00,0.07,       !- Field 7
    Until: 08:00,0.19,       !- Field 9
```

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    Until: 11:00,0.39,       !- Field 15
    Until: 12:00,0.47,       !- Field 17
    Until: 13:00,0.57,       !- Field 19
    Until: 14:00,0.54,       !- Field 21
    Until: 15:00,0.34,       !- Field 23
    Until: 16:00,0.33,       !- Field 25
    Until: 17:00,0.44,       !- Field 27
    Until: 18:00,0.26,       !- Field 29
    Until: 19:00,0.21,       !- Field 31
    Until: 20:00,0.15,       !- Field 33
    Until: 21:00,0.17,       !- Field 35
    Until: 22:00,0.08,       !- Field 37
    Until: 24:00,0.05,       !- Field 39
    For: Saturday WinterDesignDay, !- Field 41
    Until: 05:00,0.05,       !- Field 42
    Until: 06:00,0.08,       !- Field 44
    Until: 07:00,0.07,       !- Field 46
    Until: 08:00,0.11,       !- Field 48
    Until: 09:00,0.15,       !- Field 50
    Until: 10:00,0.21,       !- Field 52
    Until: 11:00,0.19,       !- Field 54
    Until: 12:00,0.23,       !- Field 56
    Until: 13:00,0.20,       !- Field 58
    Until: 14:00,0.19,       !- Field 60
    Until: 15:00,0.15,       !- Field 62
    Until: 16:00,0.13,       !- Field 64
    Until: 17:00,0.14,       !- Field 66
    Until: 21:00,0.07,       !- Field 68
    Until: 22:00,0.09,       !- Field 70
    Until: 24:00,0.05,       !- Field 72
    For: AllOtherDays,       !- Field 74
    Until: 05:00,0.04,       !- Field 75
    Until: 06:00,0.07,       !- Field 77
    Until: 11:00,0.04,       !- Field 79
    Until: 13:00,0.06,       !- Field 81
    Until: 14:00,0.09,       !- Field 83
    Until: 15:00,0.06,       !- Field 85
    Until: 21:00,0.04,       !- Field 87
    Until: 22:00,0.07,       !- Field 89
    Until: 24:00,0.04;       !- Field 91

  Schedule:Compact,
    Water Equipment Latent fract sched,  !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.05;       !- Field 3

  Schedule:Compact,
    Water Equipment Sensible fract sched,  !- Name
    Fraction,                !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,0.2;        !- Field 3

  Schedule:Compact,
    SWHSys1 Water Heater Ambient Temperature Schedule Name,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,22.0;       !- Field 3

  Schedule:Compact,
    Water Equipment Temp Sched,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
```

```
  Schedule:Compact,
    Water Equipment Hot Supply Temp Sched,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,43.3;       !- Field 3

  Schedule:Compact,
    SWHSys1 Water Heater Setpoint Temperature Schedule Name,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,60.0;       !- Field 3

  Schedule:Compact,
    SWHSys1-Loop-Temp-Schedule,  !- Name
    Temperature,             !- Schedule Type Limits Name
    Through: 12/31,          !- Field 1
    For: AllDays,            !- Field 2
    Until: 24:00,60.0;       !- Field 3

! ***ECONOMICS***
!   IN_EIAMonthlyRateGas, Source EIA historical Nov2003 thru Oct2004
! Indiana 1999 state average electricity emissions factors based on eGRID, 1065, AirData
! PSI_CS_CommercialElectricService, source http://www.cinergypsi.com/pdfs/RateCS.pdf, effective 2004-05-24
! PSI_LLF_LowLoadFactorService,source http://www.cinergypsi.com/pdfs/RATELLF.pdf, effective 2004-05-24

  UtilityCost:Tariff,
    PSI_LLF_LowLoadFactorService,  !- Name
    Electricity:Facility,    !- Output Meter Name
    kWh,                     !- Conversion Factor Choice
    ,                        !- Energy Conversion Factor
    ,                        !- Demand Conversion Factor
    ,                        !- Time of Use Period Schedule Name
    ,                        !- Season Schedule Name
    ,                        !- Month Schedule Name
    HalfHour,                !- Demand Window Length
    15.00,                   !- Monthly Charge or Variable Name
    ,                        !- Minimum Monthly Charge or Variable Name
    ,                        !- Real Time Pricing Charge Schedule Name
    ,                        !- Customer Baseline Load Schedule Name
    Comm Elect;              !- Group Name

  UtilityCost:Charge:Block,
    AnnualEnergyCharge,      !- Utility Cost Charge Block Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    ,                        !- Remaining Into Variable
    ,                        !- Block Size Multiplier Value or Variable Name
    300,                     !- Block Size 1 Value or Variable Name
    0.108222,                !- Block 1 Cost per Unit Value or Variable Name
    700,                     !- Block Size 2 Value or Variable Name
    0.087021,                !- Block 2 Cost per Unit Value or Variable Name
    1500,                    !- Block Size 3 Value or Variable Name
    0.078420,                !- Block 3 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 4 Value or Variable Name
    0.058320;                !- Block 4 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Block,
    AnnualDemandBaseCharge,  !- Utility Cost Charge Block Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
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    TotalDemand,             !- Block Size Multiplier Value or Variable Name
    190,                     !- Block Size 1 Value or Variable Name
    0.0,                     !- Block 1 Cost per Unit Value or Variable Name
    110,                     !- Block Size 2 Value or Variable Name
    0.051773,                !- Block 2 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 3 Value or Variable Name
    0.046965;                !- Block 3 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    FuelCostAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.002028;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    QualPollutionControlAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000536;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    SoxNoxRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.001127;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    DSMRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000370;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    PurchPowerTrackerAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000031;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    MidwestISOAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000216;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    CleanCoalRiderEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000833;                !- Cost per Unit Value or Variable Name

  UtilityCost:Qualify,
    MinDemand75kw,           !- Utility Cost Qualify Name
```

```
    Minimum,                 !- Qualify Type
    75,                      !- Threshold Value or Variable Name
    Annual,                  !- Season
    Count,                   !- Threshold Test
    12;                      !- Number of Months

  UtilityCost:Charge:Simple,
    TaxofeightPercent,       !- Utility Cost Charge Simple Name
    PSI_LLF_LowLoadFactorService,  !- Tariff Name
    SubTotal,                !- Source Variable
    Annual,                  !- Season
    Taxes,                   !- Category Variable Name
    0.08;                    !- Cost per Unit Value or Variable Name

!end  PSI_LLF_LowLoadFactorService

  UtilityCost:Tariff,
    PSI_CS_CommercialElectricService,  !- Name
    Electricity:Facility,    !- Output Meter Name
    kWh,                     !- Conversion Factor Choice
    ,                        !- Energy Conversion Factor
    ,                        !- Demand Conversion Factor
    ,                        !- Time of Use Period Schedule Name
    ,                        !- Season Schedule Name
    ,                        !- Month Schedule Name
    ,                        !- Demand Window Length
    9.40,                    !- Monthly Charge or Variable Name
    ,                        !- Minimum Monthly Charge or Variable Name
    ,                        !- Real Time Pricing Charge Schedule Name
    ,                        !- Customer Baseline Load Schedule Name
    Comm Elect;              !- Group Name

  UtilityCost:Charge:Block,
    AnnualEnergyCharge,      !- Utility Cost Charge Block Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    ,                        !- Remaining Into Variable
    ,                        !- Block Size Multiplier Value or Variable Name
    300,                     !- Block Size 1 Value or Variable Name
    0.082409,                !- Block 1 Cost per Unit Value or Variable Name
    700,                     !- Block Size 2 Value or Variable Name
    0.072873,                !- Block 2 Cost per Unit Value or Variable Name
    1500,                    !- Block Size 3 Value or Variable Name
    0.061696,                !- Block 3 Cost per Unit Value or Variable Name
    remaining,               !- Block Size 4 Value or Variable Name
    0.041179;                !- Block 4 Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    FuelCostAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.002028;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    QualPollutionControlAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000536;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
```

```
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.001127;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    DSMRiderAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000021;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    PurchPowerTrackerAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000034;                !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    MidwestISOAdjustEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    -0.000203;               !- Cost per Unit Value or Variable Name

  UtilityCost:Charge:Simple,
    CleanCoalRiderEnergyCharge,  !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    0.000807;                !- Cost per Unit Value or Variable Name

  UtilityCost:Qualify,
    MaxDemand75kw,           !- Utility Cost Qualify Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    TotalDemand,             !- Variable Name
    Maximum,                 !- Qualify Type
    75,                      !- Threshold Value or Variable Name
    Annual,                  !- Season
    Count,                   !- Threshold Test
    1;                       !- Number of Months

  UtilityCost:Charge:Simple,
    TaxofeightPercent,       !- Utility Cost Charge Simple Name
    PSI_CS_CommercialElectricService,  !- Tariff Name
    SubTotal,                !- Source Variable
    Annual,                  !- Season
    Taxes,                   !- Category Variable Name
    0.08;                    !- Cost per Unit Value or Variable Name

  UtilityCost:Tariff,
    IN_EIAMonthlyRateGas,    !- Name
    NaturalGas:Facility,     !- Output Meter Name
    MCF,                     !- Conversion Factor Choice
    ,                        !- Energy Conversion Factor
    ,                        !- Demand Conversion Factor
    ,                        !- Time of Use Period Schedule Name
    ,                        !- Season Schedule Name
    ,                        !- Month Schedule Name
    ,                        !- Demand Window Length
    0.0,                     !- Monthly Charge or Variable Name
```

```
    ,                        !- Customer Baseline Load Schedule Name
    Comm Gas;                !- Group Name

  UtilityCost:Charge:Simple,
    MonthlyRateGasCharge,    !- Utility Cost Charge Simple Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    totalEnergy,             !- Source Variable
    Annual,                  !- Season
    EnergyCharges,           !- Category Variable Name
    IN_MonthlyGasRates;      !- Cost per Unit Value or Variable Name

  UtilityCost:Variable,
    IN_MonthlyGasRates,      !- Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    Currency,                !- Variable Type
    8.22,                    !- January Value
    7.51,                    !- February Value
    8.97,                    !- March Value
    9.01,                    !- April Value
    9.16,                    !- May Value
    10.44,                   !- June Value
    10.32,                   !- July Value
    10.13,                   !- August Value
    9.20,                    !- September Value
    8.18,                    !- October Value
    7.83,                    !- November Value
    7.63;                    !- December Value

  UtilityCost:Charge:Simple,
    TaxofEightPercent,       !- Utility Cost Charge Simple Name
    IN_EIAMonthlyRateGas,    !- Tariff Name
    SubTotal,                !- Source Variable
    Annual,                  !- Season
    Taxes,                   !- Category Variable Name
    0.08;                    !- Cost per Unit Value or Variable Name

! ***GENERAL REPORTING***

  OutputControl:ReportingTolerances,
    0.556,                   !- Tolerance for Time Heating Setpoint Not Met {deltaC}
    0.556;                   !- Tolerance for Time Cooling Setpoint Not Met {deltaC}

  Output:SQLite,
    Simple;                  !- Option Type

  Output:VariableDictionary,IDF,Unsorted;

  Output:Surfaces:List,Details;

  Output:Surfaces:Drawing,DXF;

  Output:Constructions,Constructions;

! ***REPORT METERS/VARIABLES***

  Output:Meter,Electricity:Facility,HOURLY;

  Output:Meter,Fans:Electricity,HOURLY;

  Output:Meter,Cooling:Electricity,HOURLY;

  Output:Meter,Heating:Electricity,HOURLY;

  Output:Meter,InteriorLights:Electricity,HOURLY;

  Output:Meter,InteriorEquipment:Electricity,HOURLY;
```

```
  Output:Meter,Heating:NaturalGas,HOURLY;

  Output:Meter,InteriorEquipment:NaturalGas,HOURLY;

  Output:Meter,Water Heater:WaterSystems:NaturalGas,HOURLY;

  Output:Variable,*,Site Outdoor Air Drybulb Temperature,HOURLY;

  Output:Variable,*,Site Outdoor Air Humidity Ratio,HOURLY;

  Output:Variable,*,Site Outdoor Air Relative Humidity,HOURLY;

  Output:Variable,*,Air System Outdoor Air Flow Fraction,HOURLY;

  Output:Variable,*,Air System Simulation Cycle On Off Status,HOURLY;

  Output:Variable,*,Air System Outdoor Air Economizer Status,HOURLY;

  Output:Variable,*,Air System Total Heating Energy,HOURLY;

  Output:Variable,*,Air System Total Cooling Energy,HOURLY;

  Output:Variable,*,Air System Fan Electricity Energy,HOURLY;

  Output:Variable,*,Zone Mean Air Temperature,hourly;

! ***REPORT TABLES***

  OutputControl:Table:Style,
    HTML;                    !- Column Separator

  Output:Table:SummaryReports,
    AnnualBuildingUtilityPerformanceSummary,  !- Report 1 Name
    InputVerificationandResultsSummary,  !- Report 2 Name
    ClimaticDataSummary,     !- Report 3 Name
    EnvelopeSummary,         !- Report 4 Name
    EquipmentSummary,        !- Report 5 Name
    ComponentSizingSummary,  !- Report 6 Name
    HVACSizingSummary,       !- Report 7 Name
    SystemSummary;           !- Report 8 Name

  Output:Table:Monthly,
    Emissions Data Summary,  !- Name
    4,                       !- Digits After Decimal
    CO2:Facility,            !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    NOx:Facility,            !- Variable or Meter 2 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 2
    SO2:Facility,            !- Variable or Meter 3 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 3
    PM:Facility,             !- Variable or Meter 4 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 4
    Hg:Facility,             !- Variable or Meter 5 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 5
    WaterEnvironmentalFactors:Facility,  !- Variable or Meter 6 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 6
    Carbon Equivalent:Facility,  !- Variable or Meter 7 Name
    SumOrAverage;            !- Aggregation Type for Variable or Meter 7

  Output:Table:Monthly,
    Components of Peak Electrical Demand,  !- Name
    3,                       !- Digits After Decimal
    Electricity:Facility,    !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Electricity:Facility,    !- Variable or Meter 2 Name
```

```
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 3
    InteriorEquipment:Electricity,  !- Variable or Meter 4 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 4
    Fans:Electricity,        !- Variable or Meter 5 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 5
    Heating:Electricity,     !- Variable or Meter 6 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 6
    Cooling:Electricity,     !- Variable or Meter 7 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 7
    ExteriorLights:Electricity,  !- Variable or Meter 8 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 8
    Pumps:Electricity,       !- Variable or Meter 9 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 9
    HeatRejection:Electricity,  !- Variable or Meter 10 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 10
    ExteriorEquipment:Electricity,  !- Variable or Meter 11 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 11
    Humidification:Electricity,  !- Variable or Meter 12 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 12
    HeatRecovery:Electricity,!- Variable or Meter 13 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 13
    WaterSystems:Electricity,!- Variable or Meter 14 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 14
    Refrigeration:Electricity,  !- Variable or Meter 15 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 15
    Generators:Electricity,  !- Variable or Meter 16 Name
    ValueWhenMaximumOrMinimum,  !- Aggregation Type for Variable or Meter 16
    ElectricityProduced:Facility,  !- Variable or Meter 17 Name
    ValueWhenMaximumOrMinimum;  !- Aggregation Type for Variable or Meter 17

  Output:Table:Monthly,
    Heating Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Heating Coil NaturalGas Rate,  !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Heating Coil NaturalGas Rate,  !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:Monthly,
    Cooling Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Cooling Coil Electricity Rate,  !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Cooling Coil Electricity Rate,  !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:Monthly,
    Fan Part Load Performance,  !- Name
    0,                       !- Digits After Decimal
    Fan Electricity Rate,    !- Variable or Meter 1 Name
    SumOrAverage,            !- Aggregation Type for Variable or Meter 1
    Fan Electricity Rate,    !- Variable or Meter 2 Name
    Maximum;                 !- Aggregation Type for Variable or Meter 2

  Output:Table:TimeBins,
    *,                       !- Key Value
    Zone Air Relative Humidity,  !- Variable Name
    60,                      !- Interval Start
    10,                      !- Interval Size
    4;                       !- Interval Count

  Output:Table:TimeBins,
    *,                       !- Key Value
    Air System Outdoor Air Flow Fraction,  !- Variable Name
    0.00,                    !- Interval Start
    0.20,                    !- Interval Size
```

```
  Output:Table:TimeBins,
    *,                       !- Key Value
    Availability Manager Night Cycle Control Status,  !- Variable Name
    0,                       !- Interval Start
    1,                       !- Interval Size
    4;                       !- Interval Count

! ***ENVIRONMENTAL FACTORS REPORTING***

  Output:EnvironmentalImpactFactors,
    Monthly;                 !- Reporting Frequency

  EnvironmentalImpactFactors,
    0.663,                   !- District Heating Water Efficiency
    4.18,                    !- District Cooling COP {W/W}
    0.585,                   !- District Heating Steam Conversion Efficiency
    80.7272,                 !- Total Carbon Equivalent Emission Factor From N2O {kg/kg}
    6.2727,                  !- Total Carbon Equivalent Emission Factor From CH4 {kg/kg}
    0.2727;                  !- Total Carbon Equivalent Emission Factor From CO2 {kg/kg}

! Indiana electricity source and emission factors based on Deru and Torcellini 2007

  FuelFactors,
    Electricity,             !- Existing Fuel Resource Name
    3.546,                   !- Source Energy Factor {J/J}
    ,                        !- Source Energy Schedule Name
    3.417E+02,               !- CO2 Emission Factor {g/MJ}
    ,                        !- CO2 Emission Factor Schedule Name
    1.186E-01,               !- CO Emission Factor {g/MJ}
    ,                        !- CO Emission Factor Schedule Name
    7.472E-01,               !- CH4 Emission Factor {g/MJ}
    ,                        !- CH4 Emission Factor Schedule Name
    6.222E-01,               !- NOx Emission Factor {g/MJ}
    ,                        !- NOx Emission Factor Schedule Name
    8.028E-03,               !- N2O Emission Factor {g/MJ}
    ,                        !- N2O Emission Factor Schedule Name
    1.872E+00,               !- SO2 Emission Factor {g/MJ}
    ,                        !- SO2 Emission Factor Schedule Name
    0.0,                     !- PM Emission Factor {g/MJ}
    ,                        !- PM Emission Factor Schedule Name
    1.739E-02,               !- PM10 Emission Factor {g/MJ}
    ,                        !- PM10 Emission Factor Schedule Name
    0.0,                     !- PM2.5 Emission Factor {g/MJ}
    ,                        !- PM2.5 Emission Factor Schedule Name
    0.0,                     !- NH3 Emission Factor {g/MJ}
    ,                        !- NH3 Emission Factor Schedule Name
    1.019E-02,               !- NMVOC Emission Factor {g/MJ}
    ,                        !- NMVOC Emission Factor Schedule Name
    5.639E-06,               !- Hg Emission Factor {g/MJ}
    ,                        !- Hg Emission Factor Schedule Name
    2.778E-05,               !- Pb Emission Factor {g/MJ}
    ,                        !- Pb Emission Factor Schedule Name
    0.4309556,               !- Water Emission Factor {L/MJ}
    ,                        !- Water Emission Factor Schedule Name
    0,                       !- Nuclear High Level Emission Factor {g/MJ}
    ,                        !- Nuclear High Level Emission Factor Schedule Name
    0;                       !- Nuclear Low Level Emission Factor {m3/MJ}

! Deru and Torcellini 2007
! Source Energy and Emission Factors for Energy Use in Buildings
! NREL/TP-550-38617
! source factor and Higher Heating Values from Table 5
! post-combustion emission factors for boiler from Table 9 (with factor of 1000 correction for natural gas)

  FuelFactors,
    NaturalGas,              !- Existing Fuel Resource Name
```

```
    5.21E+01,                !- CO2 Emission Factor {g/MJ}
    ,                        !- CO2 Emission Factor Schedule Name
    3.99E-02,                !- CO Emission Factor {g/MJ}
    ,                        !- CO Emission Factor Schedule Name
    1.06E-03,                !- CH4 Emission Factor {g/MJ}
    ,                        !- CH4 Emission Factor Schedule Name
    4.73E-02,                !- NOx Emission Factor {g/MJ}
    ,                        !- NOx Emission Factor Schedule Name
    1.06E-03,                !- N2O Emission Factor {g/MJ}
    ,                        !- N2O Emission Factor Schedule Name
    2.68E-04,                !- SO2 Emission Factor {g/MJ}
    ,                        !- SO2 Emission Factor Schedule Name
    0.0,                     !- PM Emission Factor {g/MJ}
    ,                        !- PM Emission Factor Schedule Name
    3.59E-03,                !- PM10 Emission Factor {g/MJ}
    ,                        !- PM10 Emission Factor Schedule Name
    0.0,                     !- PM2.5 Emission Factor {g/MJ}
    ,                        !- PM2.5 Emission Factor Schedule Name
    0,                       !- NH3 Emission Factor {g/MJ}
    ,                        !- NH3 Emission Factor Schedule Name
    2.61E-03,                !- NMVOC Emission Factor {g/MJ}
    ,                        !- NMVOC Emission Factor Schedule Name
    1.11E-07,                !- Hg Emission Factor {g/MJ}
    ,                        !- Hg Emission Factor Schedule Name
    2.13E-07,                !- Pb Emission Factor {g/MJ}
    ,                        !- Pb Emission Factor Schedule Name
    0,                       !- Water Emission Factor {L/MJ}
    ,                        !- Water Emission Factor Schedule Name
    0,                       !- Nuclear High Level Emission Factor {g/MJ}
    ,                        !- Nuclear High Level Emission Factor Schedule Name
    0;                       !- Nuclear Low Level Emission Factor {m3/MJ}


  Output:EnergyManagementSystem,
    Verbose,                 !- Actuator Availability Dictionary Reporting
    Verbose,                 !- Internal Variable Availability Dictionary Reporting
    Verbose;                 !- EMS Runtime Language Debug Output Level

  Output:Variable,*,Zone Mean Air Temperature,Hourly;
  Output:Variable,*,Zone Air Relative Humidity,Hourly;
  Output:Variable,*,Zone Thermal Comfort Fanger Model PMV,Hourly;
  Output:Variable,*,Site Outdoor Air Drybulb Temperature,Hourly;
  Output:Variable,*,Facility Total Electricity Demand Rate,Hourly;
```